

Commutative Algebra 1

Labix

August 4, 2026

Abstract

Contents

1	Ideals Of a Commutative Ring	4
1.1	Basic Operations on Ideals	4
1.2	The Nilradical of Commutative Rings	5
1.3	The Jacobson Radical of Commutative Rings	7
1.4	The Radical of an Ideal	7
1.5	The Correspondence between Ideals and the Quotient	9
1.6	Extensions and Contractions of Ideals	11
1.7	Minimal Prime Ideals	13
2	Basic Notions of Commutative Rings	14
2.1	Noetherian Commutative Rings	14
2.2	Artinian Commutative Rings	15
2.3	Commutative Graded Rings	16
2.4	Local Rings	16
2.5	Revisiting the Polynomial Ring	18
3	Modules over a Commutative Ring	21
3.1	Cayley-Hamilton Theorem	21
3.2	Nakayama's Lemma	22
3.3	Properties of the Hom Set	23
3.4	More on Exact Sequences	24
4	Algebra Over a Commutative Ring	26
4.1	Commutative Algebras	26
4.2	Free Commutative Algebras	26
4.3	Finiteness Properties of Algebras	27
5	Localization	29
5.1	Localization of Modules	29
5.2	Localization at Single Elements and Away from Prime Ideals	31
5.3	The Localization Map	31
5.4	Ideals of a Localization	33
5.5	Localization of Graded Rings	35
5.6	Local Properties	36
6	Primary Decomposition	38
6.1	The Annihilator and Associated Primes	38
6.2	The Support of a Module	40
6.3	Primary Ideals	42
6.4	Primary Decomposition	43
6.5	The Two Uniqueness Theorems	44
6.6	Symbolic Powers	45
7	Integral Dependence	46
7.1	Integral Elements	46
7.2	Integral Closure	46
7.3	Four Main Theorems of Integral Extensions	48
7.4	Zariski's Lemma	50
7.5	Normal Domains	51
8	Introduction to Dimension Theory for Rings	55
8.1	Krull Dimension	55
8.2	Noether Normalization Lemma	56
8.3	Height of Prime Ideals	57
8.4	The Length of Modules over Commutative Rings	58
8.5	Structure Theorem for Artinian Rings	59

9	Valuation and Valuation Rings	61
9.1	Valuation Rings	61
9.2	Discrete Valuation Rings	62
10	Dedekind Domains	66
10.1	Fractional Ideals	66
10.2	Invertible Ideals	66
10.3	Dedekind Domains	68
10.4	Prime Factorization of Ideals	69

1 Ideals Of a Commutative Ring

1.1 Basic Operations on Ideals

Recall that $(R, +, \cdot)$ is a ring if the following axioms hold.

- $(R, +)$ is an abelian group.
- Multiplicative Associativity: $x \cdot (y \cdot z) = (x \cdot y) \cdot z$.
- Multiplicative Identity: There exists $1_R \in R$ such that $x \cdot 1_R = x = 1_R \cdot x$ for all $x \in R$.
- Left distributivity: $r \cdot (x + y) = r \cdot x + r \cdot y$ for all $r, x, y \in R$.
- Right distributivity: $(x + y) \cdot r = x \cdot r + y \cdot r$ for all $r, x, y \in R$.

A ring R is commutative if

$$x \cdot y = y \cdot x$$

for all $x, y \in R$.

Let R be a commutative ring. Recall that an ideal of R is a subset $I \subseteq R$ such that

- If $a, b \in I$, then $a + b \in I$.
- If $r \in R$ and $a \in I$, then $ra \in I$.

Lemma 1.1.1 Let R be a commutative ring. Let I, J be ideals of R . Let P be a prime ideal of R . Then the following are equivalent.

- $IJ \subseteq P$.
- $I \cap J \subseteq P$.
- $I \subseteq P$ or $J \subseteq P$.

Proof

- (1) $\implies$ (2): Let $f \in I \cap J$. Then $f \in I$ and $f \in J$ implies that $f^2 \in IJ \subseteq P$. Since P is prime, we conclude that $f \in P$.
- (2) $\implies$ (3): Suppose that $f \in I$ and $f \notin P$. For any $g \in J$, we have $fg \in I \cap J \subseteq P$. Since P is prime and $f \notin P$, we have $g \in P$.
- (3) $\implies$ (1): Without loss of generality suppose that $I \subseteq P$. Then $IJ \subseteq I \subseteq P$.

■

Proposition 1.1.2 (Prime Avoidance) Let R be a commutative ring. Let $I_1, \dots, I_n$ be ideals of R . Let $P_1, \dots, P_k$ be prime ideals of R .

- Let I be an ideal of R . If $I \subseteq \bigcup_{i=1}^k P_i$, then $I \subseteq P_i$ for some i .
- Let P be an ideal of R . If $\bigcap_{i=1}^n I_i \subseteq P$, then $I_i \subseteq P$ for some i .
- Let P be an ideal of R . If $P = \bigcap_{i=1}^n I_i$, then $I_i = P$ for some i .

Proof

- We prove the contrapositive by induction k . When $k = 1$, the case is clear. Suppose that $I \not\subseteq P_i$ for $1 \leq i \leq k-1$ implies $I \not\subseteq \bigcup_{i=1}^{k-1} P_i$. Now suppose that $I \not\subseteq P_i$ for $1 \leq i \leq k$. By induction hypothesis, for each i , there exists $x_j \in I$ such that $x_j \notin \bigcup_{j \neq i} P_j$. So $x_j \notin P_i$ for $j \neq i$. There are two cases. If $x_j \notin P_j$ for some j , then $x_j \notin \bigcup_{j \neq i} P_j \cup P_j = \bigcup_{j=1}^k P_j$ so we are done. If $x_j \in P_j$ for all j , then consider the element $y = \sum_{i=1}^k \prod_{j \neq i} x_j \in I$. Notice that $x_j \in P_j$ for $j \neq i$ implies that $\prod_{j \neq i} x_j$ lie in P_k for any $k \neq i$. It is not an element of P_i because P_i is prime and $x_j \notin P_i$ for $j \neq i$. Then we conclude that y does not lie in P_i for any i . Hence $y \notin \bigcup_{i=1}^k P_i$ and we are done.
- We prove the contrapositive. Suppose that $I_i \not\subseteq P$ for all i . Then for each i , there exists

$x_i \in I_i$ such that $x_i \notin P$. Then $\prod_{i=1}^n x_i \in \bigcap_{i=1}^n I_i$ is not an element of P since P is a prime ideal. Hence we are done.

- By the above, we have that $P = \bigcap_{i=1}^n I_i$ implies that $I_i \subseteq P$ for some i . Then $P = \bigcap_{i=1}^n I_i \subseteq I_i$ implies that $P = I_i$.

■

Example 1.1.3 There is an isomorphism given by

$$\frac{\mathbb{Z}[x]}{(x+1, x^2+2)} \cong \mathbb{Z}/3\mathbb{Z}$$

Proof Using the above propositions, we have that

$$\begin{aligned} \frac{\mathbb{Z}[x]}{(x+1, x^2+2)} &= \frac{\mathbb{Z}[x]}{(x+1) + (x^2+2)} \\ &\cong \frac{\mathbb{Z}[x]/(x+1)}{(3)} \end{aligned}$$

Indeed, the ideal (x^2+2) corresponds to the ideal (3) in $\frac{\mathbb{Z}[x]}{(x+1)}$ because the remainder of x^2+2 divided by $(x+1)$ is (3) . Now $\mathbb{Z}[x]/(x+1) \cong \mathbb{Z}$ by the evaluation homomorphism. Thus quotienting by the ideal (3) gives the field $\mathbb{Z}/3\mathbb{Z}$. ■

Let R be a commutative ring. Recall that two ideals I, J are coprime if $I + J = R$. In particular, this implies that $IJ = I \cap J$. Then the Chinese Remainder theorem reads as

$$\frac{R}{\prod_{i=1}^k I_i} = \frac{R}{\bigcap_{i=1}^k I_i} \cong \prod_{i=1}^k \frac{R}{I_i}$$

More generally, if I and J are not coprime, then the relation between $R/(I \cap J)$ and $R/I \times R/J$ is given by the following exact sequence:

Lemma 1.1.4

Let R be a commutative ring. Let I, J be ideals of R . Then there is a short exact sequence of the form

$$0 \longrightarrow \frac{R}{I \cap J} \xrightarrow{[a] \mapsto ([a], [a])} \frac{R}{I} \oplus \frac{R}{J} \xrightarrow{([a], [b]) \mapsto [a-b]} \frac{R}{I+J} \longrightarrow 0$$

1.2 The Nilradical of Commutative Rings

Let R be a ring. Recall that an element $r \in R$ is nilpotent if $r^n = 0_R$ for some $n \in \mathbb{N}$. When R is commutative, we can form an ideal out of nilpotent elements.

Definition 1.2.1 (Nilradicals) Let R be a commutative ring. Define the nilradical of R to be

$$N(R) = \{r \in R \mid r \text{ is nilpotent}\}$$

Note that this is different from nilpotent ideals, as nilpotency is a property of an ideal. However the Nilradical ideal is a nil ideal and every sub-ideal of the nilradical is a nil ideal.

Proposition 1.2.2 Let R be a ring and $N(R)$ its nilradical. Then the following are true.

- $N(R)$ is an ideal of R
- $N(R/N(R)) = 0$

Proof

- Suppose that r, s are nilpotent, meaning that $r^n = 0$ and $s^m = 0$. Then $(r + s)^{n+m} = 0$. Moreover, if $t \in R$ then $t \cdot r$ is also nilpotent
- Let $r \notin N(R)$. Every element $r + N(R) \in R/N(R)$ has the property that $r^n \neq 0$. Consider $(r + N(R))^n = r^n + N(R)$. If $r^n \in N(R)$ then $r^n = u$ for some nilpotent u , which means that r^n is nilpotent and thus r is nilpotent, a contradiction. This means that $r + N(R) \notin N(R/N(R))$ for all $r \notin N(R)$ and thus $N(R/N(R)) = 0$

■

Proposition 1.2.3 Let R be a commutative ring. Then we have

$$N(R) = \bigcap_{\substack{P \text{ is a prime} \\ \text{ideal of } R}} P$$

Proof Let $x \in N(R)$. Let P be an arbitrary prime ideal. Since x is nilpotent, $x^n = 0$ for some $n \in \mathbb{N}$. If $x \notin P$, then $x^2 \notin P$ since P is a prime ideal. Recursively we see that $x^k \notin P$ for all $k \in \mathbb{N} \setminus \{0\}$. But $x^n = 0 \in P$ is a contradiction. Hence $N(R) \subseteq \bigcap_{P \in \text{Spec}(R)} P$.

Now suppose that $x \in R$ is not nilpotent. Consider the set

$$\Sigma = \{I \trianglelefteq R \mid x^k \notin I \text{ for all } k \geq 1\}$$

Notice that $(0) \in \Sigma$ and hence it is non-empty. Let $I_1 \subseteq I_2 \subseteq \dots$ be a chain in Σ . Define $I = \bigcup_{k=1}^{\infty} I_k$. I claim that $I \in \Sigma$. First of all if $a, b \in I$ and $r \in R$, then $a \in I_m$ and $b \in I_n$ for some $m, n \geq 1$. Then $a, b \in I_{\max\{m, n\}}$ so that $a + b \in I_{\max\{m, n\}} \subseteq I$. Also $ra \in I_m \subseteq I$ since I_m is an ideal. Hence I itself is an ideal of R . Suppose for a contradiction that $x^n \in I$ for some n . Then $x^n \in I_k$ for some k . This is a contradiction since $I_k \in \Sigma$. Thus we know that $I \in \Sigma$. In particular, I is an upper bound of $I_1 \subseteq I_2 \subseteq \dots$. By Zorn's lemma, we conclude that Σ has a maximal element, say P .

Suppose for a contradiction that P is not a prime ideal. Let $ab \in P$ and $a, b \notin P$. Then $P \subsetneq P + (a), P + (b)$. Since P is maximal in Σ , $P + (a)$ and $P + (b)$ cannot be in Σ , and there exists $x^m \in P + (a)$ and $x^n \in P + (b)$ for some m, n . Then

$$x^{m+n} = x^m \cdot x^n \in (P + (a))(P + (b)) = P + (ab)$$

Hence $P + (ab) \notin \Sigma$. But $ab \in P$ implies that $P + (ab) = P$. We have reached a contradiction. Thus P is a prime ideal that does not contain x . We show that $x \notin N(R)$ implies $x \notin P$ for some prime ideal P . The contrapositive of this statement is $x \in P$ for all prime ideals P implies $x \in N(R)$. Hence we are done. ■

Example 1.2.4 Consider the ring

$$R = \frac{\mathbb{C}[x, y]}{(x^2 - y, xy)}$$

Then its nilradical is given by $N(R) = (x, y)$.

Proof Notice that in the ring R , $x^3 = x(x^2) = xy = 0$ and $y^3 = x^6 = (x^3)^2 = 0$ and hence x and y are both nilpotent elements of R . By definition of the nilradical, we conclude that $(x, y) \subseteq N(R)$. Now (x, y) is a maximal ideal of $\mathbb{C}[x, y]$ because $\mathbb{C}[x, y]/(x, y) \cong \mathbb{C}$. Also notice that $(x, y) \supseteq (x^2 - y, xy)$ because for any element $f(x)(x^2 - y) + g(x)(xy) \in (x^2 - y, xy)$, we have that

$$\begin{aligned} f(x)(x^2 - y) + g(x)(xy) &\in (x^2 - y, xy) = (xf(x))x - f(x)y + (g(x)xy) \\ &= (xf(x))x + (xg(x) - f(x))y \in (x, y) \end{aligned}$$

By the correspondence theorem, $(x, y)/(x^2 - y)$ is an maximal ideal of R . In particular, (x, y) is also a prime ideal. But the $N(R)$ is the intersection of all prime ideals and hence $N(R) \subseteq (x, y)$. We conclude that $N(R) = (x, y)$. ■

Definition 1.2.5 (Reduced Rings) Let R be a commutative ring. We say that R is reduced if $N(R) = 0$.

1.3 The Jacobson Radical of Commutative Rings

Let R be a commutative ring. Recall that the Jacobson radical of a ring is defined to be

$$J(R) = \bigcap_{m \text{ a maximal ideal}} m$$

since left and right maximal ideals coincide in R . Properties of the Jacobson radical include:

- $J(R/J(R)) = 0$.

Lemma 1.3.1 Let R be a commutative ring. Then $x \in J(R)$ if and only if $1 - xy \in R^\times$ for all $y \in R$.

Proof Suppose that $x \notin J(R)$. Then $x \notin m$ for some maximal ideal m . Then $R = m + (x)$ since m is maximal. Then there exists $p \in m$ and $y \in R$ such that $1 = p + xy$. Then $1 - xy = p \in m \notin R^\times$.

Suppose that $1 - xy \notin R^\times$ for some $y \in R$. Then $(1 - xy)$ is a proper ideal of R . Then there exists a maximal ideal m such that $(1 - xy) \subseteq m$. If $x \in m$ then $yx \in m$ which implies that $1 = xy + 1 - xy \in m$. This is a contradiction and so $x \notin m$. Hence $x \notin J(R)$. ■

Lemma 1.3.2 Let R be a commutative ring. Then $x \in R$ is a unit if and only if $[x] \in R/J(R)$ is a unit.

Proof Suppose that $x \in R$ is a unit. Then there exists $y \in R$ such that $xy = 1$. Then $[x][y] = [1]$ so we are done. Now suppose that $[x][y] = [1]$ for some $y \in R$. Then there exists $m \in J(R)$ such that $xy = 1 + m$. By the above lemma, $1 + m$ is a unit hence x is a unit. ■

1.4 The Radical of an Ideal

The radical of an ideal is a very different notion from the radical of module.

Definition 1.4.1 (Radical of an Ideal) Let I be an ideal of a ring R . Define the radical of I to be

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \in \mathbb{N}\}$$

Proposition 1.4.2 Let R be a commutative ring. Let I be an ideal. Then the following are true.

- $I \subseteq \sqrt{I}$
- $\sqrt{\sqrt{I}} = \sqrt{I}$
- $\sqrt{I^m} = \sqrt{I}$ for all $m \geq 1$
- $\sqrt{I} = R$ if and only if $I = R$

Proof

- Let $r \in I$. Then $r^1 \in I$ Thus by choosing $n = 1$ we shows that $r^n \in I$. Thus $r \in \sqrt{I}$.
- By the above, we already know that $\sqrt{I} \subseteq \sqrt{\sqrt{I}}$. So let $r \in \sqrt{\sqrt{I}}$. Then there exists some $n \in \mathbb{N}$ such that $r^n \in \sqrt{I}$. But $r^n \in \sqrt{I}$ means that there exists some $m \in \mathbb{N}$ such that $(r^n)^m \in I$. But $nm \in \mathbb{N}$ is a natural number such that $r^{nm} \in I$. Hence $r \in \sqrt{I}$ and so we conclude.

- Since $I^m \subseteq I$, we know that $\sqrt{I^m} \subseteq \sqrt{I}$. Let $x \in \sqrt{I}$. Then $x^n \in I$ for some $n \in \mathbb{N}$. Then we have $(x^n)^m = x^{n+m} \in I^m$ so that $x \in \sqrt{I^m}$.
- Clearly if $I = R$ then $I \subseteq \sqrt{I}$ implies that $\sqrt{I} = R$. Conversely, $\sqrt{I} = R$ implies that $1 \in \sqrt{I}$ and hence $1 \in I$. Hence $I = R$.

■

Proposition 1.4.3 Let R be a commutative ring. Let I, J be ideals of R . Then the following are true.

- If $I \subseteq J$ then $\sqrt{I} \subseteq \sqrt{J}$
- $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$
- $\sqrt{I+J} = \sqrt{\sqrt{I} + \sqrt{J}}$

Proof

- Let $x \in \sqrt{I}$. Then $x^n \in I$ for some $n \in \mathbb{N}$. Then $x^n \in J$ so $x \in \sqrt{J}$.
- Since $IJ \subseteq I \cap J \subseteq I, J$, we already have $\sqrt{IJ} \subseteq \sqrt{I \cap J} \subseteq \sqrt{I} \cap \sqrt{J}$. Let $x \in \sqrt{I} \cap \sqrt{J}$. Then there exists $n, m \in \mathbb{N}$ such that $x^n \in I$ and $x^m \in J$. Then $x^n \cdot x^m = x^{n+m} \in IJ$ implies that $x \in \sqrt{IJ}$.
- Since $I, J \subseteq I+J$, we have $\sqrt{I} + \sqrt{J} \subseteq \sqrt{I+J}$ so that $\sqrt{\sqrt{I} + \sqrt{J}} \subseteq \sqrt{I+J}$. On the other hand, $I \subseteq \sqrt{I}$ and $J \subseteq \sqrt{J}$ implies that $I+J \subseteq \sqrt{I} + \sqrt{J}$. Then $\sqrt{I+J} \subseteq \sqrt{\sqrt{I} + \sqrt{J}}$ and so we are done.

■

Lemma 1.4.4 Let R be a commutative ring. Then we have

$$N(R) = \sqrt{(0)}$$

Proof True from definitions.

■

Lemma 1.4.5 Let R be a commutative ring. Let I be an ideal of R . Let $\pi : R \rightarrow R/I$ be the quotient homomorphism. Then we have

$$\sqrt{I} = \pi^{-1} \left(N \left(\frac{R}{I} \right) \right)$$

Proof Let $x \in R$. Then we have that $x^n \in I$ if and only if $\pi(x^n) = x^n + I = I$ if and only if $x + I \in N(R/I)$.

■

Proposition 1.4.6 Let R be a commutative ring. Let I be an ideal. Then

$$\sqrt{I} = \bigcap_{\substack{p \text{ a prime ideal} \\ I \subseteq p \subseteq R}} p$$

Proof Write $\pi : R \rightarrow R/I$ the quotient homomorphism. Using prp1.2.3 and the correspondence theorem, we have that

$$\sqrt{I} = \pi^{-1} \left(\bigcap_{\substack{P \text{ is a prime} \\ \text{ideal of } R}} P \right) = \bigcap_{\substack{P \text{ is a prime} \\ \text{ideal of } R}} \pi^{-1}(P) = \bigcap_{\substack{p \text{ a prime ideal} \\ I \subseteq p \subseteq R}} p$$

■

Definition 1.4.7 (Radical Ideals) Let R be a commutative ring. Let I be an ideal of R . We say that I is radical if

$$\sqrt{I} = I$$

In particular, by the above lemma it follows that the radical of an ideal is a radical ideal.

Lemma 1.4.8 Let R be a ring. Let P be a prime ideal of R . Then P is radical.

Proof We already know that $P \subseteq \sqrt{P}$. Let $x \in \sqrt{P}$. Then $x^n \in P$ for some $n \in \mathbb{N}$. Since P is prime, by inducting downwards we deduce that $x \in P$. Thus P is radical. ■

We conclude that there is an inclusion of types of ideal in which each inclusion is strict:

$$\text{Maximal ideals} \subset \text{Prime ideals} \subset \text{Radical ideals}$$

Proposition 1.4.9 Let R be a commutative ring. Let I be an ideal of R . Then R/I is reduced if and only if I is a radical ideal.

So radical, prime and maximal ideals all have characterizations using the quotient ring:

- I is maximal if and only if R/I is a field.
- I is prime if and only if R/I is an integral domain.
- I is radical if and only if R/I is reduced.

1.5 The Correspondence between Ideals and the Quotient

Definition 1.5.1 (Max Spectrum of a Ring) Let A be a commutative ring. Define the max spectrum of A to be

$$\text{maxSpec}(A) = \{m \subseteq A \mid m \text{ is a maximal ideal of } A\}$$

Definition 1.5.2 (Spectrum of a Ring) Let A be a commutative ring. Define the spectrum of A to be

$$\text{Spec}(A) = \{p \subseteq A \mid p \text{ is a prime ideal of } A\}$$

Example 1.5.3 Consider the following commutative rings.

- $\text{Spec}(\mathbb{Z}/6\mathbb{Z}) = \{(2 + 6\mathbb{Z}), (3 + 6\mathbb{Z})\}$
- $\text{Spec}(\mathbb{Z}/8\mathbb{Z}) = \{(2 + 8\mathbb{Z})\}$
- $\text{Spec}(\mathbb{Z}/24\mathbb{Z}) = \{(2 + 24\mathbb{Z}), (3 + 24\mathbb{Z})\}$
- $\text{Spec}(\mathbb{R}[x]) = \{(f) \mid f \text{ is irreducible}\}$

Proof

- The only ideals of $\mathbb{Z}/6\mathbb{Z}$ are $(2 + 6\mathbb{Z})$ and $(3 + 6\mathbb{Z})$. We need to find which ones are prime ideals. Now $\mathbb{Z}/6\mathbb{Z} \setminus (2 + 6\mathbb{Z})$ consists of $1 + 6\mathbb{Z}$, $3 + 6\mathbb{Z}$ and $5 + 6\mathbb{Z}$. No multiplication of these elements give an element of $(2 + 6\mathbb{Z})$. So any two elements in $\mathbb{Z}/6\mathbb{Z}$ which multiply to an element of $(2 + 6\mathbb{Z})$ must contain one element that lie in $(2 + 6\mathbb{Z})$. Hence $(2 + 6\mathbb{Z})$ is prime. This is similar for $(3 + 6\mathbb{Z})$. Hence $\text{Spec}(\mathbb{Z}/6\mathbb{Z}) = \{(2 + 6\mathbb{Z}), (3 + 6\mathbb{Z})\}$.
- The only ideals of $\mathbb{Z}/8\mathbb{Z}$ are $(2 + 8\mathbb{Z})$ and $(4 + 8\mathbb{Z})$. A similar argument as above shows that $(2 + 8\mathbb{Z})$ is a prime ideal. However, $6 + 8\mathbb{Z} \notin (4 + 8\mathbb{Z})$ while $(6 + 8\mathbb{Z})^2 = 4 + 8\mathbb{Z} \in (4 + 8\mathbb{Z})$ which shows that $(4 + 8\mathbb{Z})$ is not a prime ideal.
- A similar proof as above ensues.
- Recall that $\mathbb{R}[x]$ is a principal ideal domain. Let $I = (f)$ be a prime ideal of $\mathbb{R}[x]$. Then f is irreducible. Thus every prime ideal of $\mathbb{R}[x]$ is of the form (f) for f an irreducible polynomial.

Lemma 1.5.4 Let R, S be commutative rings. Let $f_1 : R \times S \rightarrow R$ and $f_2 : R \times S \rightarrow S$ denote the projection maps. Then the map

$$f_1^* \amalg f_2^* : \text{Spec}(R) \amalg \text{Spec}(S) \rightarrow \text{Spec}(R \times S)$$

is a bijection.

Proof The core of the proof is the fact that P is a prime ideal of $R \times S$ if and only if $P = R \times Q$ or $P = V \times S$ for either a prime ideal Q of R or a prime ideal V of S . It is clear that if Q is a prime ideal of S and V is a prime ideal of R , then $R \times Q$ and $V \times S$ are both prime ideals of $R \times S$.

So suppose that P is a prime ideal in $R \times S$. Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Since $P \neq R \times S$, at least one of e_1 or e_2 is not in P . Without loss of generality assume that $e_1 \notin P$. But $e_1 e_2 = 0 \in P$ and P being prime implies that $e_2 \in P$. Since e_2 is the identity of $\{0\} \times S \cong S$, we conclude that $\{0\} \times S \subseteq P$. By the correspondence theorem, the projection map $f_1 : R \times S \rightarrow R$ gives a bijection between prime ideals of $R \times S$ that contain $\{0\} \times S$ and prime ideals of R . So $f_1(P)$ is a prime ideal of R . Thus $P = f_1(P) \times S$ which is exactly what we wanted.

Now the bijection is clear. $f_1^* \amalg f_2^*$ sends a prime ideal P of R to $P \times S$ and it sends a prime ideal Q of S to $R \times Q$. This map is surjective by the above argument. It is injective by inspection. ■

Theorem 1.5.5 Let R be a commutative ring. Let I be an ideal of R . Denote φ to be the inclusion preserving one-to-one bijection

$$\{\text{Ideals of } R \text{ containing } I\} \xrightarrow{1:1} \{\text{Ideals of } R/I\}$$

from the correspondence theorem for rings. In other words, $\varphi(A) = A/I$. Let $J \subseteq R$ be an ideal containing I . Then the following are true.

- J is a radical ideal if and only if $\varphi(J) = J/I$ is a radical ideal.
- J is a prime ideal if and only if $\varphi(J) = J/I$ is a prime ideal and so φ induces an inclusion preserving bijection

$$\text{Spec}(R/I) \xrightarrow{1:1} \{P \in \text{Spec}(R) \mid I \subseteq P\}$$

- J is a maximal ideal if and only if $\varphi(J) = J/I$ is a maximal ideal and so φ induces an inclusion preserving bijection

$$\text{maxSpec}(R/I) \xrightarrow{1:1} \{m \in \text{maxSpec}(R) \mid I \subseteq m\}$$

Proof

- Let J be a radical ideal. Suppose that $r + I \in \sqrt{J/I}$. This means that $(r + I)^n = r^n + I \in J/I$ for some $n \in \mathbb{N}$. But this means that $r^n \in J$. This implies that $r \in \sqrt{J} = J$. Thus $r + I \in J/I$ and we conclude that $\sqrt{J/I} \subseteq J/I$. Since we also have $J/I \subseteq \sqrt{J/I}$, we conclude.

Now suppose that J/I is a radical ideal. Let $r \in \sqrt{J}$. This means that $r^n \in J$ for some $n \in \mathbb{N}$. Now $r^n + I = (r + I)^n \in J/I$ implies that $r + I \in \sqrt{J/I} = J/I$. Hence $r \in J$ and so $\sqrt{J} \subseteq J$. Since we also have that $J \subseteq \sqrt{J}$, we conclude.

- Let J be a prime ideal. Then R/J is an integral domain. By the second isomorphism theorem, we have that $R/J \cong (R/I)/(J/I)$ and hence $(R/I)/(J/I)$ is also an integral domain. Hence J/I is a prime ideal. The converse is also true.
- Let J be a maximal ideal. Then R/J is a field. By the second isomorphism theorem, we have that $R/J \cong (R/I)/(J/I)$ and hence $(R/I)/(J/I)$ is also a field. Hence J/I is a maximal ideal.

The converse is also true.

1.6 Extensions and Contractions of Ideals

Definition 1.6.1 (Extension of Ideals) Let R, S be commutative rings. Let $f : R \rightarrow S$ be a ring homomorphism. Let I be an ideal of R . Define the extension I^e of I to S to be the ideal

$$I^e = \langle f(i) \mid i \in I \rangle$$

Proposition 1.6.2 Let R, S be commutative rings. Let $f : R \rightarrow S$ be a ring homomorphism. Let I, I_1, I_2 be an ideal of R . Then the following are true regarding the extension of ideals.

- If $I_1 \subseteq I_2$, then $I_1^e \subseteq I_2^e$.
- Closed under sum: $(I_1 + I_2)^e = I_1^e + I_2^e$
- $(I_1 \cap I_2)^e \subseteq I_1^e \cap I_2^e$
- Closed under products: $(I_1 I_2)^e = I_1^e I_2^e$
- $(\sqrt{I})^e \subseteq \sqrt{I^e}$

Proof

- Let $x \in I_1^e$. Then $x = \sum s_k f(i_k)$ for some $i_k \in I_1$. Then $i_k \in I_2$ implies that $x \in I_2^e$.
- Since $I_1, I_2 \subseteq I_1 + I_2$, we have $I_1^e + I_2^e \subseteq (I_1 + I_2)^e$. Conversely, let $x \in (I_1 + I_2)^e$. Then $x = \sum s_k f(i_k)$ for $i_k \in I_1 + I_2$. Then we have

$$x = \sum_{i_k \in I_1} s_k f(i_k) + \sum_{i_k \in I_2} s_k f(i_k) \in I_1^e + I_2^e$$

so we conclude.

- Since $I_1 \cap I_2 \subseteq I_1, I_2$ we are done.
- It suffices to check the generators lie in each other. Let $x \in I_1 I_2$. Then $x = \sum i_k j_k$ for some $i_k \in I_1$ and $j_k \in I_2$. Then $f(x) = \sum f(i_k) f(j_k)$. Since $f(i_k) \in I_1^e$ and $f(j_k) \in I_2^e$, then $f(x) \in I_1^e I_2^e$ so we conclude that $(I_1 I_2)^e \subseteq I_1^e I_2^e$. Conversely, suppose that $x \in I_1^e I_2^e$. Then $x = \sum f(i_k) f(j_k)$ for $i_k \in I_1$ and $j_k \in I_2$. Since f is a ring homomorphism, we have that

$$x = \sum f(i_k) f(j_k) = f\left(\sum i_k j_k\right)$$

Since $\sum i_k j_k \in I_1 I_2$, we conclude that $x \in I_1^e I_2^e$.

- We have that

$$(\sqrt{I})^e = \left(f(i) \mid i \in \bigcap_{\substack{P \text{ prime} \\ I \subseteq P}} P \right) \subseteq f\left(\bigcap_{\substack{P \text{ prime} \\ I \subseteq P}} f(P) \right) \subseteq f\left(\bigcap_{\substack{Q \text{ prime} \\ I^e \subseteq Q}} f(f^{-1}(Q)) \right)$$

The last inclusion follows since for $I^e \subseteq Q$, we must have that $I \subseteq f^{-1}(Q)$. Then we have that

$$(\sqrt{I})^e = f\left(\bigcap_{\substack{Q \text{ prime} \\ I^e \subseteq Q}} Q \right) = \sqrt{I^e}$$

and so we are done.

Definition 1.6.3 (Contraction of Ideals) Let R, S be commutative rings. Let $f : R \rightarrow S$ be a ring homomorphism. Let J be an ideal of S . Define the contraction J^c of J to R to be the ideal

$$J^c = f^{-1}(J)$$

Proposition 1.6.4 Let R, S be commutative rings. Let $f : R \rightarrow S$ be a ring homomorphism. Let J, J_1, J_2 be an ideal of S . Then the following are true regarding the extension of ideals.

- If $J_1 \subseteq J_2$, then $J_1^c \subseteq J_2^c$.
- $(J_1 + J_2)^c \supseteq J_1^c + J_2^c$
- Closed under intersections: $(J_1 \cap J_2)^c = J_1^c \cap J_2^c$
- $(J_1 J_2)^c \supseteq J_1^c J_2^c$
- Closed under taking radicals: $\text{rad}(J)^c = \text{rad}(J^c)$

Proof

- Clear since $f^{-1}(J_1) \subseteq f^{-1}(J_2)$ for $J_1 \subseteq J_2$.
- Since $J_1, J_2 \subseteq J_1 + J_2$, we have that $J_1^c + J_2^c \subseteq (J_1 + J_2)^c$.
- Since $J_1 \cap J_2 \subseteq J_1, J_2$, we have that $(J_1 \cap J_2)^c \subseteq J_1^c \cap J_2^c$. Let $x \in J_1^c \cap J_2^c$. Then we have $f(x) \in J_1, J_2$ so that $f(x) \in J_1 \cap J_2$. Hence $x \in (J_1 \cap J_2)^c$.
- Suppose that $x \in J_1^c$ and $y \in J_2^c$. Then $f(xy) = f(x)f(y) \in J_1^c J_2^c$. Hence $xy \in J_1^c J_2^c$.
-

■

Proposition 1.6.5 Let R, S be commutative rings. Let $f : R \rightarrow S$ be a ring homomorphism. Let I be an ideal of R and let J be an ideal of S . Then the following are true.

- $I \subseteq I^{ec}$
- $J^{ce} \subseteq J$
- $I^e = I^{ece}$
- $J^c = J^{cec}$

Proof

- Let $x \in I$. Then $f(x) \in I^e$. Thus $x \in f^{-1}(I^e)$.
- Since J^{ce} is generated by $f(x)$ for all $x \in J^c$, it suffices to check that $f(x) \in J$ for all $x \in J^c$. But $x \in J^c$ implies that $f(x) \in J$ so we are done.
- Since $I \subseteq I^{ec}$, we know that $I^e \subseteq I^{ece}$. Also, from the second item we take $J = I^e$ to get $I^{ece} \subseteq I^e$.
- From the first item, take $I = J^c$ to get $J^c \subseteq J^{cec}$. Also, since $J^{ce} \subseteq J$, we have that $J^{cec} \subseteq J^c$.

■

Example 1.6.6 Let S be a commutative ring and let $R \subseteq S$ be a subring. Let $f : R \rightarrow S$ be the inclusion map. Let $I \subseteq R$ be an ideal of R and let $J \subseteq S$ be an ideal of S . Then the following are true.

- $I^e = S \cdot I$.
- $J^c = J \cap R$.

1.7 Minimal Prime Ideals

Definition 1.7.1 (Minimal Prime Ideals) Let R be a commutative ring. Let I be an ideal of R . Let P be a prime ideal of R . We say that P is a minimal prime ideal over I if for any other prime ideal $Q \supseteq I$ containing I , we have $P \subseteq Q$.

Proposition 1.7.2 Let R be a commutative ring. Let I be an ideal of R . Then a minimal prime ideal over I exists.

2 Basic Notions of Commutative Rings

2.1 Noetherian Commutative Rings

We recall some facts about Noetherian rings. In the following, let R be a commutative ring, although they are also true if R is non-commutative if we take all modules defined below to be left (right) R -modules.

- If we have a short exact sequence of R -modules:

$$0 \longrightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$$

Then M_2 is Noetherian if and only if M_1 and M_3 are Noetherian.

- If M and N are R -modules, then $M \oplus N$ is Noetherian if and only if M and N are Noetherian.
- If M is an R -module and N is an R -submodule of M , then M is Noetherian if and only if N and M/N are Noetherian.
- If R is Noetherian and I is an ideal of R , then R/I is Noetherian.
- Later when once has seen localization, we can also prove that: If R is Noetherian then $S^{-1}R$ is Noetherian for any multiplicative subset S of R .

Proposition 2.1.1 Let R be a Noetherian commutative ring. Let I be an ideal of R . Then there exists $n \in \mathbb{N}$ such that

$$\sqrt{I}^n \subseteq I \subseteq \sqrt{I}$$

Proof It is clear that $I \subseteq \sqrt{I}$. Since R is Noetherian, $\sqrt{I}$ is finitely generated by say $x_1, \dots, x_n$. Then $x_i^{n_i} \in I$ for some $n_i \in \mathbb{N}$. Let $m = 1 + \sum_{i=1}^n (n_i - 1)$. Then $\sqrt{I}^m$ is generated by $x_1^{r_1} \cdots x_n^{r_n}$ for $\sum_{i=1}^n r_i = m$. If $r_i < n_i$ for i then

$$m = \sum_{i=1}^n r_i \leq \sum_{i=1}^n (n_i - 1) < m$$

is a contradiction. Hence there exists some i for which $r_i \geq n_i$. Thus $x_1^{r_1} \cdots x_n^{r_n} \in I$. Thus $\sqrt{I}^m \subseteq I$. ■

Proposition 2.1.2 Let R be a Noetherian commutative ring. Then $N(R)$ is a nilpotent ideal.

Proof By the above, there exists $n \in \mathbb{N}$ such that $(N(R))^n = \sqrt{(0)}^n \subseteq (0) \subseteq \sqrt{(0)}$. Hence $(N(R))^n = (0)$ for some $n \in \mathbb{N}$. ■

Proposition 2.1.3

Let R be a Noetherian commutative ring. Let I be an ideal of R . Then I has only finitely many minimal prime ideals.

Proof

Define the set

$$\mathcal{F} = \{I \subseteq R \mid I \text{ is an ideal of } R \text{ that has infinitely many minimal prime ideals}\}$$

Assume for a contradiction that the set is non-empty. Since R is Noetherian, every chain in $\mathcal{F}$ has a maximal element. Let I be such a maximal element of the set. I is not a prime ideal because prime ideals only have one minimal prime ideal by definition. This means that there exists $a, b \notin I$ such that $ab \in I$. Now $I + (a)$ and $I + (b)$ both contain I , and since $ab \in I$ we have that $(I + (a))(I + (b)) \subseteq I$. For any minimal prime ideal P of I , we have that $(I + (a))(I + (b)) \subseteq I \subseteq P$ implies that $(I + (a)) \subseteq P$ or $(I + (b)) \subseteq P$. This shows that the set of minimal prime ideals of I is a subset of the union of the set of minimal prime ideals of $I + (a)$ and $I + (b)$, and so without loss of generality, $I + (a)$ has infinitely many minimal prime ideals. But then $I + (a)$ strictly contains I and $I + (a) \in \mathcal{F}$. This is a contradiction since I is assumed to be maximal. ■

2.2 Artinian Commutative Rings

Let R be a commutative ring. Recall that R is Artinian if any descending chain of ideals

$$I_1 \supseteq I_2 \supseteq \cdots$$

terminates at finitely many steps, meaning $I_k = I_{k+n}$ for some $k \in \mathbb{N}$.

Generally, if R is Artinian then the following are true.

- $J(R)$ is a nilpotent ideal.
- R is Noetherian.
- R has finite length.

There are also properties of Artinian rings that only commutative rings can realize.

Proposition 2.2.1 Let R be an integral domain. Then R is Artinian if and only if R is a field.

Proof It is clear that every field is Artinian. Conversely, let R be Artinian. Consider the following descending chain of ideals in R :

$$R \supseteq (x) \supseteq (x^2) \supseteq$$

for any $0 \neq x \in R$. Since R is Artinian, the chain terminates and $(x^n) = (x^{n+1})$ for some $n \in \mathbb{N}$. Then there exists $y \in R$ such that $x^n = yx^{n+1}$. This means that $x^n(1 - yx) = 0$. Since R is an integral domain, R has no nilpotents. Hence x^n is non-zero and $1 = xy$. Thus x has an inverse so that R is a field. ■

Proposition 2.2.2 Let R be an Artinian commutative ring. Then the following are true.

- $\text{Spec}(R) = \text{maxSpec}(R)$.
- $N(R) = J(R)$

Proof Let P be a prime ideal. Since quotients of Artinian rings are Artinian, R/P is Artinian. Since R/P is also an integral domain, we conclude by the above that R/P is a field. Hence P is maximal.

Since every prime ideal in R is maximal, we have that

$$N(R) = \bigcap_{P \text{ a prime ideal}} P = \bigcap_{P \text{ a maximal ideal}} P = J(R)$$

and so we conclude. ■

Proposition 2.2.3 Let R be a commutative ring. If R is Artinian, then R has finitely many maximal ideals.

Proof Consider the collection

$$\{m_1 \cap \cdots \cap m_k \mid m_1, \dots, m_k \text{ are maximal ideals of } R\}$$

of R -submodules of R . Since R is Artinian, every collection of R -submodules of R has a minimal element. Hence this collection also has a minimal element, say $m_1 \cap \cdots \cap m_k$. Let m be another maximal ideal of R . Then

$$m \cap m_1 \cap \cdots \cap m_k \subseteq m_1 \cap \cdots \cap m_k$$

Since $m_1 \cap \cdots \cap m_k$ is minimal, they are equal. By prime avoidance, we conclude that $m \supseteq m_i$ for some i . Since they are maximal, we have $m = m_i$. Hence $m_1, \dots, m_k$ gives the full list of distinct maximal ideals of R . ■

2.3 Commutative Graded Rings

Not to be confused with graded commutative rings, commutative graded rings are commutative rings that admit a grading.

Lemma 2.3.1

Let R be a commutative graded ring. Let I be a homogeneous ideal of R . Then $\sqrt{I}$ is also homogeneous.

Let $R = \bigoplus_{n=0}^{\infty} R_n$ be a commutative graded ring. Recall that the irrelevant ideal is defined to be

$$R_+ = \bigoplus_{n=1}^{\infty} R_n$$

Moreover, for a homogeneous ideal $I \subseteq R$, we defined $I_+ = I \cap R_+$.

Lemma 2.3.2

Let R be a graded ring. Let $I \subseteq R$ be a homogeneous ideal. Then we have

$$\sqrt{I}_+ = \bigcap_{I \subseteq P \text{ is relevant}} P$$

Proof We have seen that $I \subseteq P$, then $\sqrt{I} \subseteq P$. Now let $f \notin \sqrt{I}$. Define

$$S = \{I \subseteq S_+ \mid I \text{ is a homogeneous ideal such that } I \cap \{f^n \mid n \in \mathbb{N}\} = \emptyset \text{ and } I \subseteq P\}$$

(Apply Zorn's lemma then prove that the maximal element is prime, then f is not contained in the intersection of all relevant ideals containing I). ■

2.4 Local Rings

Definition 2.4.1 (Local Rings) Let R be a commutative ring. We say that R is a local ring if it has a unique maximal ideal m . In this case, we say that R/m is the residue field of R .

Example 2.4.2 Consider the following commutative rings.

- $\mathbb{Z}/6\mathbb{Z}$ is not a local ring.
- $\mathbb{Z}/8\mathbb{Z}$ is a local ring.
- $\mathbb{Z}/24\mathbb{Z}$ is not a local ring.
- $\mathbb{R}[x]$ is not a local ring.

Proof

- The only ideals of $\mathbb{Z}/6\mathbb{Z}$ are $(2 + 6\mathbb{Z})$ and $(3 + 6\mathbb{Z})$. They do not contain each other and so they are both maximal.
- The only ideals of $\mathbb{Z}/8\mathbb{Z}$ are $(2 + 8\mathbb{Z})$ and $(4 + 8\mathbb{Z})$. But $(2 + 8\mathbb{Z}) \supseteq (4 + 8\mathbb{Z})$. Hence $\mathbb{Z}/8\mathbb{Z}$ has a unique maximal ideal.
- A similar proof as above ensues.
- Any irreducible polynomial $f \in \mathbb{R}[x]$ is such that (f) is a maximal ideal. Indeed the evaluation homomorphism gives an isomorphism $\frac{\mathbb{R}[x]}{(f)} \cong \mathbb{R}$. ■

Proposition 2.4.3 Let R be a ring and I an ideal of R . Then I is the unique maximal ideal of R if and only if I is the set containing all non-units of R .

Proof Let I be the unique maximal ideal of R . Clearly I does not contain any unit else $I = R$. Now suppose that r is a non-unit. Suppose that $r \notin I$. Define $J = \{sr | s \in R\}$. Clearly J is an ideal. It must be contained in some maximal ideal. Since I is the unique maximal ideal, $J \subseteq I$. But this means that $r \in I$, a contradiction. Thus every non-unit is in I . Suppose that I contains all non-units of R . Let $r \notin I$. Then there exists $s \notin I$ such that $rs = 1$. Then $(r + I)(s + I) = 1 + I$ in R/I . This means that every element of R/I has a multiplicative inverse which means that R/I is a field and thus I is a maximal ideal. Now let $J \neq I$ be another maximal ideal. Then J contains some unit r . This implies that $J = R$ and thus I is the unique maximal ideal. ■

Example 2.4.4 Let k be a field. Then the ring of power series $k[[x]]$ is a local ring.

Proof Let M be the set of all non-units of $k[[x]]$. I first show that $f \in M$ if and only if the constant term of f is non-zero. Let g be a power series. Then the n th coefficient of $f \cdot g$ is given by

$$c_n = \sum_{k=0}^n a_k b_{n-k}$$

If the constant term of f is 0, then $c_0 = 0$ and so $f \cdot g \neq 1$. Now if the constant term of f is $a_0 \neq 0$, then set $b_0 = \frac{1}{a_0}$. Now we can use the formula $0 = c_n$ to deduce

$$b_n = -\frac{\sum_{k=1}^n a_k b_{n-k}}{a_0}$$

This is such that $a_n \cdot b_n = 0$. Define $g = \sum_{k=0}^{\infty} b_k x^k$. Then $f \cdot g = 1$. Thus f is a unit.

By the above proposition, we conclude that M is the unique maximal ideal of $k[[x]]$. ■

Proposition 2.4.5 Let R be a commutative ring. Then the following are equivalent.

- R has exactly one prime ideal. (It is given by $N(R)$).
- Every element of R is either a unit or nilpotent.
- $N(R)$ is a maximal ideal.

Under these equivalent assumptions, $(R, N(R))$ is a local ring.

Proof

- (1) $\implies$ (2): We know that $N(R)$ is a prime ideal, hence it is the unique prime ideal and unique maximal ideal. Thus R is a local ring. By the above, elements of $R \setminus N(R)$ are units and element of $N(R)$ are nilpotent.
- (2) $\implies$ (3): It is clear that every nilpotent is a non-unit. By assumption, non-units of R are nilpotents. Hence $N(R)$ is the set of all non-units. Since $N(R)$ is an ideal, by the above we conclude that $(R, N(R))$ is a local ring. In particular, $N(R)$ is the unique maximal ideal of R .
- (3) $\implies$ (1): Suppose that $N(R)$ is a maximal ideal. Let $P \neq R$ be a prime ideal of R . Since $N(R)$ is the intersection of all prime ideals, we have $N(R) \subseteq P$. By the correspondence theorem, P corresponds to a prime ideal of $R/N(R)$. But $R/N(R)$ is a field, and since $P \neq R$ we must have that $P = N(R)$. Thus $N(R)$ is the unique prime ideal of R . ■

Proposition 2.4.6 Let R be a Noetherian commutative ring. Then the following are equivalent.

- R is an Artinian local ring.
- R has a nilpotent maximal ideal.

- R has a unique proper radical ideal.
- R has a unique prime ideal.
- $N(R)$ is a maximal ideal of R .

Proof

- (1) $\implies$ (2): Let R be Artinian and local. By 2.1.4 we have $N(R) = J(R) = m$ since $J(R)$ is the intersection of all maximal ideals. Since R is Noetherian, by 2.1.3 $N(R) = m$ is nilpotent.

■

Since every Artinian ring is Noetherian, the above proposition implies the following.

Corollary 2.4.7 Let R be an Artinian local ring. Then the following are true.

- $N(R)$ is the unique maximal / prime / radical ideal of R .
- $N(R)$ is a nilpotent ideal.

We will discuss more of local rings in the topic of localizations.

2.5 Revisiting the Polynomial Ring

Lemma 2.5.1 Let R be a commutative ring. Then $R[x]$ has infinitely many irreducible polynomials.

Proof If not, then there exists a finite list of irreducible polynomials $f_1, \dots, f_k$. Then $1 + f_1, \dots, f_k$ is not divisible by $f_1, \dots, f_k$ and so must contain a monic irreducible factor not equal to $f_1, \dots, f_k$. This is a contradiction. ■

Proposition 2.5.2 Let R be a commutative ring. Then we have

$$N(R[x]) = N(R)[x]$$

Proof Let $f = \sum_{k=0}^n a_k x^k \in N(R)[x]$. Then each a_k is nilpotent in R , and there exists $n_k \in \mathbb{N}$ such that $a_k^{n_k} = 0$. This also proves that $a_k x^k$ is nilpotent. Since the sum of nilpotents is a nilpotent, we conclude that f is nilpotent.

Now suppose that $f \in N(R[x])$. We induct on the degree of f . Let $\deg(f) = 0$. Then f is nilpotent and f lies in R . Thus $f \in N(R)[x]$. Now suppose that the claim is true for $\deg(f) \leq n - 1$. Let $\deg(g) = n$ with leading coefficient b_n . Since g is nilpotent in $R[x]$, there exists $m \in \mathbb{N}$ such that $g^m = 0$. Then in particular, $b_n^m = 0$ so that b_n is nilpotent. Then $b_n x^n$ is also nilpotent. Now since $N(R[x])$ is an ideal of $R[x]$, we have that $g - b_n x^n \in N(R[x])$. By inductive hypothesis, $g - b_n x^n \in N(R)[x]$. Since $N(R)$ is an ideal of R , we have that $N(R)[x]$ is an ideal of $R[x]$. So $g = (g - b_n x^n) + b_n x^n \in N(R)[x]$. Thus we are done. ■

Theorem 2.5.3 (Hilbert's Basis Theorem) Let R be a commutative ring. If R is Noetherian, then $R[x]$ is a Noetherian ring.

Proof It suffices to show that every ideal of $R[x]$ is finitely generated. Let I be an ideal of $R[x]$. Let $I^{\leq n}$ be the ideal generated by

$$I^{\leq n} = (f \in I \mid \deg(f) \leq n)$$

Notice that $I^{\leq n}$ is an R -submodule of $\bigoplus_{i=0}^n R \cdot x^i$. Since R is Noetherian, $I^{\leq n}$ is finitely generated as an R -module. In particular, $I^{\leq n}$ is finitely generated as an $R[x]$ -module with the same finite generating set.

I claim that the chain of ideals

$$I^{\leq 0} \subseteq I^{\leq 1} \subseteq \dots \subseteq I^{\leq k} \subseteq I = \bigcup_{i=0}^{\infty} I^{\leq i}$$

of $R[x]$ eventually stabilizes. Let $LC(f)$ be the leading coefficient of $f \in R[x]$. The define

$$LC(I) = \{LC(f) \mid f \in I\}$$

Notice that $LC(I)$ is an ideal of R . Since R is Noetherian, $LC(I)$ is finitely generated as an R -module by say $a_1, \dots, a_r$. This means that there exists $f_1, \dots, f_r \in R[x]$ such that $LC(f_i) = a_i$. Let $d = \max\{\deg(f_1), \dots, \deg(f_r)\}$. Without loss of assumption we can replace f_i with $x^{d-\deg(f_i)} f_i$ so that $f_1, \dots, f_r$ have the same degree d .

I claim that $I^{\leq n} = I^{\leq n+1}$ for $n \geq d$. $I^{\leq n} \subseteq I^{\leq n+1}$ is trivial. Suppose that $f \in I^{\leq n+1}$. If $\deg(f) \leq n$ then we are done. So suppose that $\deg(f) = n+1$. Then the leading coefficient of f is a linear combination of the leading coefficients of $f_1, \dots, f_r$. So there exists $b_1, \dots, b_r \in R$ such that $LC(f) = \sum_{i=1}^r b_i LC(f_i)$. Then $f - (\sum_{i=1}^r b_i f_i) x^{n+1-d} \in I^{\leq n}$. Since $\sum_{i=1}^r b_i f_i \in I^{\leq d} \subseteq I^{\leq n}$, we conclude that $f \in I^{\leq n}$. We conclude. ■

Proposition 2.5.4

Let $R = \bigoplus_{n \in \mathbb{N}} R_n$ be a graded ring where R_0 is a commutative ring. Then the following are equivalent.

- R is Noetherian.
- R_0 is Noetherian and R_+ is finitely generated.
- R_0 is Noetherian and R is a finitely generated R_0 -algebra.

Proof

- (1) $\implies$ (2): Suppose that R is Noetherian. Then $R_0 \cong R/R_+$ is a quotient of R and so is Noetherian. Since R_+ is a homogeneous ideal of R , it is finitely generated.
- (2) $\implies$ (3): Suppose that R_+ is finitely generated by $x_1, \dots, x_r$ where each x_i is of degree d_i . Then we have

$$R_n = \sum_{i=1}^r x_i R_{n-d_i}$$

Indeed, the expression on the right lies in R_n . Conversely, given $x \in R_n$, there exists $f_1, \dots, f_r \in R$ such that $x = \sum_{i=1}^r x_i f_i$. If g_i is the homogeneous term of f_i of degree d_i , then we have $x = \sum_{i=1}^r x_i g_i$. Hence R_n is a subset of the expression on the right. This implies that $R = R_0[x_1, \dots, x_r]$ and so R is a finitely generated R_0 -algebra.

- (3) $\implies$ (1): By Hilbert's basis theorem we know that R is Noetherian. ■

Some more important results from Groups and Rings and Rings and Modules include:

- If R is an integral domain, then $R[x]$ is an integral domain.
- R is a UFD if and only if $R[x]$ is a UFD
- If F is a field, then $F[x]$ is an Euclidean domain, a PID and a UFD
- If F is a field, then the ideal generated by p is maximal if and only if p is irreducible.

Regarding ideals of the polynomial ring, the following maybe useful:

- $I[x]$ is an ideal of R

- There is an isomorphism $\frac{R[x]}{I[x]} \cong \frac{R}{I}[x]$ given by the map

$$\left(f = \sum_{k=0}^n a_k x^k + I[x] \right) \mapsto \left(\sum_{k=0}^n (a_k + I) x^k \right)$$

- If I is a prime ideal of R , then $I[x]$ is a prime ideal of $R[x]$.

3 Modules over a Commutative Ring

Recall from Rings and Modules that a module consists of an abelian group M and a ring R such that there is a binary operation $\cdot : R \times M \rightarrow M$ that mimic the notion of a group action:

- For $r, s \in R$, $s \cdot (r \cdot m) = (sr) \cdot m$ for all $m \in M$.
- For $1_R \in R$ the multiplicative identity, $1_R \cdot m = m$ for all $m \in M$.

When R is a commutative ring, the first axiom is relaxed so that the resulting element of M makes no difference whether you apply r first or s first. This makes module act even more similarly than fields (although one still need the notion of a basis, which appears in free modules). Therefore the first section concerns transferring techniques in linear algebra such as the Cayley Hamilton theorem to module over a ring that mimic the notion of vector spaces.

3.1 Cayley-Hamilton Theorem

Definition 3.1.1 (Characteristic Polynomial) Let R be a commutative ring. Let $A \in M_{n \times n}(R)$ be a matrix. Define the characteristic polynomial of A to be the polynomial

$$c_A(x) = \det(A - xI)$$

Theorem 3.1.2 (Cayley-Hamilton Theorem for Rings) Let R be a commutative ring. Let $A \in M_{n \times n}(R)$ be a matrix. Then $c_A(A) = 0$.

Theorem 3.1.3 (Cayley-Hamilton Theorem for Modules) Let R be a commutative ring. Let M be a finitely generated R -module. Let I be an ideal of R . Let $\varphi \in \text{End}_R(M)$. If $\varphi(M) \subseteq IM$, then there exists $a_1, \dots, a_{n-1} \in I$ such that

$$\varphi^n + a_1\varphi^{n-1} + \dots + a_{n-1}\varphi + \text{id}_M = 0 : M \rightarrow M$$

Proof Suppose that M is generated by $x_1, \dots, x_n$. There exists a surjective map $\rho : R^n \rightarrow M$ given by $(r_1, \dots, r_n) \mapsto \sum_{k=1}^n r_k x_k$. Since $\varphi(M) \subseteq IM$, we have that

$$\varphi(x_k) = \sum_{i=1}^n r_{ki} x_i$$

for some $r_{ki} \in I$. Write A to be the matrix $A = (a_{ki})$. We now have a commutative diagram: In other words, we have the diagram:

$$\begin{array}{ccc} R^n & \xrightarrow{\rho} & M \\ A \downarrow & & \downarrow \varphi \\ R^n & \xrightarrow{\rho} & M \end{array}$$

By Cayley-Hamilton theorem, we have that $c_A(A) = 0$ is the zero function. For all $x \in R^n$, we have that

$$\begin{aligned} c_A(A)(x) &= 0 \\ c_A(Ax) &= 0 \\ \rho(c_A(Ax)) &= \rho(0) \\ c_A(\rho(Ax)) &= 0 && (\rho \text{ is } R\text{-linear}) \\ c_A(\varphi(\rho(x))) &= 0 && (\text{Diagram is commutative}) \end{aligned}$$

Since ρ is surjective, we conclude that for any $m \in M$, the above calculation gives $c_A(\varphi(m)) = 0$ so that $c_A(\varphi)$ is the zero map. ■

Proposition 3.1.4 Let R be a commutative ring. Let M be a finitely generated R -module. Let $\phi : M \rightarrow M$ be a surjective R -module homomorphism. Then ϕ is an isomorphism.

Proof Consider M as an $R[\phi]$ -module via the action $\phi \cdot m = \phi(m)$. Notice that $(\phi)M = M$ since ϕ is surjective. By the Cayley-Hamilton theorem, there exists $\alpha_1, \dots, \alpha_{n-1} \in R$ such that

$$\text{id}^n + \alpha_1 \phi \text{id}^{n-1} + \dots + \alpha_{n-1} \phi \text{id} + \text{id} = 0 : M \rightarrow M$$

This simplifies to the equation

$$(\alpha_1 + \dots + \alpha_{n-1})\phi(m) + m = 0$$

for all $m \in M$.

We want to show that ϕ is injective. Suppose that $\phi(m) = 0$ for some $m \in M$. From the above equation, we see that $m = 0$. Hence ϕ is an isomorphism. ■

3.2 Nakayama's Lemma

Lemma 3.2.1 (Nakayama's Lemma I) Let R be a commutative ring. Let M be a finitely generated R -module. Let I be an ideal of R . If $IM = M$, then there exists $r \in R$ such that $rM = 0$ and $r - 1 \in I$.

Proof Choose $\varphi = \text{id}_M$. Then φ is surjective so that $M = \varphi(M) \subseteq IM$. By cor 4.1.3, there exists $r_1, \dots, r_n \in I$ such that $(1 + r_1 + \dots + r_n)M = 0$. By choosing $r = 1 + r_1 + \dots + r_n$, we see that $rM = 0$ and $r - 1 \in I$ so that we conclude. ■

Lemma 3.2.2 (Nakayama's Lemma II) Let R be a commutative ring. Let M be a finitely generated R -module. Let I be an ideal of R such that $I \subseteq J(R)$ and $IM = M$. Then $M = 0$.

Proof By Nakayama's lemma I, there exists $r \in R$ such that $rM = 0$ and $r - 1 \in I \subseteq J(R)$. By 2.3.8, we have that $1 - (r - 1)(-1) = r \in R^\times$. This means that r is invertible. Hence $rM = 0$ implies $M = r^{-1}rM = 0$. ■

Corollary 3.2.3 Let R be a commutative ring. Let M be a finitely generated R -module. Let I be an ideal of R such that $I \subseteq J(R)$. Let N be an R -submodule of M . If

$$M = IM + N$$

then $M = N$.

Proof Since quotients of finitely generated modules are finitely generated, we know that M/N is finitely generated. Define the map

$$\phi : IM + N \rightarrow I \frac{M}{N}$$

by $\phi(im + n) = i(m + N)$. This map is clearly surjective. Now I claim that $\ker(\phi) = N$. For any $im + n \in \ker(\phi)$, we see that $i(m + N) = N$ means that $im \in N$. Hence $im + n \in N$. On the other hand, if $im + n \in N$ then $im \in N$. But this means that $im + N = N$. Hence $im + n \in \ker(\phi)$. By the first isomorphism theorem for modules, we conclude that

$$\frac{M}{N} = \frac{IM + N}{N} \cong I \frac{M}{N}$$

We can now apply Nakayama's lemma II to conclude that $M/N = 0$ so that $M = N$. ■

Corollary 3.2.4 Let (R, m) be a local ring. Let m be a maximal ideal of R . Let M be a finitely generated R -module. Then the following are true.

- M/mM is a finite dimensional vector space over R/m .
- $a_1, \dots, a_n \in M$ generates M as an R -module if and only if $a_1 + mM, \dots, a_n + mM$ generates M/mM as a R/m vector space.
- $a_1, \dots, a_n \in M$ is a minimal set of generators of M as an R -module if and only if $a_1 + mM, \dots, a_n + mM$ is a basis for M/mM as a R/m vector space.

Proof Since the projection map $\pi : M \rightarrow M/mM$ is surjective, clearly any set of generators of M is a set of generators for M/mM . This also shows that if M is finitely generated then M/mM is a finite dimensional R/m -vector space.

For the other direction, suppose that $a_1 + mM, \dots, a_n + mM$ generates M/mM as an R/m -vector space. Define $N = Ra_1 + \dots + Ra_n \leq M$. Set $I = J(R) = m$. We want to show that $M = IM + N$. It is clear that $IM + N \leq M$. If $x \in M$, then there exists $r_k \in R$ such that $x + mM = r_1(a_1 + mM) + \dots + r_n(a_n + mM)$. In particular, this means that

$$x - \sum_{k=1}^n r_k a_k \in mM$$

Hence $x \in IM + N$. We can now apply the above corollary to deduce that $M = N = Ra_1 + \dots + Ra_n$ so that M is generated by $a_1, \dots, a_n$. And so we are done.

Suppose that $a_1, \dots, a_n$ generate M . The above shows that $a_1 + mM, \dots, a_n + mM$ spans M/mM . So suppose for a contradiction that $a_1, \dots, a_n$ is a minimal generating set but $a_1 + mM, \dots, a_n + mM$ is not a basis for M/mM . This means that after relabelling, $a_1 + mM, \dots, a_{n-1} + mM$ spans M/mM . By the above, this means that $a_1, \dots, a_{n-1}$ generate M . This is a contradiction of the minimality of the generating set $a_1, \dots, a_n$. Hence $a_1 + mM, \dots, a_n + mM$ is a basis for M/mM .

Now suppose that $a_1 + mM, \dots, a_n + mM$ is a basis for M/mM . We have seen above that $a_1, \dots, a_n$ generate M . If this is not minimal, then there is some smaller generating set $b_1, \dots, b_k$ that still generates M where $k < n$. By the above, $b_1 + mM, \dots, b_k + mM$ spans M/mM hence $n = \dim_{R/m}(M/mM) \leq k$. This is a contradiction since $k < n$. Hence we are done. ■

This is a key property of local rings. Let R be a commutative ring. Let M be an R -module. When M is finitely generated, it makes sense to ask for a minimal set of generators for M as an R -module (A generating set S is minimal if $S \setminus \{x\}$ does not generate the module for all $x \in S$). In general, this number is not well defined. For example, consider $R = \frac{k[x,y]}{(x^2, xy, y^2, x+y-1)}$ for k a field and R as a module over itself. Then $\{1\}$ and $\{x, y\}$ are both minimal generating sets.

However, the corollaries for Nakayama's lemma show that the cardinality for any minimal set of generators is well behaved since it is governed by the dimension of M/mM as a R/m -vector space.

3.3 Properties of the Hom Set

Let R be a ring. Let M, N be R -modules. Recall that in Rings and Modules that $\text{Hom}_R(M, N)$ is a $Z(R)$ -modules. When R is commutative, $Z(R) = R$ so that the Hom set becomes an R -module.

Proposition 3.3.1 Let R be a commutative ring. Let M, N be R -modules. Then

$$\text{Hom}_R(M, N)$$

is an R -module with the following binary operations.

- For $\phi, \varphi : M \rightarrow N$ two R -module homomorphisms, define $\phi + \varphi : M \rightarrow N$ by $(\phi + \varphi)(m) = \phi(m) + \varphi(m)$ for all $m \in M$
- For $\phi : M \rightarrow N$ an R -module homomorphism and $r \in R$, define $r\phi : M \rightarrow N$ by $(r\phi)(m) =$

$r \cdot \phi(m)$ for all $m \in M$.

Proof We first show that the addition operation gives the structure of a group.

- Since M is associative as an additive group, associativity follows
- Clearly the zero map $0 \in \text{Hom}_R(M, N)$ acts as the additive inverse since for any $\phi \in \text{Hom}_R(M, N)$, we have that $\phi(m) + 0 = 0 + \phi(m) = \phi(m)$ since 0 is the additive identity for M
- For every $\phi \in \text{Hom}_R(M, N)$, the map taking m to $-\phi(m)$ also lies in $\text{Hom}_R(M, N)$. Since $-\phi(m)$ is the inverse of $\phi(m)$ in M for each $m \in M$, we have that $-\phi$ is the inverse of ϕ

We now show that

- Let $r, s \in R$, we have that $((sr)\phi)(m) = (sr) \cdot \phi(m) = s \cdot (r \cdot \phi(m)) = s(r(\phi))(m)$ and hence we showed associativity.
- It is clear that $1_R \in R$ acts as the identity of the operation.

Thus we are done. ■

Proposition 3.3.2 Let R be a ring. Let I be an indexing set. Let M_i, N be R -modules for $i \in I$. Then the following are true.

- There is an isomorphism

$$\text{Hom}\left(\bigoplus_{i \in I} M_i, N\right) \cong \bigoplus_{i \in I} \text{Hom}(M_i, N)$$

- There is an isomorphism

$$\text{Hom}\left(\prod_{i \in I} M_i, N\right) \cong \prod_{i \in I} \text{Hom}(M_i, N)$$

Definition 3.3.3 (Induced Map of Hom) Let R be a commutative ring. Let M_1, M_2, N be R -modules. Let $f : M_1 \rightarrow M_2$ be an R -module homomorphism. Define the induced map

$$f^* : \text{Hom}_R(M_2, N) \rightarrow \text{Hom}(M_1, N)$$

by the formula $\varphi \mapsto \varphi \circ f$

Lemma 3.3.4 Let R be a commutative ring. Let M_1, M_2, N be R -modules. Let $f : M_1 \rightarrow M_2$ be an R -module homomorphism. Then the induced map

$$f^* : \text{Hom}(M_2, N) \rightarrow \text{Hom}(M_1, N)$$

is an R -module homomorphism.

3.4 More on Exact Sequences

Proposition 3.4.1 Let R be a commutative ring. Let the following be an exact sequence of R -modules.

$$0 \longrightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$$

Let N be an R -module. Then the following two sequences

$$0 \longrightarrow \text{Hom}_R(M_3, N) \longrightarrow \text{Hom}_R(M_2, N) \longrightarrow \text{Hom}_R(M_1, N)$$

$$\text{Hom}_R(N, M_1) \longrightarrow \text{Hom}_R(N, M_2) \longrightarrow \text{Hom}_R(N, M_3) \longrightarrow 0$$

are exact.

Proof

- We first show that g^* is injective. Let $\phi, \rho \in \text{Hom}(C, G)$ such that $g^*(\phi) = g^*(\rho)$. This means that $\phi \circ g = \rho \circ g$. Let $c \in C$. Since g is surjective, there exists $b \in B$ such that $g(b) = c$. Then

$$\phi(c) = \phi(g(b)) = \rho(g(b)) = \rho(c)$$

Hence $\phi = \rho$.

Now we show that $\text{im}(g^*) \subseteq \ker(f^*)$. Let $g^*(\phi) \in \text{Hom}(B, G)$ for $\phi \in \text{Hom}(C, G)$. We want to show that $f^*(g^*(\phi)) = 0$. But we have that

$$(\phi \circ g \circ f)(a) = \phi(g(f(a))) = \phi(0) = 0$$

since $\text{im}(f) = \ker(g)$. Thus we conclude.

Finally we show that $\ker(f^*) \subseteq \text{im}(g^*)$. Let $f^*(\phi) = 0$ for $\phi \in \text{Hom}(B, G)$. This means that $\phi \circ f = 0$ or in other words, $\text{im}(f) \subseteq \ker(\phi)$. Since $\phi(k) = 0$ for all $k \in \text{im}(f)$, ϕ descends to a map $\bar{\phi} : \frac{B}{\text{im}(f)} \rightarrow G$. But $\text{im}(f) = \ker(g)$ hence this is equivalent to a map $\bar{\phi} : \frac{B}{\ker(g)} \rightarrow G$. But by the first isomorphism theorem and the fact that g is surjective, we conclude that $\bar{g} : \frac{B}{\ker(g)} \xrightarrow{g} C$, where $b + \ker(g) \mapsto g(b)$. Thus we have constructed a map $\bar{\phi} \circ \bar{g}^{-1} : C \rightarrow G$ given by $g(b) \mapsto b + \ker(g) \mapsto \phi(b)$. But now $g^*(\bar{\phi} \circ \bar{g}^{-1})$ is the map defined by

$$b \mapsto g(b) \mapsto b + \ker(g) \mapsto \phi(b)$$

and so this map is exactly ϕ . Thus $\phi \in \text{im}(g^*)$. ■

Example 3.4.2 Applying $\text{Hom}_{\mathbb{Z}}(-, \mathbb{Z}/p\mathbb{Z})$ to the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \longrightarrow \mathbb{Z}/p\mathbb{Z} \longrightarrow 0$$

does not give a sequence that is exact on the right.

Proof The new sequence is now

$$0 \longrightarrow \mathbb{Z}/p\mathbb{Z} \xrightarrow{\text{id}_{\mathbb{Z}/p\mathbb{Z}}} \mathbb{Z}/p\mathbb{Z} \xrightarrow{0} \mathbb{Z}/p\mathbb{Z}$$

Evidently the 0 map is not surjective. ■

Proposition 3.4.3 Let R be a commutative ring. Let the following be an exact sequence of R -modules.

$$0 \longrightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$$

Let N be an R -module. Then the following sequence

$$M_1 \otimes N \xrightarrow{f \otimes \text{id}_N} M_2 \otimes N \xrightarrow{g \otimes \text{id}_N} M_3 \otimes N \longrightarrow 0$$

is exact.

However, one can observe that we did not imply that $M_1 \otimes N \rightarrow M_2 \otimes N$ is injective. Indeed, this is because tensoring does not preserve injections.

4 Algebra Over a Commutative Ring

4.1 Commutative Algebras

Definition 4.1.1 (Commutative Algebras) Let R be a commutative ring. A commutative R -algebra is an R -algebra A that is commutative.

Proposition 4.1.2 Let R be a commutative ring. Then the following are equivalent characterizations of a commutative R -algebra.

- A is a commutative R -algebra
- A is a commutative ring together with a ring homomorphism $f : R \rightarrow A$

Proof Suppose that A is an R -algebra. Then define a map $f : R \rightarrow A$ by $f(r) = r \cdot 1$ where $r \cdot 1$ is the module operation on A . Then clearly this is a ring homomorphism.

Suppose that A is a commutative ring together with a ring homomorphism $f : R \rightarrow A$. Define an action $\cdot : R \times A \rightarrow A$ by $r \cdot a = f(r)a$. Then this action clearly allows A to be an R -module. ■

Under the correspondence of associative algebra, the above proposition gives a another correspondence between the first one.

$$\left\{ (A, R) \mid \begin{array}{l} A \text{ is a commutative} \\ R\text{-algebra} \end{array} \right\} \xleftrightarrow{1:1} \left\{ \phi : R \rightarrow A \mid \begin{array}{l} \phi \text{ is a ring homomorphism} \\ \text{such that } f(R) \subseteq Z(A) = A \end{array} \right\}$$

In particular, the construction above are inverses of each other so that it gives the one-to-one correspondence.

4.2 Free Commutative Algebras

Let R be a commutative ring. Let X be a set. Recall that we defined $R\langle X \rangle$ to be the free (non-commutative) R -algebra over X . Explicitly, if $W = \{x_1 \cdots x_n \mid x_1, \dots, x_n \in X\}$ is the set of words on X , then

$$R\langle X \rangle = \bigoplus_{w \in W} R \cdot w$$

together with multiplication defined by $(x_1 \cdots x_n) \cdot (y_1 \cdots y_m) = x_1 \cdots x_n \cdot y_1 \cdots y_m$.

Definition 4.2.1 (Free Commutative Algebra over a Ring) Let R be a commutative ring. Let X be a set. Define the free commutative R -algebra over X to be the quotient

$$\text{Free}_R(X) = \frac{R\langle X \rangle}{\langle x_i x_j - x_j x_i \mid x_i, x_j \in X \rangle}$$

Proposition 4.2.2 (Universal Property of Free Commutative Algebras) Let R be a commutative ring. Let X be a set. The free commutative algebra $\text{Free}_R(X)$ satisfies the following universal property.

- Universal Property: If A is a commutative R -algebra, then for every $f : X \rightarrow A$ a map of sets, there exists a unique homomorphism of algebras $\varphi : \text{Free}_R(X) \rightarrow A$ such that $\varphi(x_i) = f(x_i)$ for each $x_i \in X$. In other words, the following diagram commutes:

$$\begin{array}{ccc} X & \xhookrightarrow{\iota} & \text{Free}_R(X) \\ & \searrow f & \downarrow \exists! \varphi \\ & & A \end{array}$$

where $\iota : X \rightarrow \text{Free}_R(X)$ is the inclusion.

- $\text{Free}_R(X)$ is the unique R -algebra (up to unique isomorphism) that satisfies this property.

Proposition 4.2.3 Let R be a commutative ring. Let X be a set. Then there is an R -algebra isomorphism

$$\text{Free}_R(X) \cong R[X]$$

with the polynomial ring over X .

4.3 Finiteness Properties of Algebras

Definition 4.3.1 (Finitely Generated Algebras) Let R be a commutative ring. Let A be a commutative R -algebra. We say that A is finitely generated if there exists $a_1, \dots, a_n \in A$ such that every element $a \in A$ can be written as a polynomial in $a_1, \dots, a_n$. This means that

$$a = \sum_{i_1, \dots, i_n} r_{i_1, \dots, i_n} a_1^{i_1} \cdots a_n^{i_n}$$

Finitely generated algebras are also called algebra of finite type.

Proposition 4.3.2 Let A be a commutative algebra over a ring R . Then the following are equivalent.

- A is a finitely generated algebra over R
- There exists elements $a_1, \dots, a_n \in A$ such that the evaluation homomorphism

$$\phi : R[x_1, \dots, x_n] \rightarrow A$$

given by $\phi(f) = f(a_1, \dots, a_n)$ is a surjection

- There is an isomorphism

$$A \cong \frac{R[x_1, \dots, x_n]}{I}$$

for some ideal I

Proposition 4.3.3

Let k be a field. Let A, B be k -algebras. Let $\varphi : A \rightarrow B$ be a k -algebra homomorphism. Let m be a maximal ideal of B . If B is finitely generated over k , then $\varphi^{-1}(m)$ is a maximal ideal of A .

Definition 4.3.4 (Finitely Presented Algebra) Let R be a ring. Let $A = R[x_1, \dots, x_n]/I$ be a finitely generated algebra over R for some ideal I . We say that A is finitely presented if I is finitely generated.

Lemma 4.3.5 Let R be a ring, considered as an algebra over $\mathbb{Z}$. If R is finitely generated over $\mathbb{Z}$, then R is finitely presented.

Proof Trivial since $\mathbb{Z}$ is a principal ideal domain. ■

Definition 4.3.6 (Finite Algebras) Let R be a commutative ring. Let A be an R -algebra. We say that A is finite if A is finitely generated as an R -module.

Example 4.3.7

Let R be a commutative ring. Then $R[x]$ is a finitely generated algebra over R but is not a finite R -algebra.

Example 4.3.8

Let k be a field. Then $R = \frac{k[x, y, z, w]}{(yw - z^2, xw - yz, xz - y^2)}$ is a finitely generated free $k[x, w]$ -module.

Proof

Let $A = k[x, w] \cdot 1 \oplus k[x, w] \cdot y \oplus k[x, w] \cdot z$ be the $k[x, w]$ -module with multiplication defined by $y^2 = xz$, $yz = xw$ and $z^2 = yw$. It is easy to check that this defines a $k[x, w]$ -algebra. Define the map $k[x, w, y, z] \rightarrow A$ by sending x to x , y to y , z to z and w to w . This descends to a well defined k -algebra homomorphism $\psi : R \rightarrow A$ because $\psi(y^2 - xz) = \psi(yz - xw) = \psi(z^2 - yw) = 0$. This

is map is a left inverse to the map $\varphi : A \hookrightarrow k[x, y, z, w] \rightarrow R$ and so φ is injective. Moreover, φ is clearly surjective since R is generated as an $k[x, w]$ -module by $1, y, z$. ■

5 Localization

5.1 Localization of Modules

Definition 5.1.1 (Multiplicative Set) Let R be a commutative ring. $S \subseteq R$ is a multiplicative set if $1 \in S$ and S is closed under multiplication: $x, y \in S$ implies $xy \in S$

Definition 5.1.2 (Localization of a Module) Let R be a commutative ring and $S \subseteq R$ be a multiplicative set. Let M be a R -module. Define the ring of fractions of M with respect to S by

$$S^{-1}M = \left\{ \frac{m}{s} \mid m \in M, s \in S \right\} / \sim$$

where $\sim$ is defined by

$$\frac{m}{s} \sim \frac{m'}{s'} \text{ if and only if } \exists v \in S \text{ such that } v(mu' - m'u) = 0$$

Lemma 5.1.3 Let R be a commutative ring. Let M be an R -module. Let $S \subseteq R$ be a multiplicative subset. Then $S^{-1}M$ is a well defined $S^{-1}R$ -module with operation given by

$$\left(\frac{r}{s_1}, \frac{m}{s_2} \right) \mapsto \frac{r \cdot m}{s_1 s_2}$$

Proposition 5.1.4 Let R be a commutative ring. Let M be an R -module. Then there is an isomorphism

$$S^{-1}M \cong S^{-1}R \otimes_R M$$

of $S^{-1}R$ -modules given by $\frac{m}{s} \mapsto \frac{1}{s} \otimes m$.

So localization really is just a tensor product with the localization of a ring. Indeed, the idea of localization of a ring is to allow certain elements of a ring to be invertible, and the localization of a module is to allow these inverted elements to also act on it.

Definition 5.1.5 (Induced Map of Localization) Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Let M, N be R -modules. Let $\phi : M \rightarrow N$ be an R -module homomorphism. Define the induced map

$$S^{-1}\phi : S^{-1}M \rightarrow S^{-1}N$$

by the formula $\frac{m}{s} \mapsto \frac{\phi(m)}{s}$.

Lemma 5.1.6 Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Let M, N be R -modules. Let $\phi : M \rightarrow N$ be an R -module homomorphism. Then the induced map

$$S^{-1}\phi : S^{-1}M \rightarrow S^{-1}N$$

is a well defined ring homomorphism.

Lemma 5.1.7 Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Let M, N, K be R -modules. Let $f : M \rightarrow N$ and $g : N \rightarrow K$ be R -module homomorphisms. Then the following are true.

- Composition: $S^{-1}(g \circ f) = S^{-1}g \circ S^{-1}f : S^{-1}M \rightarrow K$.
- Identity: $S^{-1}\text{id}_M = \text{id}_{S^{-1}M}$

Proposition 5.1.8 Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Let the following be an exact sequence of R -modules.

$$M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3$$

Then the following is an exact sequence of $S^{-1}R$ -modules.

$$S^{-1}M_1 \xrightarrow{f} S^{-1}M_2 \xrightarrow{g} S^{-1}M_3$$

Proof Since $\text{im}(f) = \ker(g)$, we have that $g \circ f = 0$ which implies that $0 = S^{-1}0 = S^{-1}(g \circ f) = S^{-1}g \circ S^{-1}f$. Hence $\text{im}(S^{-1}f) \subseteq \ker(S^{-1}g)$. Conversely, let $m_2/s \in \ker(S^{-1}g)$. Then $g(m_2)/s = 0$ and so $g(tm_2) = tg(m_2) = 0$ for some $t \in S$. Since $\text{im}(f) = \ker(g)$, there exists $m_1 \in M_1$ such that $f(m_1) = tm_2$. Then we have

$$(S^{-1}f)(m_1/ts) = f(m_1)/ts = tm_2/ts = m_2/s$$

Hence $m_2/s \in \text{im}(S^{-1}(f))$. ■

Corollary 5.1.9 Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Let M be an R -module. Then the following are true.

- Localization commutes with quotients: If N is an R -submodule of M , then

$$S^{-1} \frac{M}{N} \cong \frac{S^{-1}M}{S^{-1}N}$$

as $S^{-1}R$ -modules.

- Localization commutes with products: If N is an R -module, then

$$S^{-1}(M \times N) \cong S^{-1}M \times S^{-1}N$$

as $S^{-1}R$ -modules.

- Localization commutes with internal sums: If N_1, N_2 are R -submodules of M , then

$$S^{-1}(N_1 + N_2) \cong S^{-1}N_1 + S^{-1}N_2$$

as $S^{-1}R$ -submodules of $S^{-1}M$.

- Localization commutes with intersections: If N_1, N_2 are R -submodules of M , then

$$S^{-1}(N_1 \cap N_2) = S^{-1}N_1 \cap S^{-1}N_2$$

as $S^{-1}R$ -submodules of $S^{-1}M$.

Proof Consider the exact sequences:

$$0 \longrightarrow N \xrightarrow{\text{incl.}} M \xrightarrow{\text{proj.}} M/N \longrightarrow 0$$

$$0 \longrightarrow M \xrightarrow{\text{incl.}} M \times N \xrightarrow{\text{proj.}} N \longrightarrow 0$$

$$0 \longrightarrow N_1 \xrightarrow{\text{incl.}} N_1 + N_2 \xrightarrow{\text{proj.}} N_2 \longrightarrow 0$$

$$0 \longrightarrow N_1 \cap N_2 \xrightarrow{n \mapsto (n, n)} N_1 \times N_2 \xrightarrow{(n_1, n_2) \mapsto n_1 - n_2} N_1 + N_2 \longrightarrow 0$$

respectively and apply the above proposition. ■

Lemma 5.1.10 Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset. Let M, N be R -modules. Let $\phi : M \rightarrow N$ be an R -module homomorphism. Then the following are true.

- Localization commutes with kernels:

$$S^{-1} \ker(\phi) \cong \ker(S^{-1}\phi)$$

- Localization commutes with cokernels:

$$S^{-1} \frac{N}{\text{im}(\phi)} \cong \frac{S^{-1}N}{\text{im}(S^{-1}\phi)}$$

- Localization commutes with images:

$$S^{-1}(\text{im } \phi) \cong \text{im}(S^{-1}\phi)$$

Proof Consider the exact sequences:

$$0 \longrightarrow \ker(\phi) \hookrightarrow M \xrightarrow{\phi} N$$

$$M \xrightarrow{\phi} N \longrightarrow \frac{M}{\text{im}(\phi)} \longrightarrow 0$$

$$0 \longrightarrow \ker(\phi) \longrightarrow M \longrightarrow \text{im}(\phi) \longrightarrow 0$$

respectively and apply 5.3.6. ■

Definition 5.1.11 (The Total Quotient Ring)

Let R be a commutative ring. Define the total quotient ring of R by

$$Q(R) = \{r \in R \mid r \text{ is not a zero divisor}\}^{-1}R$$

5.2 Localization at Single Elements and Away from Prime Ideals

Lemma 5.2.1 Let R be a commutative ring. Let $f \in R$ be non-zero. Then the set $\{f^n \mid n \in \mathbb{N}\}$ is a multiplicative set.

Definition 5.2.2 (Localization at an Element) Let R be a commutative ring. Let M be an R -module. Let $f \in R$ be non-zero. Define the localization of M at f to be the ring

$$M_f = \{f^n \mid n \in \mathbb{N}\}^{-1}M$$

Lemma 5.2.3 Let R be a commutative ring. Let $f \in R$ be non-zero. Then there is an R -algebra isomorphism

$$R_f \cong R \left[\frac{1}{f} \right]$$

given by $\frac{a}{f^k} \mapsto a \cdot \frac{1}{f^k}$.

Lemma 5.2.4 Let R be a commutative ring and P a prime ideal of R . Then $R \setminus P$ is a multiplicative set.

Proof By definition, $xy \in P$ implies $x \in P$ or $y \in P$, since $R \setminus P$ removes all these elements, we have that $x \notin P$ and $y \notin P$ implies that $xy \notin P$. ■

Definition 5.2.5 (Localization at Prime Ideals) Let R be a commutative ring. Let M be an R -module. Let P be a prime ideal. Denote

$$M_P = (R \setminus P)^{-1}M$$

the localization of M at P .

5.3 The Localization Map

Proposition 5.3.1 Let R be a commutative ring. Let S be a multiplicative subset of R . Then the following are true.

- $(S^{-1}R, +, \times)$ is a ring
- The map $k : R \rightarrow S^{-1}R$ defined by $r \mapsto r/1$ is a ring homomorphism, called the localization map.

Proof

- Define addition by $\frac{r}{s} + \frac{r'}{s'} = \frac{rs' + r's}{ss'}$ and multiplication by $\frac{r}{s} \cdot \frac{r'}{s'} = \frac{rr'}{ss'}$. Clearly addition is abelian, and has identity $\frac{0}{1}$ and inverse $\frac{-r}{s}$ for any $\frac{r}{s} \in S^{-1}R$. Multiplication also has identity $\frac{1}{1}$.

■

Lemma 5.3.2 Let R be a commutative ring. Let S be a multiplicative subset of R . Then localization map $R \rightarrow S^{-1}R$ is injective if and only if S does not contain zero divisors.

Proof Suppose that $R \rightarrow S^{-1}R$ is injective. Then $sr = 0$ implies $r = 0$ for all $s \in S$. Hence S does not contain zero divisors. Suppose that S does not contain zero divisors. Then $sr = 0$ implies that $r = 0$ since S has no zero divisors. Hence the localization map is injective. ■

Proposition 5.3.3 (Universal Property) Let R be a commutative ring. Let S be a multiplicative set. Then $S^{-1}R$ and the localization map $k : R \rightarrow S^{-1}R$ satisfies the following universal property.

- For any commutative ring B and ring homomorphism $\phi : R \rightarrow B$ such that $\phi(s) \in B^\times$ for all $s \in S$, there exists a unique ring homomorphism $\psi : S^{-1}R \rightarrow B$ such that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{k} & S^{-1}R \\ & \searrow \phi & \downarrow \exists! \psi \\ & & B \end{array}$$

- $S^{-1}R$ is the unique commutative ring (up to unique isomorphism) that has such a property.

Lemma 5.3.4 Let R be a commutative ring. If R is an integral domain, then the following are true.

- If S is a multiplicative subset of R such that $0 \notin S$, then $S^{-1}R$ is an integral domain.
- $\text{Frac}(R) = Q(R)$.
- The localization map induces a ring isomorphism

$$R \cong \bigcap_{m \text{ a maximal ideal}} R_m$$

Proof

- Suppose that $0 = \frac{a}{s} \cdot \frac{b}{t}$. By the equivalence relation this is the same as saying that $uab = 0$ for some $u \in S$. Since R is an integral domain and $0 \neq S$, we conclude that $u \notin S$ so that $ab = 0$. Again since R is an integral domain this implies that $a = 0$ or $b = 0$. Hence either $a/s = 0$ or $b/t = 0$ in $S^{-1}R$. Hence $S^{-1}R$ is an integral domain.
- Trivial.
- Clearly the map is well defined. Moreover, since for each maximal ideal m , $0 \notin R \setminus m$. Hence the localization map is injective. Suppose for a contradiction that the localization map is not surjective. Then there exists x in the intersection such that $x \neq r/1$ for all $r \in R$. Consider the ideal $I = \{r \in R \mid rx = s/1 \text{ for some } s \in R\}$. Since $1 \notin R$, I is a proper ideal. So there

exists a maximal ideal m containing I . But x also cannot lie in R_m and hence the intersection. Indeed, if $x \in R_m$, then $x = a/b$ for some $a \in R$ and $b \notin m$. Then $bx = a \in R$ implies that $b \in I$. This is a contradiction to $b \notin m$. Thus no such x exists. Hence the localization map is surjective. ■

Example 5.3.5

Let R be a commutative ring. Let $d \in R$ be squarefree. Then we have

$$\text{Frac}\left(\frac{R[x]}{(x^2 - d)}\right) = \frac{\text{Frac}(R)[x]}{(x^2 - d)}$$

Proof

Notice that the quotient on the right is a field since $x^2 - d$ is irreducible. Since it also contains $\frac{R[x]}{(x^2 - d)}$ and $\text{Frac}\left(\frac{R[x]}{(x^2 - d)}\right)$ is the smallest field containing $\frac{R[x]}{(x^2 - d)}$, we must have that $\text{Frac}\left(\frac{R[x]}{(x^2 - d)}\right) \subseteq \frac{\text{Frac}(R)[x]}{(x^2 - d)}$. On the other hand, notice that any element $(r_0/s_0) + (r_1/s_1)x \in \frac{\text{Frac}(R)[x]}{(x^2 - d)}$ is contained in $\text{Frac}\left(\frac{R[x]}{(x^2 - d)}\right)$ since $(r_0/s_0) + (r_1/s_1)x = \frac{s_1 r_0 + s_0 r_1 x}{s_0 s_1}$, hence both fields contain each other. ■

5.4 Ideals of a Localization

Definition 5.4.1 (Ideals Closed Under Division) Let R be a commutative ring. Let I be an ideal of R . Let $S \subseteq R$ be a multiplicative subset. We say that I is closed under division by s if for all $s \in S$ and $a \in R$ such that $sa \in I$, we have $a \in I$.

Lemma 5.4.2 Let R be a commutative ring. Let I be an ideal of R . Let $S \subseteq R$ be a multiplicative subset. Then we have

$$I^e = S^{-1}I$$

Proof Let $f : R \rightarrow S^{-1}R$ be the localization map. Then $f(I) \subseteq S^{-1}I$ implies that $I^e \subseteq S^{-1}I$. Conversely, suppose that $i/s \in S^{-1}I$. Then $i/s = (1/s) \cdot f(i) \in I^e$. Hence $I^e = S^{-1}I$. ■

Proposition 5.4.3 Let R be a commutative ring. Let S be a multiplicative subset of R . Let P be a prime ideal of R . Then the following are true.

- $S^{-1}P$ is a prime ideal of $S^{-1}R$ if and only if $S \cap P = \emptyset$.
- $S^{-1}P = S^{-1}R$ if and only if $S \cap P \neq \emptyset$.

Proof Recall that R/P is an integral domain if P is prime. Since S^{-1} commutes with quotients, we have that

$$\frac{S^{-1}R}{S^{-1}P} \cong S^{-1} \frac{R}{P}$$

If $S \cap P = \emptyset$, then $0 \in P$ implies that $0 \notin S$. This means that $0 \notin \phi(S)$. By 5.3.1 we conclude that $S^{-1}(R/P)$ is an integral domain. Hence $S^{-1}P$ is a prime ideal. If $S \cap P \neq \emptyset$, suppose that $x \in S \cap P$. Then ????? ■

Theorem 5.4.4 Let R be a commutative ring. Let I be an ideal of R . Let $S \subseteq R$ be a multiplicative subset. Let $\phi : R \rightarrow S^{-1}R$ denote the localization map. Then there is a one-to-one bijection

$$\{J \mid J \text{ is an ideal of } S^{-1}R\} \xrightarrow{1:1} \{I \mid \begin{smallmatrix} I \text{ is an ideal of } R \text{ and} \\ I \text{ is closed under division by } S \end{smallmatrix}\}$$

whose map is given by $J \mapsto J^c = \phi^{-1}(J)$ and inverse is given by $I \mapsto I^e = S^{-1}I$.

Proof We first show that our map of sets are well defined. Let J be an ideal of $S^{-1}R$. We first show that $\phi^{-1}(J)$ is closed under division by S . Suppose that $s \in S$ and $r \in R$ such that $sr \in \phi^{-1}(J)$. Then $sr/1 \in J$. Now since J is an ideal of $S^{-1}R$, we know that $1/s \cdot sr/1 \in J$. But $1/s \cdot sr/1 = r/1 = \phi(r)$. This means that $\phi(r) \in J$ and hence $r \in \phi^{-1}(J)$. Thus $\phi^{-1}(J)$ is an ideal closed under division by S .

Now let I be an ideal of R closed under division. I claim that $S^{-1}I$ is an ideal of $S^{-1}R$. Let $a/s, b/t \in S^{-1}I$. Then $a/s + b/t = (at + bs)/st$. Since I is an ideal, we know that $at + bs \in I$. Also since S is a multiplicative subset, $st \in S$. Hence $(at + bs)/st \in S^{-1}I$. Now let $a/s \in S^{-1}I$ and $r/t \in S^{-1}R$. Then $(a/s) \cdot (r/t) = ar/st$. Since I is an ideal, $ar \in I$. Thus $ar/st \in S^{-1}I$ so that I is an ideal.

It remains to show that the two maps are inverses of each other. Let J be an ideal of $S^{-1}R$. We want to show that $J = S^{-1}(\phi^{-1}(J))$. Let $a/s \in J$. Since J is an ideal, we have $\phi(a) = a/1 = 1/s \cdot a/s \in J$. Hence $a \in \phi^{-1}J$ so that $a/s \in S^{-1}\phi^{-1}(J)$. Thus $J \subseteq S^{-1}(\phi^{-1}(J))$. Now by 1.5.5 the extension of the contraction of J is a subset of J . Hence we conclude.

On the other hand, we also want to show that $I = \phi^{-1}(S^{-1}I)$. Again by 1.5.5 we know that $I \subseteq \phi^{-1}(S^{-1}I)$. Conversely, let $x \in \phi^{-1}(S^{-1}I)$. Then $\phi(x) = x/1 \in S^{-1}I$. This means that $x/1 = b/t$ for some $b \in I$ and $t \in S$. Then there exists $u \in S$ such that $uxt = ub$. Since $b \in I$, $ub \in I$ hence $uxt \in I$. Since $ut \in S$ and I is closed under division, we have $x \in I$.

This shows that $S^{-1}(-)$ and $\phi^{-1}(-)$ are mutual inverses of each others. Thus we conclude. ■

Using the theorem we conclude that every ideal of $S^{-1}R$ is of the form $S^{-1}I$ for some ideal I of R such that I is closed under division by S .

Proposition 5.4.5 Let R be a commutative ring. Let I be an ideal of R . Let $S \subseteq R$ be a multiplicative subset. Then the above bijection restricts to the following bijection

$$\text{Spec}(S^{-1}R) \xleftrightarrow{1:1} \left\{ I \mid \begin{array}{l} I \text{ is a prime ideal of } R \\ \text{and } I \cap S = \emptyset \end{array} \right\}$$

Proof Let $\phi : R \rightarrow S^{-1}R$ be the localization map. From the above we know that $Q = S^{-1}\phi^{-1}(Q)$ for any prime ideal Q of $S^{-1}R$. This implies that $S^{-1}\phi^{-1}(Q)$ is prime. By 5.4.3 this implies that $\phi^{-1}(Q) \cap S = \emptyset$. Thus the map $J \mapsto \phi^{-1}(J)$ induces a well defined map on our given sets of prime ideals.

Conversely, by 5.4.3 we know that if P is a prime ideal of R such that $S \cap P = \emptyset$, then $S^{-1}P$ is a prime ideal of $S^{-1}R$. Hence the inverse map is also well defined on our domain and codomain. By the above theorem it is already a bijection, hence we are done. ■

Proposition 5.4.6 Let R be a commutative ring. Let P be a prime ideal of R . Then the above bijection gives

$$\text{Spec}(R_P) \xleftrightarrow{1:1} \left\{ I \mid \begin{array}{l} I \text{ is a prime ideal of } R \\ \text{and } I \subseteq P \end{array} \right\}$$

Proof Notice that the condition that $I \cap S = \emptyset$ in the above proposition translates to $I \cap (R \setminus P) = \emptyset$, which is the same as saying $I \subseteq P$. ■

Proposition 5.4.7 Let R be a commutative ring and let P be a prime ideal of R . Then R_P is a

local ring with unique maximal ideal given by

$$PR_P = \left\{ \frac{r}{s} \mid r \in P, s \notin P \right\}$$

Proof We show that PR_P is the only unique maximal ideal. Suppose that I is an ideal in R_P such that I is not a subset of PR_P . Then there exists $a/s \in I$ such that $a \notin P$ and $s \notin P$. It is clear that s/a is then an element of R_P . So a/s is invertible. Hence $I = R_P$. ■

Be wary that in general localizations does not result in a local ring. This happens only when we are localizing with respect to a prime ideal. The importance of prime ideals is not explicit in the above because only using prime ideals P can $R \setminus P$ be a multiplicative set which ultimately allows localization to make sense.

Proposition 5.4.8 (Localization of a Localization) Let R be a commutative ring. Let S be a multiplicative subset of R . Let P be a prime ideal of R such that $S^{-1}P$ is a prime ideal of $S^{-1}R$. Then

$$(S^{-1}R)_{S^{-1}P} \cong R_P$$

Proof Define a map $S^{-1}R \rightarrow R_P$ by the identity map. This is well defined because if $s \in S$, then we know $S^{-1}P$ is a prime ideal implies $S \cap P = \emptyset$, so $s \notin P$. Thus r/s is a well defined fraction in R_P . Since it is just the identity map, it is a well defined ring homomorphism. Now let $r/s \in S^{-1}R \setminus S^{-1}P$. Then $r \notin P$ implies that r is invertible in R_P . Hence $r/s \cdot s/r = 1$ in R_P . Thus r/s is invertible in R_P . Thus we can invoke the universal property to obtain a unique map

$$(S^{-1}R)_{S^{-1}P} \rightarrow R_P$$

Conversely, define a map $R \rightarrow (S^{-1}R)_{S^{-1}P}$ by the identity map $r \mapsto (r/1)/(1/1)$. This is well defined because $1 \notin P$ implies $1/1 \in S^{-1}R \setminus S^{-1}P$. Clearly this is a well defined ring homomorphism. For $s \in S$, notice that $(s/1)/(1/1)$ is invertible in $(S^{-1}R)_{S^{-1}P}$ via the element $(1/s)/(1/1)$. Thus we can invoke the universal property of $S^{-1}R$ to obtain a unique map

$$S^{-1}R \rightarrow (S^{-1}R)_{S^{-1}P}$$

We now have two unique maps going both directions between $S^{-1}R$ and $(S^{-1}R)_{S^{-1}P}$. This implies that they are isomorphic. ■

Lemma 5.4.9 Let R be a commutative ring. Let $S \subseteq R$ be a multiplicative subset of R . If R is Noetherian, then $S^{-1}R$ is Noetherian.

Proof Follows from the correspondence of ideals in localizations. ■

5.5 Localization of Graded Rings

Proposition 5.5.1

Let R be a commutative $\mathbb{N}$ -graded ring. Let S be a multiplicative subset of R consisting only of homogeneous elements. Then

$$S^{-1}R$$

is also a graded ring with grading defined by $\deg(a/f^k) = \deg(a) - k$.

Proposition 5.5.2

Let R be a commutative $\mathbb{N}$ -graded ring. Let $f \in R_+$. Then there is a bijection

$$\{P \in \text{Spec}(R_f) \mid P \text{ is homogeneous}\} \xrightarrow{1:1} \text{Spec}((R_f)_0)$$

induced by the ring homomorphism $(R_f)_0 \hookrightarrow R_f$.

Proof

The above homomorphism defines the map from left to right by $P \mapsto \varphi^{-1}(P)$ where $\varphi : (R_f)_0 \hookrightarrow R_f$ is the inclusion map. This map is clearly well defined. For the reverse direction, let $P \in \text{Spec}((R_f)_0)$. Define $f : \text{Spec}((R_f)_0) \rightarrow \{P \in \text{Spec}(R_f) \mid P \text{ is homogeneous}\}$ by $f(P) = \bigoplus_{k \in \mathbb{N}} Q_k$ where $Q_k \subseteq R_k$ and $a \in Q_k$ if and only if $a^{\deg(f)}/f^k \in P$. ■

Definition 5.5.3 (Localization of a Graded Ring)

Let R be a commutative graded ring. Let P be a homogeneous prime ideal of R . Let $T = \{r \in R \setminus P \mid r \text{ is homogeneous}\}$. Define the localization of R at P as a graded ring to be

$$R_{(P)} = T^{-1}R$$

Proposition 5.5.4

Let R be a commutative graded ring. Let P be a homogeneous prime ideal of R . Then the localization R_P as a graded ring is also a graded ring with grading defined by $\deg(f/g) = \deg(f) - \deg(g)$ for $f/g \in R_P$.

Proposition 5.5.5 Let $R = \bigoplus_{i=0}^{\infty} R_i$ be a commutative ring that is graded. Let P be a homogeneous prime ideal of R . Then $(R_P)_0$ is a local ring with unique maximal ideal given by

$$(PR_P) \cap (R_P)_0$$

5.6 Local Properties

Definition 5.6.1 (Local Properties of Elements) Let R be a commutative ring. Let M be an R -module. A property of an element of M is local if the following is true. $m \in M$ has the property if and only if $m \in M_P$ has the property.

Lemma 5.6.2 Let R be a commutative ring. Let M be an R -module. Then $x \in M$ being the zero element is a local property.

Proof Suppose that $x = 0$ in M . Then clearly $x = 0$ in both M_P and M_m for all prime ideals P and maximal ideals m . Now let $x = 0$ in M_m for all maximal ideals m . This means that there exists $a_m \in R \setminus m$ such that $a_m x = 0$. Let I be the ideal

$$I = \sum_{m \text{ a maximal ideal}} a_m R \subseteq R$$

Since $a_m \in I$ but $a_m \notin m$, we must have that I is not contained in any maximal ideals. Hence $I = R$. Then there exists $r_i \in R$ such that $1 = \sum_{i=1}^n r_i a_{m_i}$ for some $a_{m_i} \in R \setminus m_i$. Then we have

$$x = \sum_{i=1}^n (r_i a_{m_i} x) = 0 \in M$$

■

Definition 5.6.3 (Local Properties of Modules) Let R be a commutative ring. A property of R -modules is local if for any R -modules M , the following are equivalent.

- M has the property
- M_P has the property for all primes ideals P
- M_m has the property for all maximal ideals m

Lemma 5.6.4 Let R be a commutative ring. Let M be an R -module. Then the module being 0 is a local property.

Proof If $M = 0$, then clearly $M_P = 0$ and $M_m = 0$ for all prime ideals P and maximal ideals m . Then using 5.6.2 we conclude that if $M_m = 0$ for all maximal ideals m , then $M = 0$. ■

Proposition 5.6.5 (Injectivity and Surjectivity are Local Properties) Let R be a commutative ring. Let M, N be R -modules. Let $\phi : M \rightarrow N$ be an R -module homomorphism. Let S be a multiplicative subset of R . Then the following are equivalent.

- ϕ is injective (surjective)
- For each prime ideal P of R , the induced map $\phi_P : S^{-1}M \rightarrow S^{-1}N$ is injective (surjective)
- For each maximal ideal m of R , the induced map $\phi_m : S^{-1}M \rightarrow S^{-1}N$ is injective (surjective)

More local properties: nilpotent

Non-local properties: freeness, domain

Proposition 5.6.6 (Exactness is Local) Let R be a commutative ring. Let M_1, M_2, M_3 be R -modules. Let $f : M_1 \rightarrow M_2$ and $g : M_2 \rightarrow M_3$ be R -module homomorphisms. Then the following conditions are equivalent.

- The following sequence is exact:

$$M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3$$

- The following sequence is exact:

$$(M_1)_P \xrightarrow{f_P} (M_2)_P \xrightarrow{g_P} (M_3)_P$$

for all prime ideals P of R .

- The following sequence is exact:

$$(M_1)_m \xrightarrow{f_m} (M_2)_m \xrightarrow{g_m} (M_3)_m$$

for all maximal ideals m of R .

Proof (1) $\implies$ (2), (3) is clear since localization preserves exact sequences. It remains to show that (3) $\implies$ (1). Let $x \in M$. Then we have that $g_m(f_m(x)) = 0$ for all maximal ideals m . Since being 0 is a local property, we conclude that $g(f(x)) = 0$. Hence $\text{im}(f) \subseteq \ker(g)$. Since kernels and images and quotients commute with localizations, we have that

$$\left(\frac{\ker(g)}{\text{im}(f)} \right)_m \cong \frac{\ker(g_m)}{\text{im}(f_m)} = 0$$

Since being a zero module is a local property, we conclude that $\text{im}(f) = \ker(g)$. ■

6 Primary Decomposition

6.1 The Annihilator and Associated Primes

Let R be a commutative ring. Let M be an R -module. Recall that we define the annihilator of a subset $S \subseteq M$ to be the ideal

$$\text{Ann}_R(S) = \{r \in R \mid rs = 0 \text{ for all } s \in S\}$$

When R is a commutative ring, the annihilator is a two sided ideal and consequently has some nice properties.

Proposition 6.1.1 Let R be a commutative ring. Let M be an R -module. Let $\text{Ann}_R(x)$ for $x \in M$ be a maximal element in the set

$$\{\text{Ann}_R(x) \mid 0 \neq x \in M\}$$

Then $\text{Ann}_R(x)$ is a prime ideal.

Proof Suppose that $ab \in \text{Ann}_R(x)$ and $b \notin \text{Ann}_R(x)$. Notice that if $rx = 0$ then $r(bx) = brx = 0$ so that r annihilates bx . Hence $\text{Ann}_R(x) \subseteq \text{Ann}_R(bx)$. Since x is non-zero and $b \notin I$, bx is also non-zero hence $\text{Ann}_R(bx)$ lies in the given set of annihilators. Since $\text{Ann}_R(x)$ is maximal we conclude that

$$\text{Ann}_R(x) = \text{Ann}_R(bx)$$

But ab annihilates x by definition so that a annihilates bx . Hence $a \in \text{Ann}_R(bx) = \text{Ann}_R(x)$. Hence $\text{Ann}_R(x)$ is prime. ■

Recall that if $S \subseteq M$ is a subset and R is not a commutative ring, then in general we only have the relation

$$\text{Ann}_R(\langle S \rangle) \subseteq \text{Ann}_R(S)$$

Proposition 6.1.2 Let R be a commutative ring. Let M be an R -module. Let $S \subseteq M$ be a subset. Then

$$\text{Ann}_R(\langle S \rangle) = \text{Ann}_R(S)$$

Definition 6.1.3 (Associated Prime) Let R be a commutative ring. Let M be an R -module. Let P be a prime ideal of R . We say that P is an associated prime of M if

$$\text{Ann}_R(m) = P$$

for some $m \in M$.

Lemma 6.1.4 Let R be a Noetherian commutative ring. Let M be an R -module. If $M \neq 0$, then $\text{Ass}(M) \neq \emptyset$.

Proof By 6.1.1, there exists $x \in M$ such that $\text{Ann}_R(x)$ is a prime ideal. ■

Lemma 6.1.5 Let R be a commutative ring. Let M be an R -module. Let P be a prime ideal of R . Then P is an associated prime of M if and only if R/P is isomorphic to a submodule of M .

Proof If P is an associated prime, then $P = \text{Ann}_R(m)$ for some $0 \neq m \in M$. Then $\langle m \rangle \cong \frac{R}{\text{Ann}_R(m)}$ so that R/P is isomorphic to a submodule of M . Conversely, if $R/P \cong N \leq M$ for some submodule N , notice that R/P is cyclic and so N is generated by one element $n \in N$. Then $P = \text{Ann}_R(n)$. ■

Definition 6.1.6 (Set of Associated Prime) Let R be a commutative ring. Let M be an R -module. Define the set of associated primes of M to be

$$\text{Ass}(M) = \{P \in \text{Spec}(R) \mid P \text{ is an associated prime of } M\}$$

Another way to think about the set of associated primes of M is that

$$\text{Ass}(M) = \{\text{Ann}_R(m) \mid \text{Ann}_R(m) \in \text{Spec}(R)\}$$

Lemma 6.1.7 Let R be a Noetherian commutative ring. Let M be an R -module. Then we have

$$\bigcup_{P \in \text{Ass}(M)} P = \{r \in R \mid r \text{ is a zero divisor of } M\} \cup \{0\}$$

Proof If $r \in R$ is a non-zero zero divisor of M , then $rm = 0$ for some $0 \neq m \in M$. Then $r \in \text{Ann}_R(m)$. By 6.1.1, r is contained some prime ideal that is an annihilator. Hence r lies in the union in the left. Conversely, if r lies in some annihilator then clearly r is a zero divisor, or $r = 0$. ■

Proposition 6.1.8

Let R be a commutative ring. Let S be a multiplicative subset of R . Let M be an $S^{-1}R$ -module. Then we have

$$\text{Ass}_{S^{-1}R}(S^{-1}M) = \text{Ass}_R(S^{-1}M)$$

Proof

Proposition 6.1.9

Let R be a Noetherian commutative ring. Let S be a multiplicative subset of R . Let M be an R -module. Then the following are true.

- Considering $\text{Spec}(S^{-1}R)$ as a subset of $\text{Spec}(R)$ by the correspondence of prime ideals of localization, we have

$$\text{Ass}_R(S^{-1}M) = \text{Ass}_R(M) \cap \text{Spec}(S^{-1}R)$$

- Let P be a prime ideal of R . Then $P \in \text{Ass}_R(M)$ if and only if $PR_P \in \text{Ass}_{R_P}(M_P)$.

Proof

Suppose that $P \in \text{Ass}_R(S^{-1}M)$. Let φ denote the localization map. Then $P = \text{Ann}_R(m/s)$ for some $m/s \in S^{-1}M$. But $p/t \in P$ annihilates m/s if and only if p annihilates m . Hence by lifting P back to R by the correspondence, $\varphi^{-1}(P) = \text{Ann}_R(m)$ and must also be a prime lying in $\text{Spec}(S^{-1}R)$.

Conversely, if $P \in \text{Spec}(S^{-1}R)$ annihilates $m \in M$, then it must also annihilate $m/1$. And $m/1$ is annihilated by $r/s \in S^{-1}R$ if r annihilates m in M . Thus we conclude. ■

Proposition 6.1.10 Let R be a commutative ring. Let the following be an exact sequence of R -modules.

$$0 \longrightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$$

Then we have

$$\text{Ass}(M_2) \subseteq \text{Ass}(M_1) \cup \text{Ass}(M_3)$$

Example 6.1.11 Consider the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

Then $\text{Ass}(\mathbb{Z}) \subset \text{Ass}(\mathbb{Z}) \cup \text{Ass}(\mathbb{Z}/2\mathbb{Z})$ is a strict subset.

Proof Clearly $(2) \subseteq \mathbb{Z}$ annihilates $\mathbb{Z}/2\mathbb{Z}$ but does not annihilate $\mathbb{Z}$. ■

Theorem 6.1.12 (Disassembly of an R -Module) Let R be a Noetherian commutative ring. Let M be a finitely generated R -module. Then there exists a chain of R -submodules

$$0 = M_0 \subset M_1 \subset \cdots \subset M_k = M$$

such that

$$\frac{M_i}{M_{i-1}} \cong \frac{R}{P_i}$$

for some prime ideal P_i of R .

Proof If M is trivial then we are done. So suppose that $M \neq \{0\}$. We define the R -submodules inductively.

- When $n = 1$, $\text{Ass}(M) \neq \emptyset$, say $P_1 \in \text{Ass}(M)$. Since P_1 is an annihilator, $M_1 = R/P_1$ is an R -submodule of M .
- Assume that $M_1 \subset \cdots \subset M_i$ is constructed. If $M_i = M$ then we are done. If not, then $M/M_i \neq \{0\}$ and $P_{i+1} \in \text{Ass}(M/M_i) \neq \emptyset$. Then $N = R/P_{i+1}$ is an R -submodule of M/M_i . By the correspondence theorem for R -modules, N corresponds to an R -submodule M_{i+1} of M containing M_i .

The process eventually terminates since M is Noetherian. ■

Proposition 6.1.13 Let R be a Noetherian commutative ring. Let M be a finitely generated R -module. Let $P_1, \dots, P_n \in \text{Spec}(R)$ be the prime ideals appearing in the disassembly of M . Then

$$\text{Ass}(M) \subseteq \{P_1, \dots, P_n\}$$

Proof We induct on the length of the disassembly. When $n = 0$ the result is trivial. Suppose that the result holds true for all R -modules whose length of disassembly is $\leq k$. Let M be an R -module whose disassembly has length $k+1$. Let $\varphi : M/M_k \rightarrow R/P_k$ be the isomorphism given in the disassembly. Let $m \in M$ be such that $\text{Ann}_R(m)$ is a prime ideal. If $m \in M_k$ then by inductive hypothesis we are done. So suppose that $m \notin M_k$. If r annihilates m , then r annihilates $\varphi(m)$ in R/P_k . Hence ■

Definition 6.1.14 (Embedded Associated Primes) Let R be a commutative ring. Let M be an R -module. Let $I \in \text{Ass}(M)$ be an associated prime. We say that I is embedded if I is not minimal in $\text{Ass}(M)$.

6.2 The Support of a Module

Definition 6.2.1 (Support of a Module) Let A be a commutative ring. Let M be an A -module. The support of M is the subset

$$\text{Supp}(M) = \{P \text{ a prime ideal of } A \mid M_P \neq 0\}$$

Lemma 6.2.2 Let R be a commutative ring. Let the following be an exact sequence of R -modules.

$$0 \longrightarrow M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \longrightarrow 0$$

Then we have

$$\text{Supp}(M_2) = \text{Supp}(M_1) \cup \text{Supp}(M_3)$$

Proposition 6.2.3 Let R be a commutative ring. Let M be a finitely generated R -module. Then

$$\text{Supp}(M) = \{P \in \text{Spec}(R) \mid \text{Ann}_R(M) \subseteq P\}$$

Proof We first show the case when M is generated by one element $m \in M$. Let $P \in \text{Supp}(M)$. Then $M_P \neq 0$ and so $m/1 \neq 0 \in M_P$. This means that for all $s \in R \setminus P$, we have $sm \neq 0$. Then $R \setminus P \cap \text{Ann}_R(m) = \emptyset$. Then $P \supseteq \text{Ann}_R(m) = \text{Ann}_R(M)$. Conversely, suppose that $P \notin \text{Supp}(M)$. Then $M_P = 0$ and so $m/1 = 0$. So there exists $s \in R \setminus P$ such that $sm = 0$. Hence $R \setminus P \cap \text{Ann}_R(m) \neq \emptyset$ and so $\text{Ann}_R(M) = \text{Ann}_R(m)$ is not a subset of P .

Now suppose that M is finitely generated by $m_1, \dots, m_k$. Then we have

$$\begin{aligned}
 \text{Supp}(M) &= \bigcup_{i=1}^k \text{Supp}(R \cdot m_i) \\
 &= \bigcup_{i=1}^k \{P \in \text{Spec}(R) \mid \text{Ann}_R(m_i) \subseteq P\} \\
 &= \bigcup_{i=1}^k \{P \in \text{Spec}(R) \mid \text{Ann}_R(m_i) \subseteq P\} \\
 &= \left\{ P \in \text{Spec}(R) \mid \bigcap_{i=1}^k \text{Ann}_R(m_i) \subseteq P \right\} \quad (\text{Imm1.1.2}) \\
 &= \{P \in \text{Spec}(R) \mid \text{Ann}_R(M) \subseteq P\}
 \end{aligned}$$

■

Lemma 6.2.4 Let R be a commutative ring. Let M be a finitely generated R -module. Let $P_1, \dots, P_n \in \text{Spec}(R)$ be a complete list of distinct minimal prime ideals over $\text{Ann}_R(M)$. Then we have

$$\text{Supp}(M) = \bigcup_{k=1}^n \{P \in \text{Spec}(R) \mid P_k \subseteq P\}$$

Proof We induct on the length of the diassembly of M . If $n = 1$, then M is simple, and $M \cong R/P$ with $P = \text{Ann}_R(M)$. Now suppose the result is true for $\leq n - 1$. Let $0 = M_0 \subset \dots \subset M_n = M$ be the diassembly of M . Then we obtain an exact sequence of the form

$$0 \longrightarrow M_{n-1} \xrightarrow{f} M \xrightarrow{g} \frac{M}{M_{n-1}} \longrightarrow 0$$

In particular, we have $\text{Supp}(M) = \text{Supp}(M/M_{n-1}) \cup \text{Supp}(M_{n-1})$. By induction, we have $\text{Supp}(M_{n-1}) = \bigcup_{i=1}^{n-1} \{P \in \text{Spec}(R) \mid P_i \subseteq P\}$, and similarly for the simple module M/M_{n-1} we have the result of the case $n = 1$. Hence we are done. ■

Let R be a commutative ring. Let M be an R -module. Let $P_1, \dots, P_n \in \text{Spec}(R)$ be the prime ideals appearing in the diassembly of M . Then summarizing the above, we have

$$\text{Ass}(M) \subseteq \{P_1, \dots, P_n\} \subseteq \text{Supp}(M) = \{P \in \text{Spec}(R) \mid \text{Ann}_R(M) \subseteq P\}$$

It turns out that the minimal primes of the four sets coincide.

Proposition 6.2.5 Let R be a Noetherian commutative ring. Let M be a finitely generated R -module. Let $P_1, \dots, P_n \in \text{Spec}(R)$ be the prime ideals appearing in the diassembly of M . Then the following sets are equal.

- $\{P \in \text{Spec}(R) \mid P \text{ is minimal in } \text{Supp}(M)\}.$
- $\{P \in \text{Spec}(R) \mid P \text{ is minimal in } \text{Ass}(M)\}.$
- $\{P \in \text{Spec}(R) \mid P \text{ is a minimal prime ideal over } \text{Ann}_R(M)\}.$
- $\{P \in \text{Spec}(R) \mid P \text{ is minimal in } \{P_1, \dots, P_n\}\}.$

Proof

- (1) = (4): By the above lemma, we have

$$\text{Supp}(M) = \bigcup_{i=1}^n \{P \in \text{Spec}(R) \mid P_i \subseteq P\} = \bigcup_{P \in \{P_1, \dots, P_n\} \text{ minimal}} \{Q \in \text{Spec}(R) \mid P_k \subseteq Q\}$$

If P is minimal in $\text{Supp}(M)$, then it is minimal in the union. Then $P \in \{Q \in \text{Spec}(R) \mid P_k \subseteq Q\}$ for some minimal P_k . If $P \neq P_k$ then evidently P is not minimal, hence $P = P_k$. The converse is similar.

- (1) = (3): By 6.2.3, $\text{Supp}(M) = \{P \in \text{Spec}(R) \mid \text{Ann}_R(M) \subseteq P\}$ means that P is a minimal prime ideal over $\text{Ann}_R(M)$ if and only if P is minimal in $\text{Supp}(M)$.

■

6.3 Primary Ideals

Definition 6.3.1 (Primary Ideals) Let R be a commutative ring. Let Q be a proper ideal of R . We say that Q is a primary ideal of R if $fg \in Q$ implies $f \in Q$ or $g^m \in Q$ for some $m > 0$.

Lemma 6.3.2 Let R be a commutative ring. Let Q be an ideal of R . Then the following are true.

- If Q is a prime ideal, then Q is a primary ideal.
- If $\sqrt{Q}$ is a maximal ideal, then Q is a primary ideal.

Proof Let $fg \in Q$. Since Q is prime, $f \in Q$ or $g \in Q$ and so we are done.

Let $fg \in Q$ and $f \notin Q$. Let $I = \{g \in R \mid fg \in Q\}$. Clearly $Q \subseteq I$. Moreover $1 \notin I$. Hence I is a proper ideal. Then we have $m = \sqrt{Q} \subseteq \sqrt{I}$. Hence $I \subseteq \sqrt{I} = m$ since m is maximal. This shows that $g \in I$ implies $g \in m = \sqrt{Q}$. Hence we are done. ■

Lemma 6.3.3 Let $\phi : R \rightarrow S$ be a ring homomorphism and Q be a primary ideal in S . Then $\phi^{-1}(Q)$ is primary in R .

Proposition 6.3.4 Let R be a commutative ring. Let Q be a proper ideal of R . Then Q is primary if and only if every zero divisor in R/Q is nilpotent.

Lemma 6.3.5 Let R be a commutative ring. Let Q be a primary ideal of R . Then the following are true.

- $\sqrt{Q}$ is a prime ideal.
- $\sqrt{Q}$ is minimal among primes that contain Q .

Definition 6.3.6 (P-Primary Ideals) Let R be a commutative ring. Let P be a prime ideal. Let Q be an ideal. We say that Q is a P -primary ideal of R if the following are true.

- Q is a primary ideal.
- $\sqrt{Q} = P$.

Lemma 6.3.7 Let R be a commutative ring. Let P be a prime ideal. Let Q_1, Q_2 be P -primary ideals. Then $Q_1 \cap Q_2$ is a P -primary ideal.

Proposition 6.3.8 Let R be a Noetherian commutative ring. Let P be a prime ideal of R . Let Q be a proper ideal. Then Q is P -primary if and only if $\text{Ass}(R/Q) = \{P\}$.

Proof Let Q be a P -primary ideal. We know that $\text{Ass}(R/Q)$ is non-empty. So let I be a prime ideal such that $I \in \text{Ass}(R/Q)$. Clearly $Q \subseteq I$. There exists $[r] \in R/Q$ where $[r] \neq 0$ such that $\text{Ann}_R([r]) = I$. Let $x \in I \setminus \{0\}$. Then $[xr] = [x] \cdot [r] = 0 \in R/Q$ implies that $[x]$ is a zero divisor of R/Q . By 6.3.4, we conclude that $[x] \in N(R/Q)$. Then by lemma 1.4.5, we have $x \in \sqrt{Q} = P$. Hence we have $Q \subseteq I \subseteq \sqrt{Q} = P$. Taking radical gives $I = P$ since I is a prime ideal. Hence $\text{Ass}(R/Q) = \{P\}$.

Now suppose that $\text{Ass}(R/Q) = \{P\}$. Let $xy \in Q$. Suppose that $x \notin Q$. Then we have $[x] \cdot [y] = [xy] = 0 \in R/Q$. Hence $y \in \text{Ann}_R([x])$. But we also have

$$\sqrt{\text{Ann}_R([x])} = \bigcap_{\substack{I \text{ is a minimal prime} \\ \text{ideal over } \text{Ann}_R([x])}} I = \bigcap_{\substack{I \text{ is minimal} \\ \text{in } \text{Ass}([x])}} I = P$$

Similarly, we know that

$$\sqrt{\text{Ann}_R(R/Q)} = \bigcap_{\substack{I \text{ is a minimal prime} \\ \text{ideal over } \text{Ann}_R(R/Q)}} I = \bigcap_{\substack{I \text{ is minimal} \\ \text{in } \text{Ass}(R/Q)}} I = P$$

Then $y \in \text{Ann}_R([x])$ implies that $y \in \sqrt{\text{Ann}_R([x])} = P = \sqrt{\text{Ann}_R(R/Q)}$. This means that $y^n \in \text{Ann}_R(R/Q)$ for some $n \in \mathbb{N}$. Hence $y^n \in Q$. ■

Lemma 6.3.9 Let R be a Noetherian commutative ring. Let P be a prime ideal. Let Q be P -primary. Then we have

$$P^n \subseteq Q \subseteq P$$

for some $n \in \mathbb{N}$.

Proof Since R is Noetherian, P is finitely generated. Suppose that $P = (f_1, \dots, f_k)$. Since $\sqrt{Q} = P$, we have $f_i^{n_i} \in Q$ for some $n_i \in \mathbb{N}$. Then for any monomial of degree $m > \sum_{i=1}^k (n_i - 1)$ is a multiple of $f_i^{n_i}$ for some $1 \leq i \leq k$. Hence $P^m \subseteq Q$. ■

Example 6.3.10 Let k be a field. Let $I = (x^2, xy) \subseteq k[x, y]$. Then we have

$$(x^2) \subseteq I \subseteq (x)$$

but I is not primary. In particular, this shows that the condition in the above lemma is not a sufficient condition for ideals to be primary.

Proof I is not primary because $xy \in I$ but $x \notin I$ and $y^n \notin I$ for any $n \in \mathbb{N}$. ■

Corollary 6.3.11 Let R be a Noetherian commutative ring. Let m be a maximal ideal of R . Let Q be a proper ideal. Then the following are equivalent.

- Q is m -primary.
- $\text{Ass}(R/Q) = \{m\}$
- There exists $n \in \mathbb{N}$ such that $m^n \subseteq Q \subseteq m$.

Proof By the above proposition we have (1) $\iff$ (2). The above lemma also shows that (1) $\implies$ (3). Finally, suppose that $m^n \subseteq Q \subseteq m$. Then taking radicals give $m = \sqrt{m^n} \subseteq \sqrt{Q} \subseteq \sqrt{m} = m$. By 6.3.3 we conclude that Q is m -primary. ■

6.4 Primary Decomposition

We want to express ideal I in R as $I = P_1^{e_1} \cdots P_n^{e_n}$ similar to a factorization of natural numbers, for some prime ideals $P_1, \dots, P_n$. However this notion fails and thus we have the following new type of ideal.

Definition 6.4.1 (Primary Decompositions) Let A be a commutative ring. Let I be an ideal of A . A primary decomposition I consists of primary ideals $Q_1, \dots, Q_r$ of A such that

$$I = Q_1 \cap \dots \cap Q_r$$

Example 6.4.2 Let k be a field. For any $\alpha \in k$, the ideal $(x^2, xy) \subseteq k[x, y]$ has a primary decomposition given by

$$(x^2, xy) = (x) \cap (x^2, y - \alpha x)$$

Proof Since (x) is a prime ideal, it is a (x) -primary ideal. ■

Definition 6.4.3 (Minimal Primary Decompositions) Let A be a commutative ring. Let I be an ideal of A . Let

$$I = Q_1 \cap \dots \cap Q_r$$

be a primary decomposition of I . We say that the decomposition is minimal if the following are true.

- Each $\sqrt{Q_i}$ are distinct for $1 \leq i \leq r$
- Removing a primary ideal changes the intersection. This means that for any i , $I \neq \bigcap_{j \neq i} Q_j$

Theorem 6.4.4 Let R be a Noetherian commutative ring. Let I be a proper ideal of R . Then I admits a minimal primary decomposition.

Definition 6.4.5 (Prime Divisors of an Ideal) Let R be a commutative ring. Let I be an ideal of R . We say that a prime ideal P of R is a prime divisor of I if $P = \sqrt{Q}$ for some ideal Q that lies in a minimal primary decomposition of I .

6.5 The Two Uniqueness Theorems

Theorem 6.5.1 (First Uniqueness Theorem)

Let R be a Noetherian commutative ring. Let I be an ideal of R . Let

$$I = \bigcap_{i=1}^k Q_i$$

be a minimal primary decomposition of I , where each Q_i is P_i -primary. Then we have

$$\text{Ass}(R/I) = \{P_1, \dots, P_k\}$$

Moreover, $\text{Ass}(R/I)$ is unique.

Theorem 6.5.2 (Second Uniqueness Theorem)

Let R be a Noetherian commutative ring. Let I be an ideal of R . Let

$$I = \bigcap_{i=1}^k Q_i$$

be a minimal primary decomposition of I , where each Q_i is P_i -primary. Let P be a minimal element of $\text{Ass}(R/I) = \{P_1, \dots, P_n\}$. Let $\varphi : R \rightarrow R_P$ be the localization map. Then we have

$$Q_i = \varphi^{-1}(I_P)$$

Moreover, the primary decomposition is uniquely determined by the minimal P in $\text{Ass}(R/I)$.

6.6 Symbolic Powers

Definition 6.6.1 (Symbolic Powers)

Let R be a Noetherian commutative ring. Denote $\varphi : R \rightarrow R_P$ the localization map with respect to the prime ideal P . Let I be an ideal of R . Define the n th symbolic power of I to be

$$I^{(n)} = \bigcap_{P \in \text{Ass}(R/I)} \varphi^{-1}(I^n R_P)$$

7 Integral Dependence

7.1 Integral Elements

Definition 7.1.1 (Integral Elements) Let B be a commutative ring and let $A \subseteq B$ be a subring. Let $b \in B$. We say that b is integral over A if there exists a monic polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in A[x]$ such that $p(b) = 0$.

When A and B are field, this is a familiar notion in Field and Galois theory.

Lemma 7.1.2 Let K be a field. Let $F \subseteq K$ be a subfield. Let $k \in K$. Then k is integral over F if and only if k is algebraic over F .

Proposition 7.1.3 Let B be a commutative ring and let $A \subseteq B$. Let $b \in B$. Then the following are equivalent.

- b is integral over A
- $A[b] \subseteq B$ is finitely generated A -submodule.
- There exists an A sub-algebra $A' \subseteq B$ such that $A[b] \subseteq A'$ and A' is finitely generated as an A -module.

Proof

- (1) $\implies$ (2): Since b is integral over A , $b^n = a_{n-1}b^{n-1} + \cdots + a_1b + a_0$. Hence $A[b] = \bigoplus_{i=0}^{n-1} A \cdot b^i$ is a finitely generated A -module.
- (2) $\implies$ (3): Choose $A' = A[b]$.
- (3) $\implies$ (1). By assumption, A' is a finitely generated A -module. Let $\phi : A' \rightarrow A'$ be the ring homomorphism defined by $\phi(x) = bx$. By Cayley-Hamilton theorem, there exists $a_1, \dots, a_{n-1} \in A$ such that

$$\phi^n + a_{n-1}\phi^{n-1} + \cdots + a_1\phi + a_0 = 0$$

Since ϕ is the multiplication by b map, we have

$$(b^n + a_{n-1}b^{n-1} + \cdots + a_1b + a_0)(y) = 0$$

for all $y \in A'$. Choosing $y = 1$, we see that b is integral over A . ■

Lemma 7.1.4 Let $A \subseteq B$ be commutative rings. Then B is a finitely generated A -module if and only if $B = A[x_1, \dots, x_n]$ for some $x_1, \dots, x_n \in B$ that is integral over A .

Proof Induct on n and use the fact that x_i is integral over A if and only if $A[x_i]$ is a finitely generated A -module, and the fact that x_i is integral over $A[x_1, \dots, x_{i-1}]$. ■

Proposition 7.1.5 Let B be a commutative ring and let $A \subseteq B$ be a subring. Let $b_1, b_2 \in B$ be integral over A . Then $b_1 + b_2$ and b_1b_2 are both integral over A .

7.2 Integral Closure

Definition 7.2.1 (Integral Closure) Let B be a commutative ring. Let $A \subseteq B$ be a subring. Define the subring

$$\overline{A} = \{b \in B \mid b \text{ is integral over } A\}$$

to be the integral closure of A in B .

Example 7.2.2 The integral closure of $\mathbb{Z} \subseteq \mathbb{Q}$ is $\mathbb{Z}$.

Proposition 7.2.3 Let B be a commutative ring. Let $A \subseteq B$ be a subring. Let S be a multiplicatively closed subset of A . Then

$$\overline{S^{-1}A} = S^{-1}\overline{A}$$

Proof

Let $r/s \in S^{-1}\overline{A}$. Then there exists $a_0, \dots, a_{n-1} \in A$ such that $r^n + \sum_{i=0}^{n-1} a_i r^i = 0$. Dividing by s^n gives

$$(r/s)^n + \sum_{i=0}^{n-1} (a_i/s^{n-i})(r/s)^i = 0$$

Hence $r/s \in \overline{S^{-1}A}$.

Conversely, suppose that $r/s \in \overline{S^{-1}A}$. Then there exists $f_i/g_i \in S^{-1}A$ such that

$$(r/s)^n + \sum_{i=0}^{n-1} (f_i/g_i)(r/s)^i = 0$$

Multiplying by $(sg_0 \cdots g_{n-1})^n$ gives

$$(g_0 \cdots g_{n-1}r)^n + \sum_{i=0}^{n-1} (f_i s^{n-i} g_0^{n-i} \cdots g_{i-1}^{n-i} \cdot g_i^{n-i-1} \cdots g_{i+1}^{n-i} \cdots g_{n-1}^{n-i})(g_0 \cdots g_{n-1} \cdot r)^i = 0$$

Hence $g_0 \cdots g_{n-1} \cdot r \in \overline{A}$. Since each $g_i \in S$, we have that $r/s = (g_0 \cdots g_{n-1} \cdot r)/(g_0 \cdots g_{n-1} \cdot s) \in S^{-1}\overline{A}$. ■

Definition 7.2.4 (Integral Extensions) Let B be a commutative ring and let $A \subseteq B$ be a subring. We say that B is integral over A if $\overline{A} = B$. We also say that B is the integral extension of A .

Lemma 7.2.5 Let $A \subseteq B \subseteq C$ be commutative rings. Then C is integral over B and B is integral over A if and only if C is integral over A .

Proposition 7.2.6 Let A, B be commutative rings such that $A \subset B$ is an integral extension. Then the following are true.

- Let J be an ideal of B . Then $\frac{B}{J}$ is integral over $\frac{A}{J \cap A}$.
- Let S be a multiplicative subset of B . Then $S^{-1}B$ is integral over $S^{-1}A$.

Proof Suppose that J is an ideal of B . Let $b + J \in B/J$. Since $b \in B$ and B is integral over A , there exists $a_0, \dots, a_{n-1} \in A$ such that

$$b^n + a_{n-1}b^{n-1} + \cdots + a_1b + a_0 = 0$$

Reduction to J gives

$$(b + J)^n + (a_{n-1} + J)(b + J)^{n-1} + \cdots + (a_1 + J)(b + J) + (a_0 + J) = J$$

This shows that $b + J$ is an integral element of $A/J \cap A$ because each $a_i + J$ is an element of $A/J \cap A$ by restriction to A .

Let $b/s \in S^{-1}B$. Since B is integral over A , there exists $a_0, \dots, a_{n-1} \in A$ such that

$$b^n + a_{n-1}b^{n-1} + \dots + a_1b + a_0 = 0$$

Dividing s^n on both sides give

$$\frac{b^n}{s^n} + \frac{a_{n-1}}{s} \frac{b^{n-1}}{s^{n-1}} + \dots + \frac{a_1}{s^{n-1}} \frac{b}{s} + \frac{a_0}{s^n} = 0$$

This shows that b/s is an integral element of $S^{-1}A$. ■

Lemma 7.2.7 Let A, B be integral domains such that $A \subset B$ is an integral extension. Then A is a field if and only if B is a field.

Proof Suppose that A is a field. Let $0 \neq b \in B$. Then there exists $a_0, \dots, a_{n-1} \in A$ such that

$$b^n + a_{n-1}b^{n-1} + \dots + a_1b + a_0 = 0$$

for smallest of such $n \in \mathbb{N}$. Rearranging gives

$$b(b^{n-1} + a_{n-1}b^{n-2} + \dots + a_1) = -a_0$$

Notice that $a_0 \neq 0$ because otherwise it contradicts the minimality of n . Since A is a field, we can divide $-a_0 \neq 0$ on both sides to find an inverse of b . Hence B is a field.

Now assume that B is a field. Let $0 \neq a \in A$. Since B is a field, $a^{-1} \in B$ is such that there exists $a_0, \dots, a_{n-1} \in A$ such that

$$a^{-n} + a_{n-1}a^{-(n-1)} + \dots + a_1a^{-1} + a_0 = 0$$

Multiplying a^{n-1} on both sides and rearranging, we get

$$a^{-1} = -(a_{n-1} + \dots + a_1a^{n-2} + a_0a^{n-1})$$

This shows that $a^{-1} \in A$. Hence A is a field. ■

Definition 7.2.8 (Integrally Closed) Let B be a commutative ring. Let $A \subseteq B$ be a subring. We say that A is integrally closed in B if $\overline{A} = A$.

Theorem 7.2.9 (Gauss's Lemma) Let B be a commutative ring. Let $A \subseteq B$ be a subring. Suppose that A is integrally closed in B . Then the following are true.

- If $f, g \in B[x]$ are monic polynomials such that $fg \in A[x]$, then $f, g \in A[x]$.
- If $f \in A[x]$ is irreducible, then f is irreducible as a polynomial in $B[x]$.

Proof Clearly the first statement implies the second. We first prove that for any monic polynomial $f \in B[x]$, there exists a ring C such that $B \subseteq C$ and f factorizes as a product of linear terms in $C[x]$. To show this, we induct on n . If $n = 1$ then we are done. Suppose that the hypothesis is true for some $k \in \mathbb{N}$. Suppose that $\deg(f) = k + 1$. ■

7.3 Four Main Theorems of Integral Extensions

We want to compare prime ideals between integral extensions.

Lemma 7.3.1 Let A, B be rings such that $A \subset B$ is an integral extension. Let Q be a prime ideal of B . Then $Q \cap A$ is a maximal ideal of A if and only if Q is a maximal ideal of B .

Proof By 7.2.6, we know that B/Q is integral over $A/Q \cap A$. By 7.2.7, B/Q is a field if and only if $A/Q \cap A$ is a field. Hence Q is a maximal ideal of B if and only if $Q \cap A$ is a maximal ideal of A . ■

Theorem 7.3.2 (The Lying Over Theorem)

Let A, B be rings such that $A \subset B$ is an integral extension. Let P be a prime ideal of A . Then there exists a prime ideal Q of B such that $P = Q \cap A$

Proof

Let $\alpha : A \rightarrow A_P$ and $\beta : B \rightarrow B_P$ be the localization maps. Consider the following commutative diagram.

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \alpha \downarrow & & \downarrow \beta \\ A_P & \hookrightarrow & B_P \end{array}$$

Since PB_P is the unique maximal ideal of B_P , we know that $PA_P = PB_P \cap A_P$ is the unique maximal ideal of A_P . On the other hand, we also know that $\beta^{-1}(PB_P)$ is a prime ideal of B . By commutativity of the diagram, we have that P is mapped to $\beta^{-1}(PB_P)$. Then by definition of extension we have that $\beta^{-1}(PB_P) \cap A = P$. ■

Theorem 7.3.3 (The Incomparability Theorem) Let A, B be rings such that $A \subset B$ is an integral extension. If $Q_1 \neq Q_2$ are distinct prime ideals of B such that $Q_1 \cap A = P = Q_2 \cap A$, then $Q_1 \not\subseteq Q_2$ and $Q_2 \not\subseteq Q_1$.

Proof

We prove the contrapositive statement. That is, if Q_1, Q_2 are prime ideals of B such that $Q_1 \cap A = P = Q_2 \cap A$ and $Q_1 \subseteq Q_2$, then $Q_1 = Q_2$.

Let $\alpha : A \rightarrow A_P$ and $\beta : B \rightarrow B_P$ be the localization maps. Consider the following commutative diagram.

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \alpha \downarrow & & \downarrow \beta \\ A_P & \hookrightarrow & B_P \end{array}$$

Since PB_P is the unique maximal ideal of B_P , we know that $PA_P = PB_P \cap A_P$ is the unique maximal ideal of A_P . On the other hand, we also know that $\beta^{-1}(PB_P)$ is a prime ideal of B . By commutativity of the diagram, we have that P is mapped to $\beta^{-1}(PB_P)$. Then by definition of extension we have that $\beta^{-1}(PB_P) \cap A = P$.

Let Q_1, Q_2 be as given. We have that

$$(Q_1 \cap A)A_P = PA_P = (Q_2 \cap A)A_P$$

is the same maximal ideal of A_P since they both contract to P in A . By the above lemma, $(Q_1 \cap A)B_P$ and $(Q_2 \cap A)B_P$ are both maximal ideals of B_P . By commutativity of the diagram, $(Q_1 \cap A)B_P = Q_1B_P$ and $(Q_2 \cap A)B_P = Q_2B_P$. Since $Q_1 \subseteq Q_2$, we have that $Q_1B_P \subseteq Q_2B_P$. Since Q_1B_P and Q_2B_P are both maximal ideals, they must be equal. Hence by contraction we deduce that $Q_1 = Q_2$. ■

Theorem 7.3.4 (The Going-Up Theorem) Let A, B be commutative rings such that $A \subset B$ is an integral extension. Let $0 \leq m < n$. Consider the following situation

$$\begin{array}{ccc} B & Q_1 \subseteq \cdots \subseteq Q_m & \text{(Prime ideals of } B) \\ \uparrow & & \\ A & P_1 \subseteq \cdots \subseteq P_m \quad \subseteq P_{m+1} \subseteq \cdots \subseteq P_n & \text{(Prime ideals of } A) \end{array}$$

where $Q_i \cap A = P_i$ for $1 \leq i \leq m$. Then there exists prime ideals $Q_{m+1}, \dots, Q_n$ of B such that the following are true.

- $Q_{m+1} \subseteq \dots \subseteq Q_n$
- $Q_i \cap A = P_i$ for $m+1 \leq i \leq n$

Moreover, if the chain of prime ideals of A is strictly increasing the new chain of prime ideals in B is strictly increasing.

Proof By induction, it suffices to prove the case $m = 1$ and $n = 2$. This means that we want to find a prime ideal Q_2 such that $Q_1 \subseteq Q_2$ and $Q_2 \cap A = P_2$. By 7.2.6, B/Q_1 is integral over A/P_1 . Since P_2/P_1 is a prime in A/P_1 by the correspondence theorem, by 7.3.2 there exists a prime ideal Q_2/Q_1 in B/Q_1 such that $Q_2/Q_1 \cap A/P_1 = P_2/P_1$. This implies that $Q_2 \cap A = P_2$.

Finally, suppose that the chain of prime ideals in A is strictly increasing. Suppose for a contradiction that $Q_i = Q_{i+1}$. Then we must have $P_i = Q_i \cap A = Q_{i+1} \cap A = P_{i+1}$, a contradiction. ■

7.4 Zariski's Lemma

Lemma 7.4.1 Let F be a field. Let $f \in F[x]$ be a polynomial. Then the localization $F[x]_f$ is not a field.

Proof By 1.8.1, $F[x]$ has infinitely many irreducible polynomials. Then there exists a monic irreducible polynomial g that does not divide f . Assume for a contradiction that $F[x]_f$ is a field. Then $g/1$ is invertible. So there exists $h \in F[x]$ and $n \in \mathbb{N}$ such that $1 = g \cdot \frac{h}{f^n}$. This means that there exists $m \in \mathbb{N}$ such that $ghf^m = f^{n+m} \in F[x]$. If $n + m = 0$, then g is a unit, a contradiction. Otherwise, g divides f^{n+m} . Since g is irreducible, g divides f and is also a contradiction. Hence $F[x]_f$ is not a field. ■

Theorem 7.4.2 (Zariski's Lemma) Let F be a field. Let K/F be a field extension. Then K/F is a finite field extension if and only if K is finitely generated as an F -algebra.

Proof Since K is finitely generated as an F -algebra, there exists $x_1, \dots, x_n \in K$ such that every element in K can be written as a polynomial in $x_1, \dots, x_n$. This means that $K = F(x_1, \dots, x_n)$ as fields. Suppose for a contradiction that K/F is not an algebraic (integral) extension. Without loss of generality, suppose that $F(x_1, \dots, x_r)/F$ is transcendental (not integral) and $K/F(x_1, \dots, x_r)$ is algebraic (integral).

Let $L = F(x_1, \dots, x_{r-1})$. Consider the transcendental (not integral) extension $L(x_r)/L$. Now K is generated as an L -algebra by the elements $x_1, \dots, x_n$. Since $K/L(x_r)$ is integral, there exists monic polynomials $p_i \in L(x_r)[y]$ such that $p_i(x_i) = 0$. Since $L(x_r)$ is the field of fractions of the polynomial ring $L[x_r]$, each coefficient of p_i can be expressed as a fraction g/h for $g, h \in L(x_r)$ and $h \neq 0$. Let f be the product of all denominators of the coefficient of p_i for all i . Then $p_i \in L[x_r]_f[y]$. So every $x_1, \dots, x_n$ satisfies a monic polynomial with coefficients in $L[x_r]_f$. Hence the $L[x_r]_f$ subalgebra of K generated by $x_1, \dots, x_n$ is integral over $L[x_r]_f$. By 7.2.7, $L[x_r]_f$ is a field. This is a contradiction to the above lemma. Hence we are done. ■

There is a correspondence between the different terms used in Field and Galois Theory and Commutative Algebra

Field Extension K/F	B an A -algebra
$x \in K$ is algebraic	$b \in B$ is integral
K/F is an algebraic extension	$A \subseteq B$ is an integral extension
The algebraic closure $F < \bar{F} < K$	The integral closure $A \subseteq \bar{A} \subseteq B$
K/F is a finite extension	S is a finitely generated R -algebra

Corollary 7.4.3 Let F be an algebraically closed field. Let K be a field that is also a finitely generated algebra over F . Then $K = F$.

Proof By Zariski's lemma, K/F is a finite field extension. Let $x \in K$. Let f be the minimal polynomial of x . Since F is algebraically closed, f is linear. Hence $x \in F$. ■

Corollary 7.4.4 Let F be an algebraically closed field. Then we have

$$\max\text{Spec}(F[x_1, \dots, x_n]) = \{(x_1 - a_1, \dots, x_n - a_n) \mid (a_1, \dots, a_n) \in F^n\}$$

Proof Let m be a maximal ideal of $F[x_1, \dots, x_n]$. Then $F[x_1, \dots, x_n]/m$ is a finitely generated F -algebra that is a field. By the above, we have that $F[x_1, \dots, x_n]/m \cong F$. Then there exists $a_i \in F$ such that a_i corresponds to $x_i + m$ by the isomorphism. This means that $a_i + m = x_i + m$, or $(x_i - a_i) \in m$. Hence $(x_1 - a_1, \dots, x_n - a_n) \subseteq m$. Since $(x_1 - a_1, \dots, x_n - a_n)$ is maximal by the evaluation homomorphism, we conclude that $m = (x_1 - a_1, \dots, x_n - a_n)$. ■

7.5 Normal Domains

We now concern ourselves with integral domains. Let R be an integral domain. A special fact about R is that the canonical homomorphism $R \rightarrow R_{(0)} = \text{Frac}(R)$ is an injection. This means that we can think of R as living inside of $\text{Frac}(R)$ while preserving all the structure of R .

Definition 7.5.1 (Normal Domains) Let R be an integral domain. We say that R is normal if R is integrally closed in $\text{Frac}(R)$.

Proposition 7.5.2 Let R be a normal domain. Let S be a multiplicative subset of R . Then $S^{-1}R$ is a normal domain.

Proof We want to show that $S^{-1}R$ is integrally closed in $\text{Frac}(R) = \text{Frac}(S^{-1}R)$. This means that we want to show $\overline{S^{-1}R} = S^{-1}R$. It is clear that $S^{-1}R \subseteq \overline{S^{-1}R}$. So let $g \in \overline{S^{-1}R}$. Suppose that $p(x) = x^n + \sum_{k=0}^{n-1} a_k x^k \in (S^{-1}R)[x]$ such that $p(g) = 0$. Choose $s \in S$ such that $sa_i \in R$ for $0 \leq i \leq n-1$. Then notice that $sg \in S^{-1}R$ satisfies the monic polynomial

$$q(x) = x^n + \sum_{k=0}^{n-1} s^{n-k} a_k x^k$$

since $q(sg) = s^n g^n + \sum_{k=0}^{n-1} s^n a_k x^k = s^n p(g) = 0$. But q is a polynomial in R since $s^{n-k} a_k \in R$. Thus we have that $sg \in \overline{R} = R$ since R is normal. This means that $g \in S^{-1}R$ and hence we conclude. ■

Proposition 7.5.3 Let R be a commutative ring. If R is a UFD, then R is normal.

Proof Let $a/b \in \text{Frac}(R)$ that is integral. Assume that a, b do not have common factors. Then there exists $r_0, \dots, r_{n-1} \in R$ such that

$$\frac{a^n}{b^n} + r_{n-1} \frac{a^{n-1}}{b^{n-1}} + \dots + r_1 \frac{a}{b} + r_0 = 0$$

Rearranging, we get

$$a^n = -b(r_{n-1}a^{n-1} + \dots + r_1a + r_0b^{n-1})$$

This shows that any irreducible element dividing b also divides a^n , and hence a . Since a and b do not have common factors, this means that no irreducible element divides b . Since R is a UFD, b must be a unit. Hence $a/b \in R$. ■

Example 7.5.4 The integral closure of $\mathbb{Z}$ in $\mathbb{Q}[i]$ is $\mathbb{Z}[i]$.

Proof If $a+bi \in \mathbb{Z}[i]$, then $p(x) = x^2 - 2ax + a^2 + b^2$ is a monic polynomial such that $p(a+bi) = 0$. Conversely, let $z \in \mathbb{Q}[i]$ lie in the integral closure of $\mathbb{Z}$. Then z is also an integral element of $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a UFD, $\mathbb{Z}[i]$ is a normal domain and so is integrally closed in $\text{Frac}(\mathbb{Z}[i]) = \mathbb{Q}[i]$. So $z \in \overline{\mathbb{Z}[i]} = \mathbb{Z}[i]$ shows that $\overline{\mathbb{Z}} \subseteq \overline{\mathbb{Z}[i]}$. ■

Proposition 7.5.5 (Normal is a Local Property) Let R be an integral domain. Then the following are equivalent.

- R is normal
- R_P is normal for all prime ideals P
- R_m is normal for all maximal ideals m .

Proof Notice that an integral domain R is normal if and only if the canonical inclusion map $R \hookrightarrow \overline{R}$ is surjective. Since surjectivity is a local property, this map is surjective if and only if for all prime ideals P of R , $R_P \hookrightarrow \overline{R}_P$ is surjective. But $\overline{R}_P = \overline{R_P}$ by the above. Hence $R \hookrightarrow \overline{R}$ is surjective if and only if $R_P \rightarrow \overline{R_P}$ is surjective. Hence R is normal if and only if R_P is normal for all prime ideals P of R . The similar holds for all maximal ideals. ■

Atiyah-Macdonald

Proposition 7.5.6 Let R be a normal domain. Then $R[x]$ is a normal domain.

Proposition 7.5.7 Let R be a normal domain. Let $\text{Frac}(R) < K$ be an algebraic extension. Let $a \in K$. Then a is integral over R if and only if the minimal polynomial $\min(\text{Frac}(R), a) \in R[x]$.

Proof Suppose that $\min(\text{Frac}(R), a) \in R[x]$. Then $\min(\text{Frac}(R), a)(a) = 0$ and $\min(\text{Frac}(R), a)$ is monic by definition. Hence a is integral over R .

Now suppose that $a \in K$ is integral over R . Let $\overline{K}$ be the algebraic closure of K . Then $\min(\text{Frac}(R), a)$ splits into monic irreducible polynomials

$$\min(\text{Frac}(R), a)(x) = (x - a_1) \cdots (x - a_n) \in \overline{K}[x]$$

for $a_1, \dots, a_n \in \overline{K}$. Since a is integral over R , there exists a monic polynomial $g \in R[x]$ such that $g(a) = 0$. By definition of the minimal polynomial, we have $\min(\text{Frac}(R), a)$ divides g . Hence $g(a_i) = 0$ for each i and that $a_1, \dots, a_n$ are integral over R . Now the coefficients of $\min(\text{Frac}(R), a)$ are sums and products of $a_1, \dots, a_n$, and hence are also integral over R . But R is a normal domain so the coefficients of $\min(\text{Frac}(R), a)$ lie in R . ■

Example 7.5.8

Let R be a UFD such that 2 $\in R$ is invertible. Let $d \in R$ be such that $x^2 - d \in R[x]$ is irreducible. If d is squarefree, then $\frac{R[x]}{(x^2-d)}$ is normal.

Proof

Suppose that $d \in R$ is squarefree. By 5.3.5 we have that

$$\text{Frac}\left(\frac{R[x]}{(x^2-d)}\right) = \frac{\text{Frac}(R)[x]}{(x^2-d)}$$

so that the fraction field of $\frac{R[x]}{(x^2-d)}$ is an algebraic extension of $\text{Frac}(R)$ of degree 2. Let $\alpha \in \frac{R[x]}{(x^2-d)}$. Then there exists a monic polynomial $f \in \frac{R[x]}{(x^2-d)}[y]$ such that $f(\alpha) = 0$. Now notice that the field extension of $\text{Frac}\left(\frac{R[x]}{(x^2-d)}\right)$ over $\text{Frac}(R)$ has degree 2, and is a galois extension since it splits the separable polynomial $x^2 - d$. Then the Galois group is given by $\{\text{id}, \sigma\}$. Now since the norm of a

Galois extension is given by the product of elements of the Galois group, we have

$$N(f) = f \cdot \sigma(f) \in \text{Frac}(R)[y]$$

Since $f \in \frac{R[x]}{(x^2-d)}[y]$, we must have that $N(f) \in R[y]$. Moreover, $N(f)$ is clearly monic and $N(f)(\alpha) = 0$.

On the other hand, α has the form $a_0 + a_1x$ for $a_0, a_1 \in \text{Frac}(R)$. Notice that the minimal polynomial of α is

$$p(y) = y^2 - 2a_0y + a_0^2 - a_1^2d \in \text{Frac}(R)[y]$$

Indeed, no linear polynomial can be a minimal polynomial for α . Then by definition of the minimal polynomial, there is a factorization $N(f) = p(y)q(y)$ for some $q \in \text{Frac}(R)[y]$. Since R is normal, we can apply 7.2.9 Gauss's lemma to see that the factorization $N(f) = p(y)q(y)$ exists in $R[y]$. Hence $p(y) \in R[y]$, and so $2a_0, a_0^2 - a_1^2d \in R$.

Since 2 is invertible in R , we must have $a_0 \in R$. Since $a_0 \in R$ and $a_0^2 - a_1^2d \in R$, we must have $a_1^2d \in R$. Now since $a_1 \in \text{Frac}(R)$, suppose that $a_1 = r/s$ for $r, s \in R$ and $s \neq 0$ and r, s in lowest terms (possible since R is a UFD). Then $\frac{r^2d}{s^2} \in R$ implies that s is invertible in R . Hence $a_1 \in R$. Thus $\alpha = a_0 + a_1x \in \frac{R[x]}{(x^2-d)}$ so that $\frac{R[x]}{(x^2-d)}$ is normal. ■

Example 7.5.9

The following rings are normal.

- $\frac{\mathbb{Z}[x]}{(x^2-n)}$ where n is squarefree and $n \equiv 2, 3 \pmod{4}$.
- $\mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right]$ where n is squarefree and $n \equiv 1 \pmod{4}$.

Proof

Let $\alpha = a + bx \in \frac{\mathbb{Z}[x]}{(x^2-n)} \subseteq \mathbb{Q}(\sqrt{n})$. The proof for the first point mimics the above until we reach that $2a \in \mathbb{Z}$ and $a^2 - b^2n \in \mathbb{Z}$. Then we have $(2a)^2 - (2b)^2n = 4(a^2 - b^2n) \in \mathbb{Z}$. Then following the above example, we must have that $2b \in \mathbb{Z}$. Now we have that

$$(2a)^2 - (2b)^2n \equiv 0 \pmod{4}$$

which gives $(2a)^2 \equiv (2b)^2n \pmod{4}$. Now recall that for any $x \in \mathbb{Z}$, we must have $x^2 \equiv 0, 1 \pmod{4}$.

If $n \equiv 2 \pmod{4}$, then $(2a)^2 \equiv (2b)^2 \equiv 0 \pmod{4}$. In this case, $2a$ and $2b$ are even numbers and so $a, b \in \mathbb{Z}$. If $n \equiv 3 \pmod{4}$, then again $(2a)^2 \equiv (2b)^2 \equiv 0 \pmod{4}$ so that $a, b \in \mathbb{Z}$. Hence $\alpha \in \frac{\mathbb{Z}[x]}{(x^2-n)}$ so that the ring is integral.

Let $\alpha = a + b\frac{1+\sqrt{n}}{2} \in \mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right] \subseteq \mathbb{Q}\left(\frac{1+\sqrt{5}}{2}\right)$. Notice that the minimal polynomial of α in $\mathbb{Q}$ is given by $x^2 - (2a+b)x + \frac{(2a+b)^2 - nb^2}{4}$. Using a similar method as above, we find that $2a+b \in \mathbb{Z}$ and $\frac{(2a+b)^2 - nb^2}{4} \in \mathbb{Z}$. This means that $(2a+b)^2 - nb^2 \in 4\mathbb{Z}$. This gives $b \in \mathbb{Z}$. Now since $n \equiv 1 \pmod{4}$, we must have $(2a+b)^2 \equiv b^2 \equiv 0, 1 \pmod{4}$. If $(2a+b)^2$ and b^2 are divisible by 4, then $2a+b$ and b is even. Hence $a \in \mathbb{Z}$. If $(2a+b)^2$ and b^2 are divisible by 1, then $2a+b$ and b are both odd so that $2a$ is even. Hence $a \in \mathbb{Z}$. Thus $\alpha \in \mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right]$ so that $\mathbb{Z}\left[\frac{1+\sqrt{n}}{2}\right]$ is normal. ■

Example 7.5.10

Let k be an algebraically closed field. The following rings are normal.

- $\frac{k[x_1, \dots, x_n]}{(x_1^2 + \dots + x_m^2)}$ where $n \geq m \geq 3$.
- $\frac{k[w, x, y, z]}{(wz - xy)}$.

Proof

I first claim that $x_1^2 + \cdots + x_k^2$ for $2 \leq k \leq n$ is irreducible. We proceed by induction. Consider the base case $k = 2$. Consider $x_1^2 + x_2^2$ as a polynomial in x_1 over $k[x_2, \dots, x_n]$. Clearly $(x_2 - 1)$ is a prime ideal since $k[x_2, \dots, x_n]/(x_2 - 1) \cong k$ or $k[x_3, \dots, x_n]$ depending on $n \geq 3$ or not. In both cases they are integral domains. Then clearly $(x_2 - 1)$ satisfies the conditions in Eisenstein's criterion so that $x_1^2 + x_2^2$ is irreducible in $k[x_1, \dots, x_n]$. Now suppose that $x_1^2 + \cdots + x_k^2$ is irreducible in $k[x_1, \dots, x_n]$ for $k \leq n - 1$. I claim that $x_1^2 + \cdots + x_{k+1}^2$ is irreducible. We consider it as a polynomial over x_1 in $k[x_2, \dots, x_n]$. By assumption, $x_2^2 + \cdots + x_{k+1}^2$ is irreducible. Since $k[x_2, \dots, x_n]$ is a UFD, $(x_2^2 + \cdots + x_{k+1}^2)$ is a prime ideal whose square does not contain $x_2^2 + \cdots + x_{k+1}^2$. Hence by Eisenstein's criterion $x_1^2 + \cdots + x_{k+1}^2$ is irreducible.

Now let $m \geq 3$. Then $-(x_2^2 + \cdots + x_m^2)$ is irreducible and hence squarefree. Then by 7.5.8 we conclude that $\frac{k[x_1, \dots, x_n]}{(x_1^2 + \cdots + x_m^2)}$ is normal.

For the second point, notice that there is a k -algebra isomorphism

$$k[w', x, y, z'] \cong k[w, x, y, z]$$

where $w = w' + z'$ and $z = w' - z'$ (possible since $2 \neq 0$ in k). This induces an isomorphism

$$\frac{k[w, x, y, z]}{(wz - xy)} \cong \frac{k[w', x, y, z']}{((w')^2 - ((z')^2 + xy))}$$

Rewriting coordinates, we may deal with the ring $\frac{k[w, x, y, z]}{w^2 - (z^2 + xy)}$. I claim that $z^2 + xy$ is irreducible. To prove this, consider $z^2 + xy$ as a polynomial in z over $k[x, y]$. Then notice that (x) is a prime ideal in $k[x, y]$ such that $xy \notin (x^2)$. Hence by Eisenstein's criterion, we have that $z^2 + xy$ is irreducible, and hence squarefree. Then we may apply 7.5.8 to conclude that our given ring is normal. ■

8 Introduction to Dimension Theory for Rings

8.1 Krull Dimension

Definition 8.1.1 (Krull Dimension) Let R be a commutative ring. Define the Krull dimension of R to be

$$\dim(R) = \max\{t \in \mathbb{N} \mid p_0 \subset \cdots \subset p_t \text{ for } p_0, \dots, p_t \text{ prime ideals}\}$$

In particular, notice that a commutative ring R has $\dim(R) = 0$ if and only if every prime ideal is maximal.

Lemma 8.1.2 Let R be a commutative ring. Let I be an ideal of R . Then we have $\dim(R) \geq \dim(R/I)$.

Lemma 8.1.3

Let R, S be commutative rings such that $R \subseteq S$ is an integral extension. Then we have

$$\dim(R) = \dim(S)$$

Proof Let $P_0 \subset \cdots \subset P_n$ be a strictly increasing chain of prime ideals of R . By the going up theorem, we obtain an increasing chain of prime ideals $Q_0 \subseteq \cdots \subseteq Q_n$ in S . Hence $\dim(R) \leq \dim(S)$.

Conversely, suppose that $Q_0 \subset \cdots \subset Q_n$ is a strictly increasing chain of prime ideals in S . Then $\varphi^{-1}(Q_0) \subset \cdots \subset \varphi^{-1}(Q_n)$ is an increasing chain of prime ideals of R . Hence $\dim(S) \leq \dim(R)$. ■

Lemma 8.1.4 Let R be a commutative ring. Then the following are true.

- If R is a field, then $\dim(R) = 0$
- If R is Artinian, then $\dim(R) = 0$

Proof Let R be a field. Then the only proper prime ideal of R is (0) . In particular, (0) forms the only chain of prime ideals in R . Hence $\dim(R) = 0$.

Now let R be Artinian. Let P be a prime ideal of R . Then R/P is an integral domain. Moreover, every quotient of an Artinian ring is Artinian. Hence R/P is Artinian. By prp1.3.1, we conclude that R/P is a field. Hence P is a maximal ideal. Any chain of prime ideals of R must terminate at the first prime ideal since it is maximal. Hence $\dim(R) = 0$. ■

Definition 8.1.5 (Dimension of Modules) Let R be a commutative ring. Let M be an R -module. Define the dimension of M to be

$$\dim(M) = \dim\left(\frac{R}{\text{Ann}_R(M)}\right)$$

Proposition 8.1.6

Let R be a commutative ring. Let M be an R -module. Then we have

$$\dim(M) = \sup\{\dim(R/P) \mid P \in \text{Ass}(M)\}$$

Proof We know that the set of all minimal prime ideals of $\text{Ass}(M)$ is equal to the set of all minimal prime ideals over $\text{Ann}_R(M)$ by 6.2.5. Any maximal chain of prime ideals over $\text{Ann}_R(M)$ must also start at one of these minimal prime ideals. Hence by the correspondence theorem we have $\sup\{\dim(R/P) \mid P \in \text{Ass}(M)\} = \dim(R/\text{Ann}_R(M))$. ■

8.2 Noether Normalization Lemma

Lemma 8.2.1

Let k be a field. Let T be a finite set of monomials in $k[x_1, \dots, x_n]$. Then there exists $w_1, \dots, w_{n-1}$ such that for any two $a_{k_1, \dots, k_n} x_1^{k_1} \cdots x_n^{k_n}, b_{r_1, \dots, r_n} x_1^{r_1} \cdots x_n^{r_n} \in T$, $(k_1, \dots, k_n) \neq (r_1, \dots, r_n)$ implies that

$$\sum_{i=1}^{n-1} w_i k_i + k_n \neq \sum_{i=1}^{n-1} w_i r_i + r_n$$

Proof We induct on n . When $n = 1$ then we are done. When $n \geq 2$, we can write each monomial in T as a monomial in $x_2, \dots, x_n$ to apply the inductive hypothesis and obtain $w_2, \dots, w_n$. Then set $w_1 > \max\{\sum_{i=2}^{n-1} w_i k_i + k_n \mid (k_1, \dots, k_n) \text{ represents a monomial in } T\}$. For $(k_1, \dots, k_n) \neq (r_1, \dots, r_n)$ there are two cases. If $(k_2, \dots, k_n) \neq (r_2, \dots, r_n)$, then by inductive hypothesis we have $\sum_{i=2}^{n-1} w_i k_i + k_{n-1} \neq \sum_{i=2}^{n-1} w_i r_i + r_{n-1}$. Since w_1 is larger than these two numbers, we must have that $\sum_{i=1}^{n-1} w_i k_i + k_n \neq \sum_{i=1}^{n-1} w_i r_i + r_n$. Otherwise, if $(k_2, \dots, k_n) = (r_2, \dots, r_n)$ but $k_1 \neq r_1$, then $w_1 k_1 \neq w_1 r_1$ and so we are done. $\blacksquare$

Theorem 8.2.2 (Noether Normalization Lemma)

Let k be a field. Let R be a finitely generated k -algebra with the isomorphism $R \cong \frac{k[x_1, \dots, x_n]}{I}$ for some ideal I . Then there exists $y_1, \dots, y_d \in R$ for some $0 \leq d \leq n$ such that the following are true.

- $y_1, \dots, y_d$ are algebraically independent.
- R is finitely generated over $k[y_1, \dots, y_d]$.
- $d = \dim(R)$.

Proof

Since R is a finitely generated k -algebra, we have $R \cong \frac{k[x_1, \dots, x_n]}{I}$ for some $n \in \mathbb{N}$ and some ideal I . We induct on n . When $n = 1$, we have $R \cong \frac{k[x]}{I}$. Since $k[x]$ is a PID, $I = (f)$ for some $f \in k[x]$. If $f = 0$, then we are done. So suppose that $f \neq 0$. Then R is generated as a k -vector space by $1, x, \dots, x^{\deg(f)-1}$. Thus R is finitely generated over $k[x]$.

Now suppose that $n \geq 2$. If each $x_1, \dots, x_n$ are algebraically independent then we are done. So suppose not. Then I claim that there is a k -subalgebra S of R generated by $n - 1$ elements such that R is finitely generated over S . Since $x_1, \dots, x_n$ are not algebraically independent, there exists a polynomial in k such that $f(x_1, \dots, x_n) = 0$. For $r_1, \dots, r_{n-1} \in \mathbb{N}$, write $z_i = x_i - x_n^{r_i}$. Then we have

$$f(z_1 + x_n^{r_1}, \dots, z_{n-1} + x_n^{r_{n-1}}, x_n) = 0$$

If the polynomial f has coefficients $a_{k_1, \dots, k_n}$, then expanding out the expression, we have

$$\sum a_{k_1, \dots, k_n} \prod_{i=1}^{n-1} (z_i + x_n^{r_i})^{k_i} x_n^{k_n} = 0$$

Each term in the summand has a unique term with highest order of x_n after expanding, which is given by $a_{k_1, \dots, k_n} x_n^{\sum_{i=1}^{n-1} r_i k_i + k_n}$. By the above lemma, we can find $r_1, \dots, r_{n-1}$ such that $(k_1, \dots, k_n) \neq (j_1, \dots, j_n)$ implies that $\sum_{i=1}^{n-1} r_i k_i + k_n \neq \sum_{i=1}^{n-1} r_i j_i + j_n$. Then this means that there is a unique term that achieves the maximum of $\sum_{i=1}^{n-1} r_i k_i + k_n$. This proves that the highest term of f as a polynomial in y_n has coefficient $a_{k_1, \dots, k_n}$ for some $k_1, \dots, k_n$. Dividing out f by this coefficient, we deduce that x_n is an integral element over $k[z_1, \dots, z_{n-1}]$. This proves that $S = k[z_1, \dots, z_{n-1}]$ is a k -subalgebra of R such that R is finitely generated over S .

By the inductive hypothesis, there exists $y_1, \dots, y_d$ for some $0 \leq d \leq n - 1$ such that $y_1, \dots, y_d$ are algebraically independent and S is finitely generated over $k[y_1, \dots, y_d]$. Since R is finitely generated over S and S is finitely generated over $k[y_1, \dots, y_d]$, we conclude that R is finitely generated over $k[y_1, \dots, y_d]$.

It remains to show that $d = \dim(R)$. Since R is integral over $S = k[y_1, \dots, y_d]$, we have $\dim(R) =$

$\dim(S)$ and so it suffices to show that $d = \dim(S)$. We induct on d . When $d = 0$, then $S = k$ and so $\dim(S) = 0$. Now let $d > 0$. Then $0 \subset (y_1) \subset (y_1, y_2) \subset \cdots \subset (y_1, \dots, y_d)$ is a chain of prime ideals in S so that $d \leq \dim(S)$. Conversely, suppose that $0 \subset P_1 \subset \cdots \subset P_n$ is a strictly increasing chain of prime ideals in S . Let $0 \neq u \in P_1$. By the Noether normalization lemma, because first variable is always trivially algebraically independent, there exists $t_2, \dots, t_d \in S$ such that S is integral over $T = k[u, t_2, \dots, t_d]$. By the inductive hypothesis, we have $d - 1 = \dim(T/(u))$. By the going up theorem, $0 \subset P_1 \cap T \subset \cdots \subset P_n \cap T$ is a strictly increasing chain of prime ideals in T . Then in $T/(P_1 \cap T)$ the chain becomes a strictly increasing chain of prime ideals of length $n - 1$. Since $\dim(T/(P_1 \cap T)) \leq \dim(T/(u))$, we have that $n - 1 \leq d - 1$ and hence $\dim(S) \leq d$. ■

Proposition 8.2.3

Let k be a field. Let R be a finitely generated k -algebra that is an integral domain. Then we have

$$\dim(R) = \text{trdeg}_k(\text{Frac}(R))$$

Proof By the Noether normalization lemma, there exists algebraically independent elements $y_1, \dots, y_{\dim(R)}$ such that R is finitely generated over $k[y_1, \dots, y_{\dim(R)}]$. Then $\text{Frac}(R)$ is algebraic over $\text{Frac}(k[y_1, \dots, y_{\dim(R)}]) = k(y_1, \dots, y_{\dim(R)})$. Hence $\text{trdeg}_k(\text{Frac}(R)) = \dim(R)$. ■

Corollary 8.2.4

Let k be a field. Then we have

$$\dim(k[x_1, \dots, x_n]) = n$$

Proof The transcendence degree of $\text{Frac}(k[x_1, \dots, x_n]) = k(x_1, \dots, x_n)$ is n . ■

8.3 Height of Prime Ideals

Definition 8.3.1 (Height of a Prime Ideal) Let R be a commutative ring. Let p be a prime ideal of R . Define the height of p to be

$$\text{ht}(p) = \max\{t \in \mathbb{N} \mid p_0 \subset \cdots \subset p_t = p \text{ for } p_0, \dots, p_t \text{ prime ideals}\}$$

Lemma 8.3.2 Let R be a commutative ring. Then

$$\dim(R) = \max\{\text{ht}(P) \mid P \in \text{Spec}(R)\}$$

Lemma 8.3.3 Let R be a commutative ring. Let P be a prime ideal of R . Then

$$\text{ht}(P) = \dim(R_P)$$

Proof Let $\dim(R_P) = n$. Then there exists a strict chain of prime ideals of R_P of length n (and no chain of prime ideals of length $> n$). By prp5.4.6, prime ideals of R_P are in bijection with prime ideals of R that P contains. Hence the maximal chain of prime ideals of length n correspond to a chain of prime ideals in R that contain P , of length n . Hence $\dim(R_P) = n \leq \text{ht}(P)$. Conversely, let $m = \text{ht}(P)$. Then there exists a strict chain of prime ideals that are subsets of P , that are of length m . By the same correspondence, the chain of prime ideals correspond to a chain of prime ideals in R_P of length m . Hence $\text{ht}(P) = m \leq \dim(R_P)$.

The two inequalities combine to show that $\dim(R_P) = \text{ht}(P)$. ■

Lemma 8.3.4 Let R be a commutative ring. Let P be a prime ideal of R . Then

$$\dim(R) \geq \dim(R/P) + \text{ht}_R(P)$$

Proposition 8.3.5 Let k be a field. Let R be a finitely generated k -algebra that is an integral domain. Then we have

$$\dim(A) = \dim(A/P) + \text{ht}_A(P)$$

Proposition 8.3.6 (Dimension is a Local Concept) Let R be a commutative ring. Then the following numbers are equal.

- The Krull dimension $\dim(R)$
- The supremum $\sup\{\dim(R_m) \mid m \text{ is a maximal ideal of } R\}$
- The supremum $\sup\{\text{ht}_R(m) \mid m \text{ is a maximal ideal of } R\}$

Corollary 8.3.7 Let (R, m) be a local ring. Then

$$\dim(R) = \dim(R_m) = \text{ht}_R(m)$$

Theorem 8.3.8 (Krull's Principal Ideal Theorem) Let R be a Noetherian ring. Let I be a proper and principal ideal of R . Let p be the smallest prime ideal containing I . Then

$$\text{ht}_R(p) \leq 1$$

8.4 The Length of Modules over Commutative Rings

Let R be a ring. Recall that the length of an R -module M is defined to be the supremum

$$l_R(M) = \sup\{n \in \mathbb{N} \mid 0 = M_0 \subset M_1 \subset \cdots \subset M_n = M\}$$

Lemma 8.4.1 Let (R, m) be a local ring and let M be an R -module. If $mM = 0$, then

$$l_A(M) = \dim_{A/m}(M)$$

Example 8.4.2

Let (R, m) be a local ring. Then we have

$$l_R\left(\frac{m^n}{m^{n+1}}\right) = \dim_{R/m}\left(\frac{m^n}{m^{n+1}}\right)$$

Proof

Follows from above. ■

Lemma 8.4.3

Let R be a commutative ring. Let M be a finitely generated R -module. Let m be a maximal ideal such that $m^{n-1}M \neq 0$ and $m^n M = 0$. Then $l_R(M) < \infty$.

Proof

For $1 \leq i \leq n$, consider the exact sequence

$$0 \longrightarrow m^i M \longrightarrow m^{i-1} M \longrightarrow \frac{m^{i-1} M}{m^i M} \longrightarrow 0$$

We have that $l_R(m^{i-1} M / m^i M) = l_R(m^{i-1} M) - l_R(m^i M)$. Then summing over i gives

$$\sum_{i=1}^n l_R\left(\frac{m^{i-1} M}{m^i M}\right) = l_R(M) - l_R(m^n M) = l_R(M)$$

Since $m^i M$ is R -submodule of M it is finitely generated. Hence $m^i M / m^{i+1} M$ is a finite dimensional R/m vector space. Hence $l_R(M)$ is also finite. ■

Lemma 8.4.4

Let R be a commutative ring. Let M be an R -module. Let S be a multiplicative subset of R . Then we have

$$l_R(M) \geq l_R(S^{-1}M)$$

8.5 Structure Theorem for Artinian Rings

Let R be a ring. Let M be an R -module. Recall that a composition series for M is a sequence of R -submodules

$$0 = M_0 \subset M_1 \subset \cdots \subset M_k = M$$

such that $\frac{M_{i+1}}{M_i}$ is a simple R -module for $1 \leq i < k$.

Proposition 8.5.1 Let $R \neq 0$ be a commutative ring. Then R is Artinian if and only if R is Noetherian and $\dim(R) = 0$.

Proof Let R be Artinian. In Rings and Modules, the Akizuki-Hopkins-Levitzki theorem proves that R is Noetherian. Moreover, lmm8.1.4 shows that $\dim(R) = 0$.

Now let R be Noetherian and $\dim(R) = 0$. This means that every prime ideal of R is maximal. Let S be the set of all ideals of R that admit a composition series. I claim that S is non-empty. Let $T = \{\text{Ann}_R(x) \mid 0 \neq x \in R\}$. By 6.1.1, the maximal element $\text{Ann}_R(x)$ in T is a prime ideal. Since $\dim(R) = 0$ we have $\text{Ann}(x)$ is a maximal ideal. $R/\text{Ann}(x)$ is a field (and hence a simple R -module). The multiplication map $r \mapsto rx$ has kernel $\text{Ann}(x)$. Hence the induced map $R/\text{Ann}(x) \rightarrow R$ is injective, and we can consider $R/\text{Ann}(x)$ as a subring of R . Together with the fact that it is a simple R -module makes it an R -submodule with composition series length of 1. Hence S is non-empty.

Let $N_1 \subseteq N_2 \subseteq \cdots$ be a chain in S . Since R is Noetherian, the chain terminates with some ideal $I \in S$. If $I = R$, then R has a composition series. If $I \neq R$, then R/I is non-zero. Choose a prime ideal P of R such that $I \subseteq P \neq R$ (this always exists since we can choose maximal ideals). Then we have $0 \neq R/P \subseteq R/I$. Let $p : R \rightarrow R/I$ be the projection map. Let $T = p^{-1}(R/P)$. Then we have that $N \subset T \subseteq M$ and $T/N \cong R/P$. Since $\dim(R) = 0$, P is maximal hence R/P is a field (and a simple R -module). This proves that $T \in S$. But this contradicts the maximality of N . Hence $N = R \in T$. Thus R has a composition series. From Rings and Modules we know that this implies R is Noetherian. Hence we conclude. ■

Example 8.5.2 Let k be a field. The $k[x]$ -module $\frac{k[x, x^{-1}]}{k[x]}$ is Artinian but not Noetherian.

Proof It is not Noetherian because it is not finitely generated. Write $M = \frac{k[x, x^{-1}]}{k[x]}$. For the Artinian result, we first show that if $N \leq \frac{k[x, x^{-1}]}{k[x]}$ and for all $n \in \mathbb{N}$ there exists $f + k[x] \in \frac{k[x, x^{-1}]}{k[x]}$ such that f contains the term $1/x^n$, then $N = \frac{k[x, x^{-1}]}{k[x]}$.

By assumption, for any $n \in \mathbb{N}$, there exists $f \in k[x, x^{-1}]$ such that f contains the term $1/x^n$. If $\deg(f) < -n$, then we can multiply f with $x^{\deg(f)+n}$ to get a polynomial g such that $\deg(g) = -n$. So denote $f_n + N$ the element in N such that $\deg(f_n) = -n$. Then by multiplying with a suitable coefficient α , $f_n - \alpha f_{n-1}$ contains only $1/x^n$. Hence N contains $1/x^n$ for all $n \in \mathbb{N}$ as $k[x]$ -module. Since these elements generate $\frac{k[x, x^{-1}]}{k[x]}$ as a k -module, they also generate as a $k[x]$ -module. Hence $N = \frac{k[x, x^{-1}]}{k[x]}$.

This means that if N is a proper sub-module, there exists a minimal $n \in \mathbb{N}$ such that $1/x^n \in N$ and $1/x^{n+1} \notin N$. Hence every N is a finitely generated k -module, or in other words, N is a finite dimensional vector space. Thus any decreasing chain of $k[x]$ -submodules must terminate by a

dimension argument. ■

Theorem 8.5.3 (Structure Theorem for Commutative Artinian Rings) Let R be an Artinian commutative ring. Then R decomposes into a direct product of Artinian local rings

$$R \cong \bigoplus_{i=1}^k R_i$$

Moreover, the decomposition is unique up to reordering of the direct product.

Proof Let $m_1, \dots, m_k$ be the full list of distinct maximal ideals of R . Then

$$\prod_{i=1}^k m_i^n = 0$$

for some $n \in \mathbb{N} \setminus \{0\}$. The ideals m_i^n and m_j^n are pairwise coprime for $i \neq j$. Hence by the Chinese Remainder Theorem we obtain ring isomorphisms

$$\begin{aligned} R &\cong \frac{R}{0} \\ &\cong \frac{R}{\prod_{i=1}^k m_i^n} \\ &\cong \frac{R}{\bigcap_{i=1}^k m_i^n} && (m_i^n \text{ and } m_j^n \text{ pairwise coprime}) \\ &\cong \bigoplus_{i=1}^k \frac{R}{m_i^n} && (\text{CRT}) \end{aligned}$$

By the correspondence of maximal ideals, R/m_i^n has a unique maximal ideal m_i/m_i^n . Hence it is local. Also since R is Artinian, R/m_i^n is Artinian. Thus we are done. ■

9 Valuation and Valuation Rings

9.1 Valuation Rings

Definition 9.1.1 (Valuation Rings) Let R be an integral domain. We say that R is a valuation ring if for all $x \in \text{Frac}(R)$ and $x \neq 0$, then either x or x^{-1} is in R .

Lemma 9.1.2 Let R be an integral domain. Then R is a valuation ring if and only if the ideals of R are totally ordered by inclusion.

Proof Let R be a valuation ring. Let I, J be ideals of R . If I is not a subset of J , there exists $x \in I$ such that $x \notin J$. Then for any $0 \neq y \in J$, $x/y \in \text{Frac}(R) \setminus R$ since otherwise y is a unit in J so that $J = R$ and $I \subseteq R$. Then $y/x \in R$ so that $y = x(y/x) \in I$. Hence $J \subseteq I$.

Now suppose that the ideals of R are totally ordered by inclusion. ■

Lemma 9.1.3 Let R be a valuation ring. Then the following are true.

- R is a local ring.
- R is normal.

Proof Since all ideals of R are totally ordered, there is only one unique maximal ideal.

Let $x \in \text{Frac}(R)$ be integral over R . Then

$$x^n + r_{n-1}x^{n-1} + \cdots + r_1x + r_0 = 0$$

for some $r_0, \dots, r_{n-1} \in R$. If $x \in R$ then we are done. If $x \notin R$ then since R is a valuation ring, $x^{-1} \in R$. Then

$$x = -(r_1 + r_2x^{-1} + \cdots + r_nx^{1-n}) \in R$$

so that R is normal. ■

Definition 9.1.4 (Totally Ordered Group) Let G be an abelian group. We say that G is a totally ordered group if there is a total order " $\leq$ " on G such that $a \leq b$ implies $ca \leq cb$ for all $a, b, c \in G$.

Definition 9.1.5 (Valuation on a Field) Let K be a field. Let G be a totally ordered abelian group. A valuation on K with values in G is a map $v : K^\times \rightarrow G$ such that for all $x, y \in K^*$, we have

- $v(xy) = v(x) + v(y)$ (v is a group homomorphism)
- $v(x + y) \geq \min\{v(x), v(y)\}$

We use the convention that $v(0) = \infty$.

Definition 9.1.6 (Associated Valuation Ring) Let K be a field and $v : K \rightarrow \mathbb{Z}$ a discrete valuation. Define the associated valuation ring of K to be the subring

$$R_v = \{x \in K \mid v(x) \geq 0\}$$

Lemma 9.1.7 Let K be a field. Let v be a discrete valuation on K . Then R_v is a valuation ring.

Definition 9.1.8 (Discrete Valuations) Let K be a field. A discrete valuation on K is a valuation $v : K^\times \rightarrow \mathbb{Z}$.

Definition 9.1.9 (Normalized Discrete Valuations) Let (K, v) be a discrete valuation ring. We say that it is normalized if v is surjective.

Lemma 9.1.10 Let K be a field with a discrete valuation v . Then $v(K^\times) = n\mathbb{Z}$ for some $n \in \mathbb{N}$.

Lemma 9.1.11 (Normalization of a Discrete Valuation) Let K be a field with a discrete valuation v such that $v(K^\times) = n\mathbb{Z}$ for some $n \in \mathbb{N}$. Define the normalization of v to be the valuation $v_N : K^\times \rightarrow \mathbb{Z}$ defined by

$$v_N(k) = \frac{1}{n}v(k)$$

for all $k \in K^\times$.

Therefore we always work on normalized discrete valuations.

9.2 Discrete Valuation Rings

Definition 9.2.1 (Discrete Valuation Rings) Let R be a commutative ring. We say that R is a discrete valuation ring if there exists a field K and a discrete valuation v on K such that

$$R = R_v$$

is the associated valuation ring of K .

Lemma 9.2.2 Let R be a discrete valuation ring with valuation v . Then $0 \neq u \in R$ is a unit if and only if $v(u) = 0$. In particular, the maximal ideal of R is given by

$$\{r \in R \mid v(r) > 0\}$$

Proof Let R be a discrete valuation ring. Suppose that $x \in R$ is a unit. Then $v(x^{-1}) = -v(x)$. Then $-v(x), v(x) \geq 0$ implies $v(x) = 0$. Now if $v(y) > 0$, suppose for contradiction that $u \in R$ is an inverse of y , then

$$0 = v(1) = v(uy) = v(u) + v(y)$$

But $v(y) > 0$ implies that $v(u) < 0$ which implies that $u \notin R$, a contradiction. ■

Example 9.2.3 Let $n \in \mathbb{N}$. Define $\text{ord}_n : \mathbb{Q} \rightarrow \mathbb{Z}$ as follows. For $p/q \in \mathbb{Q}$, let $p = p'n^i$ and $q = q'n^j$ such that $\gcd(p', n) = \gcd(q', n) = 1$. Then define

$$\text{ord}_n\left(\frac{p}{q}\right) = \text{ord}_n\left(n^{i-j}\frac{p'}{q'}\right) = i - j$$

Then ord_n is a discrete valuation if and only if n is prime. In this case, the valuation ring of ord_n is given by

$$R_{\text{ord}_n} = \mathbb{Z}_n$$

Proof Suppose that n is a prime. Let $n^s p_1/q_1 \in \mathbb{Q}$ and $n^t p_2/q_2$ be in lowest terms. Then $n^{s+t}(p_1 p_2/q_1 q_2)$ is in lowest terms since n is prime. Then we have

$$\text{ord}_n(n^{s+t}(p_1 p_2/q_1 q_2)) = s + t = v(n^s p_1/q_1) + v(n^t p_2/q_2)$$

Without loss of generality, suppose that $s \leq t$. Then $n^s p_1/q_1 + n^t p_2/q_2 = n^s(p_1/q_1 + n^{t-s} p_2/q_2)$ is in lowest terms since n is prime. Then we have

$$v(n^s p_1/q_1 + n^t p_2/q_2) = v(n^s(p_1/q_1 + n^{t-s} p_2/q_2)) = s = \min\{v(n^s p_1/q_1), v(n^t p_2/q_2)\}$$

Thus ord_n is a discrete valuation.

If n is composite, without loss of generality suppose that $n = pq$ for p and q primes.

The valuation ring of ord_n for n prime is given by

$$R_{\text{ord}_n} = \left\{ \frac{p}{q} \in \mathbb{Q} \mid n \text{ does not divide } q \right\}$$

Hence $R_{\text{ord}_n} = \mathbb{Z}_n$. ■

Definition 9.2.4 (Uniformizing Parameter) Let R be a discrete valuation ring with valuation v . A uniformizing parameter for R is an element $t \in R$ such that $v(t) = 1$.

Proposition 9.2.5 Let R be a discrete valuation ring with valuation v . Let $t \in R$ be a uniformizing parameter of R . Then the following are true.

- Every $r \in R \setminus \{0\}$ can be written in the form

$$r = ut^n$$

for some unit u and $n \geq 0$.

- The valuation of any element $r = ut^n \in R$ is given by

$$v(ut^n) = n$$

- The set of all ideals of R is given by

$$\{(t^n) \mid n \in \mathbb{N} \setminus \{0\}\}$$

In particular, the unique maximal ideal of R is (t) .

Proof

- If $x \in R$ is a unit then we are done. If not, then consider the element $u = t^{-n}x$ for $n = v(x)$. Then we have

$$v(u) = v(t^{-n}x) = -n + v(x) = 0$$

Hence u is a unit. Multiplying t^n on both sides of $u = t^{-n}x$ proves that $x = ut^n$ for some unit u and $n \in \mathbb{N}$.

- It follows that the valuation of $r = ut^n$ is n .
- Let I be an ideal of R . Let $n = \min\{v(x) \mid x \in I\}$. or all $x \in I$, we can write x as $x = ut^k$ for some unit u and $k \geq n$. Hence $I \subseteq (t^n)$. Since n is a minimum, there exists $x \in I$ such that $x = ut^n$ for some unit u and $n \in \mathbb{N}$. Then $u^{-1}x = t^n \in I$ since I is an ideal. Hence $I = (t^n)$. It follows that the unique maximal ideal of R is given by (t) . ■

The rest of the section devotes efforts to recognizing discrete valuation rings.

Proposition 9.2.6 (Equivalent Characterizations of DVRs I) Let R be an integral domain. Then the following are equivalent.

- R is a discrete valuation ring.
- R is local, a PID and not a field.
- R is Noetherian, local, $\dim(R) = 1$ and normal.
- R is Noetherian, local, $\dim(R) > 0$ and the unique maximal ideal m is principal.
- R is a UFD with a unique irreducible element up to multiplication of a unit

Proof

- (1) $\implies$ (2): We have seen that valuation rings are local. It is a PID by 9.2.5. It is not a field

since R is a local ring with non-trivial unique maximal ideal.

- (2) $\implies$ (3): Every PID is Noetherian and normal and every prime ideal is maximal. But local rings have a unique maximal ideal. The maximal ideal is non-trivial since R is not a field. Hence $\dim(R) = 1$.
- (3) $\implies$ (4): By Nakayama's lemma, $m \neq m^2$. I claim that any $x \in m \setminus m^2$ generates m . Since $\dim(R) = 1$, we have $\text{Spec}(R) = \{(0), m\}$. Assume for a contradiction that $m/(x) \neq \{0\}$. By Imm6.2.4, we have $\text{Ass}(m/(x)) \neq \{0\}$. By our assumption for contradiction, we can only have $\text{Ass}(m/(x)) = \{m\}$. By definition, this means that there exists $0 \neq [y] \in m/(x)$ such that $\text{Ann}_R([y]) = m$. In other words, $ym \subseteq (x)$. Considering everything inside $\text{Frac}(R)$, we have $y/x \in \text{Frac}(R)$ is such that $y/x \notin R$ and $y/x \cdot m \subseteq R$. There are now two cases.

Case 1: $y/x \cdot m = R$.

Then $1 = yt/x$ for some $t \in m$, which means that $x = yt$ and $x \in ym \subseteq m^2$. This is a contradiction.

Case 2: $y/x \cdot m = m$. Then the multiplication map $z \mapsto y/x \cdot z$ satisfies the hypothesis of the Cayley-Hamilton theorem, and there exists $a_0, \dots, a_{n-1} \in R$ such that

$$(y/x)^n + a_{n-1}(y/x)^{n-1} + \dots + a_1(y/x) + a_0 = 0$$

But then this proves that y/x is integral over R . Since R is normal, $y/x \in R$. This is also a contradiction.

Thus m is a PID.

- (4) $\implies$ (1): Suppose that $m = (x)$ for some $x \in R$. If x is nilpotent, then $\dim(R) = 0$ and a contradiction. I claim that $\bigcap_{i=1}^{\infty} (x^i) = \{0\}$. Suppose that t lies in the intersection. Then $t = yx$ for some $y \in R$. If y is not in the intersection, then there exists $n \in \mathbb{N}$ such that y is non-zero in $(x^n)/(x^{n+1})$. By Nakayama's lemma, y generates (x^n) and so t generates (x^{n+1}) . Then $t \notin (x^{n+2})$ is a contradiction. In particular, there for any $y \in R$, we have $y \in (x^n) \setminus (x^{n+1})$ for some $n \in \mathbb{N}$. This means that $y = ux^n$ for some $u \notin (x)$. In particular, u is a unit. Similarly, $z = vx^m$ for v a unit. Then $yz = uvx^{n+m}$ is non-zero. Hence R is an integral domain. Then the map $ux^n \mapsto n$ is a valuation.
- (5) $\implies$ (1): Let t be the unique irreducible element. Define a map $v : \text{Frac}(R) \rightarrow \mathbb{Z}$ as follows. Since R is a UFD, every element in R can be uniquely written as zt^n for z a unit and $n \in \mathbb{N}$. Also, every element in $\text{Frac}(R)$ can be uniquely written as zt^n for z a unit in $n \in \mathbb{Z}$. Then define $v(zt^n) = n$. It is clear that v is a valuation. Its associated valuation ring is then precisely R .

■

Proposition 9.2.7 (Equivalent Characterizations of DVRs II) Let R be an integral domain that is Noetherian and local with unique maximal ideal m . Then the following are equivalent.

- R is a discrete valuation ring.
- $\dim(R) = 1$ and R is normal.
- R is not a field and m is principal.
- $\dim(R) = 1$ and $\dim_{R/m}(m/m^2) = 1$ (R is a regular local ring)
- $I = m^k$ for all non-zero ideals I of R
- There exists $t \in R$ and $k > 0$ such that $I = (t^k)$ for all non-zero ideal I of R

Proof The proposition is an immediate consequence of the above.

■

Proposition 9.2.8 Let R be a Noetherian integral domain and $\dim(R) = 1$. Then R is normal if and only if R_m is a discrete valuation ring for all maximal ideals m .

In summary, if R is a discrete valuation ring, then R has the following properties.

- R is integrally closed and in particular is normal.
- R is a PID and in particular is a UFD and an integral domain.
- R is Noetherian and local
- R has Krull dimension 1.
- $\dim_{R/m}(m/m^2) = 1$ (these are called regular local rings as we will see in Commutative Algebra 2)
- Every ideal I of R is equal to the power m^k of the maximal ideal m . In particular if m is generated by the uniformizing parameter t , then $I = (t^k)$ in this case.
- Such a t is an irreducible element (that is unique up to multiplication by a unit), and every element of R can be written as ut^n for u a unit and $n \in \mathbb{N}$.

There is a simple diagram of relationships between DVRs and some other standard types of commutative rings.

$$\text{DVRs} \subset \text{PIDs} \subset \text{UFDs} \subset \text{Normal Domains} \subset \text{Integral Domains}$$

10 Dedekind Domains

10.1 Fractional Ideals

Definition 10.1.1 (Fractional Ideal) Let R be an integral domain. Let I be a R -submodule of $\text{Frac}(R)$. We say that I is a fractional ideal of R if there exists $r \in R \setminus \{0\}$ such that $rI \subseteq R$.

While I is not exactly an ideal of R , we can think of it as if it were an ideal because it is isomorphic to an actual ideal of R .

Lemma 10.1.2 Let R be an integral domain. Let I be a fractional ideal of R where $rI \subseteq R$ for some $r \in R \setminus \{0\}$. Then there is an R -module isomorphism

$$I \cong rI \subseteq R$$

given by $i \mapsto ri$.

Proof I claim that there is an R -module isomorphism $I \cong rI$ for $rI \subseteq R$ given by $i \mapsto ri$. The kernel of this R -module homomorphism is given by $\{i \in I \mid ri = 0\}$. But $ri = 0$ if and only if $r = 0$ or $i = 0$. Since $r \neq 0$ we must have $i = 0$ so that the kernel is trivial. Moreover, this R -module homomorphism is surjective since for any $k \in rI$ it can be written as $k = ri$ for some i . Then $i \in I$ maps to ri under the morphism. Hence $I \cong rI$ as R -modules. ■

Example 10.1.3 The $\mathbb{Z}$ -submodule $\mathbb{Z} \cdot \frac{1}{2}$ of $\mathbb{Q}$ is a fractional ideal.

Proof Indeed, we have $2(\mathbb{Z} \cdot \frac{1}{2}) = \mathbb{Z}$, and we think of $\mathbb{Z} \cdot \frac{1}{2}$ as a $\mathbb{Z}$ -module isomorphic to $\mathbb{Z}$. ■

Lemma 10.1.4 Let R be an integral domain. Let I be a fractional ideal of R . If R is Noetherian, then I is finitely generated.

Proof Let R be Noetherian. Since I is isomorphic to rI for some non-zero $r \in R$, and rI is an ideal of R , R being Noetherian implies that rI is finitely generated and hence I is finitely generated. ■

10.2 Invertible Ideals

Definition 10.2.1 (Inverse of an Ideal) Let R be an integral domain. Let I be an R -submodule of $\text{Frac}(R)$. Define

$$I^{-1} = \{s \in \text{Frac}(R) \mid sI \subseteq R\}$$

Lemma 10.2.2 Let R be an integral domain. Let I be an R -submodule of $\text{Frac}(R)$. Then there is an R -module isomorphism

$$I^{-1} \cong \text{Hom}_R(I, R)$$

given by $s \mapsto (r \mapsto sr)$.

Proof Denote $\varphi_s : I \rightarrow R$ the multiplication by s map for $s \in I^{-1}$. It is clear that the given map is an R -module homomorphism. The map is injective since R is an integral domain. It remains to show that the map is surjective. Let $\varphi \in \text{Hom}_R(I, R)$. For any $r \in R$ and $i \in I$, we have

$$\varphi(r \cdot i) = r \cdot \varphi(i)$$

Definition 10.2.3 (Invertible Ideals) Let R be an integral domain. Let I be an R -submodule of $\text{Frac}(R)$. We say that I is invertible if there exists an R -submodule of J of R such that $JI = R$.

Lemma 10.2.4 Let R be an integral domain. Let $I \subseteq R$ be a subset. Then I is an ideal if and only if I is a fractional ideal.

Proof Clearly if I is a fractional ideal, then I is an ideal. Conversely, if I is an ideal then $rI \subseteq R$ for all $r \in R$ implies that I is a fractional ideal. ■

Proposition 10.2.5 Let R be an integral domain. Let I be an R -submodule of $\text{Frac}(R)$. Then I is invertible if and only if $I^{-1}I = R$.

Proof Clearly if $I^{-1}I = R$ then I is invertible. Now suppose that $JI = R$ for some R -submodule J of $\text{Frac}(R)$. Then we have

$$R = JI \subseteq I^{-1}I = R$$

by definition of I^{-1} . Hence $JI = I^{-1}I$. Multiplying J on both sides and using the fact that R is commutative, we have that $J = I^{-1}$. ■

Lemma 10.2.6 Let R be an integral domain. Let I be an invertible ideal of R . Then for any prime ideal P of R , the ideal IR_P of R_P is a principal ideal.

Proof Since $I^{-1}I = R$, write $1 = \sum_{i=1}^k s_i a_i$ for $s_i \in I^{-1}$ and $a_i \in I$. Since $1 \notin P$, at least one of $s_i a_i$ is not in P . Then $s_i a_i$ is a unit in PR_P and so a_i generates IR_P . ■

Proposition 10.2.7 Let R be an integral domain. Let I be an R -submodule of $\text{Frac}(R)$. Then the following are true.

- If I is a non-zero principal ideal of R , then I is invertible.
- If I is invertible, then I is fractional.
- If I is invertible, then I is finitely generated.

Proof

- Suppose that $I = (a)$ for $a \in R$. Then clearly we have $(1/a)(a) = R$.
- Let I be invertible. Since $I^{-1}I = R$, we can write $1 = \sum_{i=1}^n s_i a_i$ for $s_i \in I^{-1}$ and $a_i \in I$. Then for any $r \in R$, we have $b = \sum_{i=1}^n s_i (a_i b)$ where $a_i b \in R$. Let s be the product of the denominators of s_i . Then $sb \in R$. Hence I is a fractional ideal.
- Let I be invertible. Since $I^{-1}I = R$, we can write $1 = \sum_{i=1}^n s_i a_i$ for $s_i \in I^{-1}$ and $a_i \in I$. Then for any $x \in R$, we have $x = \sum_{i=1}^n (s_i x) a_i$. Since $s_i \in I^{-1}$ and $x \in R$, we have $s_i x \in R$. Hence x can be written as a R -linear combination of $a_1, \dots, a_n$. Hence I is finitely generated. ■

Proposition 10.2.8 Let R be an integral domain. Let I be an R -submodule of $\text{Frac}(R)$. Then I is invertible if and only if the following are true.

- I is fractional.
- I is finitely generated.
- For any prime ideal P of R , IR_P is a principal ideal of R_P .

Proof We have seen the forward direction already. Now suppose that I satisfies the three listed conditions. I claim that $(I^{-1})_P = (I_P)^{-1}$. Let $r/s \in (I^{-1})_P$ and $a/b \in I_P$. Then clearly $r/s \cdot a/b \in R_P$ so that $r/s \in (I_P)^{-1}$. Conversely, suppose that $I = R(a_1, \dots, a_n)$. Let $x \in (I_P)^{-1}$. Then $xa_i \in R_P$. This means that there exists $c_i \in R \setminus P$ such that $xa_i c_i \in R$. Set $c = c_1 \cdots c_n$. Then clearly $cx \in I^{-1}$ so that $x \in (I^{-1})_P$.

Suppose that $I^{-1}I \neq R$. Since $I^{-1}I$ is a proper ideal of R , there exists a maximal ideal m of R containing $I^{-1}I$. By the correspondence of ideals for localization, we have $(I^{-1})_m I_m = (I_m)^{-1} I_m \subseteq$

mR_m . This is a contradiction since the above proposition together with the fact that IR_m is a principal ideal of R_m should imply that $(I_m)^{-1}I_m = R_m$. ■

Proposition 10.2.9 Let R be an integral domain. Let P be a non-zero prime ideal of R . If R is Noetherian and P is invertible, then R_P is a discrete valuation ring.

Proof Let R be a Noetherian integral domain and P a non-zero invertible prime ideal. We know that PR_P is the unique maximal ideal of the local ring R_P . By the above prp, PR_P is a principal ideal. Thus R_P is now a Noetherian local ring with principal maximal ideal. By prp10.4.6 in Commutative Algebra 1, we conclude that R_P is a discrete valuation ring. ■

10.3 Dedekind Domains

Definition 10.3.1 (Dedekind Domains) Let R be an integral domain. We say that R is a dedekind domain if every non-zero ideal I of R is invertible.

Dedekind sought for an integral domain whose ideals can be factorized uniquely as a product of primes.

Proposition 10.3.2 Let R be an integral domain that is not a field. Then the following are equivalent.

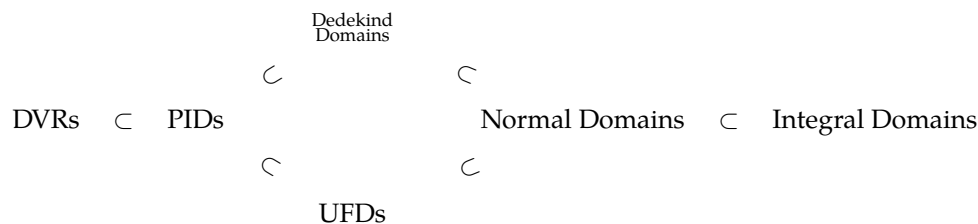
- Every non-zero ideal I of R is invertible ($I^{-1}I = R$).
- R is Noetherian, $\dim(R) = 1$ and normal
- R is Noetherian, $\dim(R) = 1$ and for any non-zero maximal ideal m of R , R_m is a discrete valuation ring.
- R is Noetherian, $\dim(R) = 1$ and every primary ideal in R is a prime power.

Proof

- (1) $\implies$ (2): For any ideal I of R , I is invertible. By 10.2.6, I is finitely generated. Then every R -submodule of R is finitely generated and so R is Noetherian. For any prime ideal P of R , 10.2.8 implies that R_P is a discrete valuation ring since P by assumption is invertible. Then R_P is a normal domain for any prime ideal P . Since normality is a local condition, we conclude that R is a normal domain.
- (2) $\implies$ (3): For any maximal ideal m of R , R_m is Noetherian since localization preserves Noetherianity. Also, R_m is local. Since normality is a local condition, R_m is also normal. Finally, we have $\dim(R_m) = \dim(R) = 1$. Hence by the equivalent characterizations of DVRs, we conclude that R_m is a DVR.
- (3) $\implies$ (1): Let $I \subseteq R$ be a fractional ideal of R . We know by 10.1.4 that I is finitely generated. Since R_m is a normal Noetherian local ring of dimension 1, the ideal I_m of R_m must be principal. By 10.2.7 we conclude that I is invertible. ■

By virtue of the fourth item, we can think of Dedekind domains as a patching up of local discrete valuation rings.

We summarize the relation between Dedekind domains and other types of domains in the following diagram:



In particular, DVRs, PIDs and Dedekind domains are 1-dimensional. Moreover, notice that the only difference between DVRs and Dedekind domains is that DVRs are local rings. They both share the fact that they are Noetherian, $\dim(R) = 1$ and normal.

10.4 Prime Factorization of Ideals

Definition 10.4.1 (Prime Factorization of Ideals) Let R be a commutative ring. Let I be an ideal of R . A prime factorization of I consists of maximal ideals $P_1, \dots, P_k$ such that the following are true.

- For some $n_1, \dots, n_k \in \mathbb{N} \setminus \{0\}$, we have

$$I = P_1^{n_1} \cdots P_k^{n_k}$$

- Each $P_1, \dots, P_n \in \text{Ass}(I)$ is an associated prime ideal of I .
- The factorization is unique up to permutation.

Proposition 10.4.2 Let R be an integral domain. Then R is a Dedekind domain if and only if every ideal of R has a prime factorization.

Proposition 10.4.3 Let R be a Dedekind domain. For any prime ideal P of R , denote $v_i : \text{Frac}(R_P) \rightarrow \mathbb{Z}$ the discrete valuation of R_P . Then for any $a \in R \setminus \{0\}$, we have

$$(a) = P_1^{v_1(a)} \cdots P_n^{v_n(a)}$$

for $P_1, \dots, P_n \in \text{Ass}((a))$.

Proposition 10.4.4 Let R be a Dedekind domain. Let I and J be ideals of R whose prime factorization is given by

$$I = P_1^{a_1} \times \cdots \times P_n^{a_n} \quad \text{and} \quad J = P_1^{b_1} \times \cdots \times P_n^{b_n}$$

for $P_1, \dots, P_n$ distinct prime ideals of R . Then the following are true.

- $I + J = P_1^{\min\{a_1, b_1\}} \times \cdots \times P_n^{\min\{a_n, b_n\}}$
- $I \cap J = P_1^{\max\{a_1, b_1\}} \times \cdots \times P_n^{\max\{a_n, b_n\}}$
- $IJ = P_1^{a_1+b_1} \times \cdots \times P_n^{a_n+b_n}$

Proposition 10.4.5 Let R be a Dedekind domain. Let I be an ideal of R . Then the following are true.

- For any $a \in I$, there exists $b \in R$ such that $I = (a, b)$.
- I is can be finitely generated by two elements.